Dell PowerFlex Manager 5.0.x

User Guide

October 2025
Rev. 1.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1

Introduction

This document provides procedures on how to manage PowerFlex rack, PowerFlex appliance, or Powerflex software through PowerFlex Manager.

The target audience for this document includes Dell Technologies customers and services.

SupportAssist is now known as Dell Technologies connectivity services. References in the documentation will be updated over time.

For additional PowerFlex software documentation, go to PowerFlex Software: Technical Documentation.

Common PowerFlex terms

The following table identifies common PowerFlex terms and associated acronyms.

Table 1. Common PowerFlex terms

Term	Acronym
device gateway target	DGWT
management virtual machine	MVM
metadata manager	MDM
PowerFlex data server	PDS
PowerFlex management cluster	PFMC
PowerFlex management platform	PFMP
storage data client	SDC
storage data target	SDT

See the PowerFlex glossary for more information about PowerFlex terms and acronyms.

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Revision history

Table 2. Revisions

Date	Document revision	Description of changes
October 2025	1.0	Initial release

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Welcome to PowerFlex Manager

This section introduces the main features of PowerFlex Manager.

PowerFlex Manager provides a level of IT operations management based on your chosen consumption model (Dell PowerFlex rack, Dell PowerFlex appliance, custom node, or just PowerFlex software). It increases efficiency by reducing time-consuming manual tasks that are required to manage system operations. Use PowerFlex Manager to deploy and manage new and existing environments.

PowerFlex Manager includes the following features:

- A dashboard that provides system configuration details and communicates health status for infrastructure elements and resource groups
- Firmware and software compliance monitoring and reporting
- Compliance remediation for nodes, switches, VMware ESXi, PowerFlex, and CloudLink
- Hardware monitoring and alerting through either Dell Technologies connectivity services, email, or SNMP and syslog to Dell Technologies Support
- Aggregated logging for troubleshooting with the ability to send logs through Dell Technologies connectivity services to Dell Technologies Support

Use PowerFlex Manager to define and capture infrastructure requirements to support a specific business outcome. Once defined, the process can be repeated through automation.

New features in 5.0

This PowerFlex release introduces the following new features:

Table 3. New features

Feature	Details
Erasure coding architecture	In this release, erasure coding is introduced to PowerFlex. Erasure coding is a data protection method that splits data into multiple fragments, or data strips, encodes it, and distributes the strips across different storage nodes in multiple locations. Erasure coding then creates additional fragments, or parity strips, that are stored on separate parity nodes.
Protection scheme support	The PowerFlex erasure coding architecture supports two protection schemes: • 2 + 2 • 8 + 2
Support for storage node	In this release, the ability to add a storage node to your PowerFlex system is introduced. You can now scale up your storage without having to add compute resources. The storage node enables data to be distributed across multiple nodes using a high-speed network providing fault tolerance. If one node fails, the system can continue operating with minimal impact and protection against loss of data. The storage node can be accessed and managed under the **Lifecycle** tab.
Device groups	In this release, a new object is introduced: device group. The device group can be viewed under the **Block** tab in PowerFlex Manager. The device group is responsible for grouping devices of a similar type together making it easier to manage your devices. PowerFlex supports two different types of device groups: storage devices and persistent memory devices (SDPMs and NVDIMMs).
Thin clones	In this release, support for thin clones is added and these can be managed using PowerFlex Manager. Thin clones allow for writable copies of snapshots without consuming much storage space, because they share data blocks with the parent snapshot. You can modify data within thin clones while preserving the original snapshot, making it possible to create multiple versions of data over time. Also,

Table 3. New features (continued)

Feature	Details
	thin clones remain unaffected even if the parent snapshot is deleted, ensuring data integrity and continuity.
Unlimited over-provisioning control	In this release, over-provisioning is unlimited by default. Enabling over-provisioning control on storage pools allows PowerFlex Manager to limit the size of the volumes and snapshots that are created by percentages. For example, if the storage pool is 10 TB and the over-provisioning factor is 200%, you can create thin volumes up to 20 TB.
Allocation granularity of 1 GB	The minimum size of a volume is 1 GB. In this release, you can increase the volume size in increments of 1 GB. For example, if you have a volume that is 10 GB, you can increase it to 11 GB.
Support for write cache with persistent memory	In this release, write cache is written to a persistent memory device (battery-backed write cache (BBWC)), enhancing performance and reducing the wear leveling of the storage device.
Support for compression with or without persistent memory	In this release, compression is enabled by default and is no longer dependent on persistent memory.
Usability enhancements	In this release, the PowerFlex Manager user interface has undergone several usability enhancements: ● Restructuring of the **Block** tab, which makes it easier to manage the storage system. The storage system entities are now under one storage system tab. ● Enhanced navigation through the topology section. ● More detailed metrics in the dashboard. ● Enhanced reporting of events and actionable alerts with the remediation steps.
Operating system support	This release of PowerFlex is supported on the latest operating systems. You can access the supported operating systems by selecting **PowerFlex** in the Dell PowerFlex Simple Support Matrix.
Single component upgrade	In this release, there is support for upgrading individual components, such as applying security patches without performing a full RCM/IC upgrade.

Terminology

The following table describes important PowerFlex Manager terms and concepts.

Table 4. Terminology

Term	How used in PowerFlex Manager
Compliance	Compliance is the comparison between current firmware and software levels and target compliance versions, which contain the baseline firmware and software versions. PowerFlex Manager shows any deviation from the baseline in the compliance status of the resources. You can use PowerFlex Manager to initiate updates to bring the resources to a compliant state. To facilitate upgrades of the management software, and compliance upgrades. PowerFlex Manager steers you toward valid upgrade paths. If the current version of the software is incompatible with the target version, when you attempt an upgrade, PowerFlex Manager displays a warning. PowerFlex Manager also warns you if any of the RCM or IC versions that are loaded are incompatible with the target compliance versions.
Destination	A destination defines the external location where PowerFlex Manager sends events and alerts to, for example, SNMP.
Device group	A device group consists of devices that come from five or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same media type.

Table 4. Terminology (continued)

Term	How used in PowerFlex Manager
Device	A storage device or persistent memory device that is added to a storage system.
Device Gateway Target	A Device Gateway Target (DGWT) is a distributed capacity service running on the storage node. The DGWT service processes and saves the data to the devices. The component is accountable for placing data on the storage drives using erasure coding protection schemes.
Host	A host is an entity that consumes PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.
NVMe target	A frontend component that translates NVMe over Fabric protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.
PowerFlex data server	A PowerFlex data server(PDS) is a storage service that is responsible for processing host I/Os. The PDS service organizes the data within the storage node and divides it into smaller chunks of data of equal size.
Protection domain	A protection domain is a logical entity that consists of and manages a group of storage nodes each of which contains devices. A storage node contains both the PowerFlex Data server (PDS) and a Device Gateway Target (DGWT). The PDS and DGWT services enable the storage and processing of data to be separated. Data is logically isolated into specific datasets and performance capabilities within a specified protection domain. The effect of any specific device failure is limited.
Repository	The location of firmware and software packages from which updates can be performed. You can create more than one repository, selecting one as the default. This default repository is used to calculate compliance for all undeployed resources. PowerFlex Manager lets you specify other repositories for deployed resources. The operating system is included with the compliance version, and you can select the operating system that you want to use from the compliance version.
Resource	The PowerFlex nodes, network switches, and hypervisors in the system. When PowerFlex Manager discovers resources, it identifies each component and how to talk to its management interface.
Resource group	A group of resources that are managed as a cluster. It is the state of your deployed configuration. PowerFlex Manager performs health monitoring, compliance monitoring and remediation, and the ability to add or remove resources. A resource group is the finished outcome.
Snapshot	Snapshots are read-only copies of volumes that are used for local data protection. They provide a method of recovery for data that has been corrupted or accidentally deleted.
Software management	Software management is the term that is used to deploy and configure PowerFlex software on a set of nodes on an existing operating system. The term is also referred to as software-only or just PowerFlex software.
Source	A source defines what PowerFlex Manager should listen for in order to forward events and alerts to a destination, for example, PowerFlex alerts or SNMP forwarding. In PowerFlex Manager, PowerFlex internal sources are treated as sources within the context of notification policy rules.
Storage pool	A logical structure where volumes are created.
Template	A template contains the configuration requirements that must be applied to a resource or group of resources during deployment. These requirements include firmware and software, operating system and or hypervisor, and PowerFlex. A template represents the wanted state of your deployed configuration.

Table 4. Terminology (continued)

Term	How used in PowerFlex Manager
	PowerFlex Manager provides sample templates that represent the standard configuration options. You can clone a sample and use it for your implementation. You can also define a custom template if necessary.
Thin clone	A thin clone is a read/write copy of a volume or a snapshot that allows changes to content using writes. Thin clones are thin copies of the object from which they were created.
Volume	A unit of storage that can be identified by a unique ID or name. PowerFlex objects are assigned a unique ID that can be used to identify the object. The default name for each volume object is its ID. Dell highly recommends giving each volume a meaningful name that is associated with its operational role.

User roles

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

The following table summarizes the activities that can be performed for each user role:

Table 5. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage lifecycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources • Manage lifecycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block systems, such as create volume. **NOTE:** Operations such as create storage pool cannot be performed by the StorageAdmin	• Manage storage resources • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)

Table 5. User roles and activities (continued)

Role	Description	Activities
	role, but can be performed by the LifecycleAdmin role.	
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage lifecycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
SnapshotManager	SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations required to set up and manage snapshots.	• View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
DriveReplacer	This is a subset of the Technician role. The DriveReplacer can perform only operations required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Monitor	The Monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The Support role is a special kind of SystemAdmin (all activities except for user/security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special, often dangerous, commands for advanced troubleshooting.	• Manage storage resources • Manage lifecycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Special Dell Technologies Support operations

LDAP users and security permissions

LDAP security permissions are managed through group memberships, allowing permissions to be assigned to groups instead of individual users. This streamlines administration and improves security by ensuring only authorized users can access specific resources.

You cannot assign multiple roles to a group. All roles in a group have a proximate hierarchy.

Table 6. LDAP hierarchy

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
1	N/A (local user)	N/A (local user)

Table 6. LDAP hierarchy (continued)

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
2	Admin	Super-User
3	Admin	Super-User
4	Admin	Super-User
5	StorageAdmin or LifecycleAdmin	Super-User
6	Monitor	SecurityAdmin
Not ranked	SecurityAdmin	SecurityAdmin

4

Initial configuration and setup

This section includes tasks you need to perform when you first begin using PowerFlex Manager.

Initial configuration

The first time that you log in to PowerFlex Manager, you are prompted with an **Initial Configuration Wizard**, which prompts you to configure the basic settings that are required to start using PowerFlex Manager.

Before you begin know the following:

- The Dell Technologies connectivity services configuration details.
- Whether you intend to use a Release Certification Matrix (RCM) or Intelligent Catalog (IC).
- The type of installation you want to perform, including details about your existing PowerFlex instance when you intend to import from another PowerFlex instance.

To configure the basic settings:

1. On the **Welcome** page, read the instructions and click **Next**.
2. On the **SupportAssist** page, optionally enable Dell Technologies connectivity services and specify Dell Technologies connectivity services connection settings, and click **Next**.

   NOTE: If you are enabling Dell Technologies connectivity services the secure connect gateway and policy manager must be deployed and configured before you can enable Dell Technologies connectivity services,

3. On the **Installation Type** page, specify whether you want to deploy a new instance of PowerFlex or import an existing instance, and click **Next**.
4. On the **Summary** page, verify all settings for Dell Technologies connectivity services and installation type. Click **Finish** to complete the initial configuration.

After completing the **Initial Configuration Wizard**, you can get started using PowerFlex Manager from the **Getting Started** page.

Enabling Dell Technologies connectivity services

Dell Technologies connectivity services is a secure support technology for the data center. Dell Technologies connectivity services refers to secure connect gateway, which is used for call home functionality and remote connectivity. You can enable Dell Technologies connectivity services as part of the initial configuration wizard. Alternatively, you can enable it later by adding it as a destination to a notification policy in Events and Alerts.

Ensure that secure connect gateway and policy manager are deployed and configured.

PowerFlex Manager provides support through integration with the secure connect gateway, ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. Dell Technologies connectivity services is the functionality that enables this connection.

When you enable Dell Technologies connectivity services, you can take advantage of the following benefits, depending on the service agreement on your device:

- Automated issue detection - Dell Technologies connectivity services monitors your Dell Technologies devices and automatically detects hardware issues, both proactively and predictively.
- Automated case creation - When an issue is detected, Dell Technologies connectivity services automatically opens a support case with Dell Technologies Support.
- Automated diagnostic collection - Dell Technologies connectivity services automatically collects system state information from your devices and uploads it securely to Dell Technologies. Dell Technologies Support uses this information to troubleshoot the issue.
- Proactive contact - A Dell Technologies Support agent contacts you about the support case and helps you resolve the issue.

The first time that you access the initial configuration wizard, the connection status displays as not configured.

Ensure that you have the details about your Dell Technologies connectivity services configuration.

If you do not enable Dell Technologies connectivity services in the initial setup, you can add it later as a destination when adding a notification policy through **Settings** > **Events and Alerts** > **Notification Policies**.

1. Click **Enable SupportAssist**.
2. From the **Connection Type** tab, there are two options:

If you want to:	Do the following:
Connect directly to Dell Technologies.	Click **Connect Directly**.
Connect to a Dell Technologies connectivity services gateway server that connects to Dell Technologies by remote access.	Click **Connect via Gateway Server** and provide the following details: • The IP and port number that is assigned to the primary and secondary gateway servers. The port number is auto populated to **9443**. • Optionally, click **Add Gateway** to include an additional gateway server. The maximum number of gateways is eight. • If your environment has a proxy server, enter the IP address, port number, username, and password for the proxy server. • Enter the access key and PIN for authentication to Dell Technologies connectivity services. • Optionally, click **Test connection** to test the connection status. • Optionally, click **Send Test Alert** to send a test alert.

3. From the **Support Contacts** tab, identify the primary support contact by providing the contact's first name, last name, email, and phone details. Optionally, you can add up to two additional support contacts.
4. From the **Device Registration** tab, register your device by completing the following steps:
   a. Select the type of device registration from the **Type** menu.
   b. Depending on the device registration type, you may have to enter information:

Table 7. Device registration

Device registration type	Information required
• PowerFlex rack • PowerFlex appliance	• Enterprise License Management Systems (ELMS) Software Unique ID (eSWUID) • Solution serial number
PowerFlex software	N/A

5. From the **Customize Connection to SupportAssist** tab, you can:
   a. Click **Connect to CloudIQ** to enable Dell Technologies to send telemetry data, alerts, and analytics through Dell Technologies connectivity services.
   b. Click **Enable Remote Support** to enable authorized Dell Technologies support engineers to troubleshoot your system remotely.
6. From the **Re-authenticate SupportAssist** tab, click **request a New Access Key and PIN** to generate an access key and pin. The **AccessKey Portal** opens.
   a. From the menu, select **SERIALNUMBER**.
   b. Enter the serial number of your device or PowerFlex software and click **Submit**.
   c. Click **Generate New Access key**.
7. From the **Re-authenticate SupportAssist** tab, enter the access key and pin.
8. Click **Next**.

After you complete the initial configuration, go to the **Settings** page and verify that Dell Technologies connectivity services is turned on. You may need to configure the Dell Technologies connectivity services policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.

Related information

Events
Viewing an event
Alerts
License management

Add a notification policy

Specifying the installation type

If you have an existing PowerFlex deployment that you would like to import, you can specify details about this deployment as part of the initial configuration. Alternatively, if you would like to deploy a new instance, you can bypass the import step as part of the initial configuration.

The initial configuration wizard supports two different installation workflows; a new instance of PowerFlex or the import of a PowerFlex instance.

1. Specify the type of installation that you want to perform:
   - If you do not have an existing PowerFlex deployment and would like to bypass the import step, click **I want to deploy a new instance of PowerFlex**.
   - if you would like to import an existing PowerFlex directly from an existing PowerFlex Manager virtual appliance, click **I have a PowerFlex instance managed by PowerFlex Manager to import**.

     Provide the following details about the existing PowerFlex Manager virtual appliance:
     - IP address or DNS name for the virtual appliance
     - Username and password for the virtual appliance
2. Provide the following details about the existing PowerFlex instance:
   - IP addresses for all nodes with a primary and secondary MDM
   - System ID for the cluster
   - LIA password
3. Click **Next** to proceed.

Deploy a PowerFlex cluster using a CSV topology file

Use the following procedure to deploy a PowerFlex cluster using a CSV topology file.

1. Log in to PowerFlex Manager.
2. On the **Getting Started** page, click **Deploy With Installation File** to provide an installation topology file for deployment of nodes.
3. In the **PowerFlex Installer** window, click **Browse** to upload the appropriate artifacts (RPMs) according to the operating system of your PowerFlex operating system machines.
4. Click **Upload Files** and click **Next** when the RPMs have finished uploading.
5. Click **Browse** to upload an installation CSV topology file that describes the desired topology of PowerFlex.

   To view sample files, click the links to the following two example files: **PowerFlex_config_example_all fields.csv (all fields)** and **PowerFlex_basic_example_minimum fields.csv (minimum fields).**
6. Click **Upload Installation CSV**.
7. Select the **I accept the terms of the End User License Agreement (mandatory)** check box.
8. Click **Next**.
9. In the **Advanced Settings** tab, you can select how you would like to install PowerFlex 5.0.x.
   - You can skip options to upload, install, configure, and / or validate a device.
   - To overwrite data, select **Force overwrite of existing data on devices used for PowerFlex storage**.

     PowerFlex can be installed on new or clean disks (Storage Node Devices). If the installation is done on a previously deployed PowerFlex cluster, or used disks, you must tell the installer to overwrite the older data.

     NOTE: If you select this option, the data is erased.
   - The device awareness check is enabled by default. If you disable this setting, it may cause potential device failures not to be detected early enough. To disable the diagnostic, turn **Enable Diagnostics Check** to off.
10. Click **Next**.
11. In the **Summary** tab, verify all the configurations and click **Finish**.
12. Click the **Go to Events Page** button to track the installation progress.

Verifying the initial configuration

Before you complete the initial configuration, you can review the settings that you provided for Dell Technologies connectivity services, compliance, and installation type.

1. On the **Summary** page, verify the settings that you configured on the previous pages.
2. To edit any information, click **Back** or click the corresponding page name in the left pane.
3. If you are importing an existing PowerFlex instance from PowerFlex Manager, type **IMPORT POWERFLEX MANAGER**.
4. If the information is correct, click **Finish** to complete the initial configuration.

   If you are importing an existing environment, PowerFlex Manager displays a message indicating that the import operation is in progress. When the import operation is complete, PowerFlex Manager displays the **Getting Started** page.

After completing the **Initial Configuration Wizard**, you must perform some additional steps. The steps vary depending on which installation type you selected:

- If you clicked **I want to deploy a new instance of PowerFlex** on the **Installation Type** page, you now must deploy a new instance of PowerFlex. The **Getting Started** page provides more information about what to do next.
- If you clicked **I have a PowerFlex instance managed by PowerFlex Manager to import** on the **Installation Type** page, you must perform these steps:
  1. On the **Settings** page, upload the compatibility matrix file and upload the latest repository catalog (RCM or IC).
  2. On the **Resources** page, select the PowerFlex entry, and perform a nondisruptive update.
  3. On the **Resource Groups** page, perform an RCM/IC upgrade on any migrated service that must be upgraded.

  CAUTION: Check the Alerts page before performing the upgrade. Look for major and critical alerts that are related to PowerFlex Block to be sure that the MDM cluster is healthy before proceeding.

Getting started

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

The following table describes each step:

Table 8. Getting Started steps

Step	Description
Upload Compliance File	Provide compliance file location and authentication information for use within PowerFlex Manager. The compliance file defines the specific hardware components and software version combinations that are tested and certified by Dell for Dell products. This step enables you to choose a default compliance version for compliance or add new compliance versions. You can also click **Settings** > **Repositories** > **Compliance Versions**. **NOTE:** Before you make an RCM or IC the default compliance version, you must first upload a suitable compatibility management file under **Settings** > **Repositories** > **Compatibility Management**.
Define Networks	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Enter detailed information about the available networks in the environment. This information is used later during deployments that are based on templates and resource groups. These deployments use the network information to configure nodes and switches to have the right network connectivity. PowerFlex Manager uses the defined networks in templates to specify the networks or VLANs that are configured on nodes and switches for your resource groups.

Table 8. Getting Started steps (continued)

Step	Description
	This step is enabled immediately after you perform an initial configuration for PowerFlex Manager. You can also click **Settings** > **Networking** > **Networks**.
Discover Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Grant PowerFlex Manager access to resources (nodes, switches, virtual machine managers) in the environment by providing the management IP and credential for the resources to be discovered. This step is not enabled until you define your networks. You can also click **Resources** > **Discover Resources**.
Manage Deployed Resources (Optional)	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Add an existing resource group for a cluster that is already deployed and manage the resources within PowerFlex Manager. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Resource Groups** > **Add Existing Resource Group**.
Deploy Resources	If you plan to perform an advanced CSV-based deployment, this step can be skipped. Create a template with requirements that must be followed during a deployment. Templates enable you to automate the process of configuring and deploying infrastructure and workloads. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and make modifications as needed. Choose the sample template that is most appropriate for your environment. For a two-layer deployment, clone the compute-only templates. Then clone one of the storage templates. This step is not enabled until you define your networks. You can also click **Lifecycle** > **Templates**.
Manage PowerFlex License	Configure licensing for PowerFlex. You can also click **Settings** > **License Management**.

To revisit the **Getting Started** page, click **Getting Started** on the help menu.

Related information

Networking
Discover a resource
Adding an existing resource group
Templates
License management
Add a storage node

Changing your password

When you first log in to PowerFlex Manager, you must set your password. You can also change your password at any time after the first login.

1. Click the user icon in the upper right corner of PowerFlex Manager.
2. Click **Change password**.
3. Type the password in the **New Password** field.

   Password rules are:

   - Includes a minimum of eight and a maximum of 32 characters
   - Includes only alphanumeric and punctuation characters
4. Type the password again in the **Verify Password** field.
5. Click **Apply**.

5

Performing common tasks

This section includes tasks that you perform regularly when using PowerFlex Manager.

Deploying and provisioning

The following table describes common tasks for deploying and provisioning the system and what steps to take in PowerFlex Manager to initiate each task:

Table 9. Common tasks

If you want to...	Do this in PowerFlex Manager...
Define a resource group and infrastructure requirements	1. If you are managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks that you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. For example, for a two-layer deployment, clone one of the PowerFlex storage-only templates. Then clone one of the PowerFlex compute-only templates. You can also create a template. However, for most environments, you can clone one of the sample templates that are provided with PowerFlex Manager.
Deploy a new resource group	Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. You can only deploy a resource group using a published template.
Perform an advanced PowerFlex CSV deployment	If you would like to use the legacy PowerFlex Installer to provide an installation topology file, click **Deploy With Installation File** on the **Getting Started** page. See the topic on deploying a PowerFlex cluster using a CSV topology file.

Related information

Repositories
Networking
Resources
Preparing a template for a resource group deployment
Deploying a resource configuration in a resource group

Configuring block storage

The following table describes common tasks for deploying and managing block storage and what steps to take in PowerFlex Manager to initiate each task:

Table 10. Common tasks

If you want to...	Do this in PowerFlex Manager...
Deploy a new block storage configuration	1. If you are managing and updating components other than PowerFlex, upload the compliance file on the **Settings** > **Repositories** > **Compliance Versions** page. 2. Define the networks that you need on the **Settings** > **Networking** > **Networks** page. 3. Discover the resources required for your deployment on the **Resources** > **Discover Resources** page. 4. Clone a sample template and make changes as needed. Click **Lifecycle** > **Templates** and click **Sample templates**. Open a sample template and click **More Actions** > **Clone** on the right side of the screen. 5. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, click **Deploy New Resource Group**. For a two-layer deployment, clone one of the PowerFlex storage-only templates. Then, clone one of the PowerFlex compute-only templates. You can also create a template. However, for most environments, you can clone one of the sample templates that are provided with PowerFlex Manager.
Import an existing block storage configuration	Click **Lifecycle** > **Resource Groups** and click **+Add Existing Resource Group**. Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Manage block storage	On the menu bar, click **Block**, and choose the type of block storage components you want to manage: • **Storage System** • **Volumes** • **Hosts** If you create objects on the **Block** tab, you must update the inventory on the **Resources** page, and then click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

Related information

Repositories
Networking
Discover a resource
Clone a template
Deploying a resource configuration in a resource group
Managing block storage

Managing components

Once PowerFlex Manager is configured, you can use it to manage the system.

The following table describes common tasks for managing system components and what steps to take in PowerFlex Manager to initiate each task:

Table 11. Common tasks

If you want to...	Do this in PowerFlex Manager...
Add an existing resource group	Click **Lifecycle** > **Resource Groups** and click **+Add Existing Resource Group**.

Table 11. Common tasks (continued)

If you want to...	Do this in PowerFlex Manager...
	Be sure to upload a compliance file (if necessary), define the networks, and discover resources before adding the existing resource group.
Enter maintenance mode	1. Click **Lifecycle** > **Storage Nodes**. 2. From the list of storage nodes, select the checkbox beside the storage node that you want to enter into maintenance mode and click **More Actions** > **Enter Maintenance Mode**. 3. In the **Enter Storage Node into Maintenance Mode** dialog box, verify that you are selecting the wanted storage node, and click **Enter Maintenance Mode**.
Exit maintenance mode	1. Click **Lifecycle** > **Storage Nodes**. 2. From the list of storage nodes, select the checkbox beside the storage node that you want to exit from maintenance mode and click **More Actions** > **Exit Maintenance Mode**. 3. In the **Exit Maintenance Mode on Storage Node** dialog box, verify that you are selecting the wanted storage node, and click **Exit Maintenance Mode**.
Enter service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Enter Service Mode**.
Exit service mode (applicable for PowerFlex appliance and PowerFlex rack only)	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Exit Service Mode**.
Perform node expansion	1. Click **Lifecycle** > **Resource Groups**. On the **Resource Groups** page, select a resource group. 2. On the **Resource Group Details** tab, under **Add Resources**, click **Add Nodes**. The procedure is the same for new resource groups and existing resource groups.
Remove a node	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, under **More Actions**, click **Remove Resource**. 4. Select **Delete Resource** for the **Resource removal type**.
Replace a drive	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. Under **Physical Nodes**, click **Drive Replacement**.
Reconfigure MDM roles	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, click **Reconfigure MDM Roles** under **More Actions**. You can also reconfigure MDM roles from the **Resources** page. Select a PowerFlex gateway and click **View Details**. Then, click **Reconfigure MDM Roles**.
Run inventory (nodes, switches, PowerFlex gateway, and VMware vCenter cluster)	1. Click **Resources** and click the **All Resources** tab. 2. Click the checkbox for the resource you want to update and then click **Run Inventory**. 3. After running the inventory, click **Update Resource Group Details** under **More Actions** on the **Resource Groups** page for any resource group that requires the updated resource data.
View network topology	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** tab, click the **Port View** tab.

Related information

Lifecycle
Resource groups
Templates
Resources

Monitoring system health

Once PowerFlex Manager is configured, you can monitor system health.

The following table describes common tasks for monitoring system health and managing software and firmware compliance and what steps to take in PowerFlex Manager to initiate each task:

Table 12. Common tasks

If you want to...	Do this in PowerFlex Manager...
Download a report that lists compliance details for all resources	1. Click **Resources**. 2. Click **Export Report** and download a compliance report (PDF or CSV) or a configuration report (PDF).
Generate a troubleshooting bundle	1. Click **Settings** and click **Serviceability**. 2. Click **Generate Troubleshooting Bundle**. You can also generate a troubleshooting bundle from the **Resource Groups** page: 1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **Generate Troubleshooting Bundle**.
Monitor software and firmware compliance	1. Click **Lifecycle** > **Resource Groups**. 2. On the **Resource Groups** page, select a resource group. 3. On the **Resource Group Details** page, click **View Compliance Report**.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, **Raw Capacity**, **Data Savings**, and **Resources / Inventory** sections. For information about which resource groups are healthy and in compliance and which are not, look at the **Resource Groups** section.
Monitor system resources and health	On the **Dashboard**, look at the **Overall Performance**, **Usable Capacity**, ,**Data Savings** sections. For information about which resource groups are healthy and in compliance, and which are not, look at the **Resource Groups** section.
Perform software and firmware remediation	From the compliance report, view the firmware or software components. Click **Update Resources** to update non-compliant resources.
View alerts	On the menu bar, click **Monitoring** > **Alerts**.

Related information

Lifecycle
Resources

Managing external changes

If you make manual changes outside of PowerFlex Manager, you might need to perform some steps within PowerFlex Manager to ensure that the external changes are reflected within the user interface and the environment is kept in a healthy state.

The following table describes common tasks for managing external changes and what steps to take in PowerFlex Manager for each task:

Table 13. Common tasks

If you have...	Do this in PowerFlex Manager...
Created objects on the **Block** tab	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.
Manually added nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated nodes.
Manually added VMware NSX to a cluster outside of PowerFlex Manager	Click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with VMware NSX, which will be in life cycle mode.
Manually changed the IP address for a switch	1. Remove the switch. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the switch, and click **Remove**. 2. Rediscover the switch. On the **Resources** page, click the **All Resources** tab and click **Discover**. 3. If the IP address is in use, remove the resource group and add it again. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**.
Manually created additional VLANs on the VDS in vCenter outside of PowerFlex Manager	1. Remove the resource group. On the **Resource Group Details** page, click **Remove Resource Group** under **More Actions**. When you click **Remove Resource Group**, do not select **Delete Resource Group** as the **Resource Group Removal Type**. This action completely tears down the resource group and makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Instead, select **Remove Resource Group** as the **Resource Group Removal Type**. Then, to keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and choose **Managed**. 2. Rediscover the resource group. On the **Lifecycle** > **Resource Groups** page, click **+Add Existing Resource Group**. If you do not choose the correct options when you remove the resource group, you could tear down the resource group and destroy it, or leave the servers in an unmanaged state, and not be able to add the existing resource group.
Manually created volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updated volumes.
Manually deleted volumes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires updated information about volumes deleted. PowerFlex Manager displays a message indicating that the volumes have been removed from the resource group.
Manually removed nodes outside of PowerFlex Manager	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Resource Groups** page for any resource group that requires updated information about nodes deleted. Then, manually remove the resource on the **Resources** page by clicking the **All Resources** tab, selecting the resource, and clicking **Remove**.

Table 13. Common tasks (continued)

If you have...	Do this in PowerFlex Manager...
Manually replaced a node outside of PowerFlex Manager for a resource group in PowerFlex Manager	1. Remove the node from the resource group. On the **Lifecycle** > **Resource Groups** page, select a resource group. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource**. 2. Remove the node from the list of resources. On the **Resources** page, click the **All Resources** tab. From the list of resources, select the resource, and click **Remove**. 3. Discover the new node. On the **Resources** page, click **Discover** on the **All Resources** tab. 4. Update the resource group details. On the **Lifecycle** > **Resource Groups** page, click **Update Resource Group Details**.
Renamed objects such as VMware ESXi host, volume, datastore, VDS, port group, data center, cluster, and so forth	Click **Run Inventory** on the **Resources** page to update the inventory. Then, click **Update Resource Group Details** on the **Lifecycle** > **Resource Groups** page for any resource group that requires the updates.

Related information

Lifecycle
Resource groups

6

Displaying system information at a glance

This section introduces the dashboard, which displays a graphical view of the overall health of the system.

The dashboard displays at-a-glance information about overall performance, usable and raw capacity, data savings, resources and inventory, resource groups, and alerts.

Interpreting overall performance and bandwidth

This section displays a graph that shows the overall performance (Input/Output Operations Per Second (IOPS) and bandwidth) for a block configuration within the system.

IOPS measures how many read and write operations the system can handle per second. Bandwidth measures the total amount of data that is transferred per second in GB/s.

The relationship between IOPS and Bandwidth can be summarized as follows:

- High IOPS, Low Bandwidth: This scenario (which is shown in Figure 1: Overall performance) occurs when there are many small I/O operations. Each operation transfers a small amount of data, resulting in high IOPS (7265 K) but relatively low bandwidth (4.84 GB/s).
- Low IOPS, High Bandwidth: This scenario occurs when there are fewer, larger I/O operations. Each operation transfers a large amount of data, leading to lower IOPS but higher bandwidth.

You can hover over the graph at any time to get the exact metrics for that point in time.

The following graph is an example of how overall performance (IOPS and bandwidth) for a block configuration is represented in the dashboard:

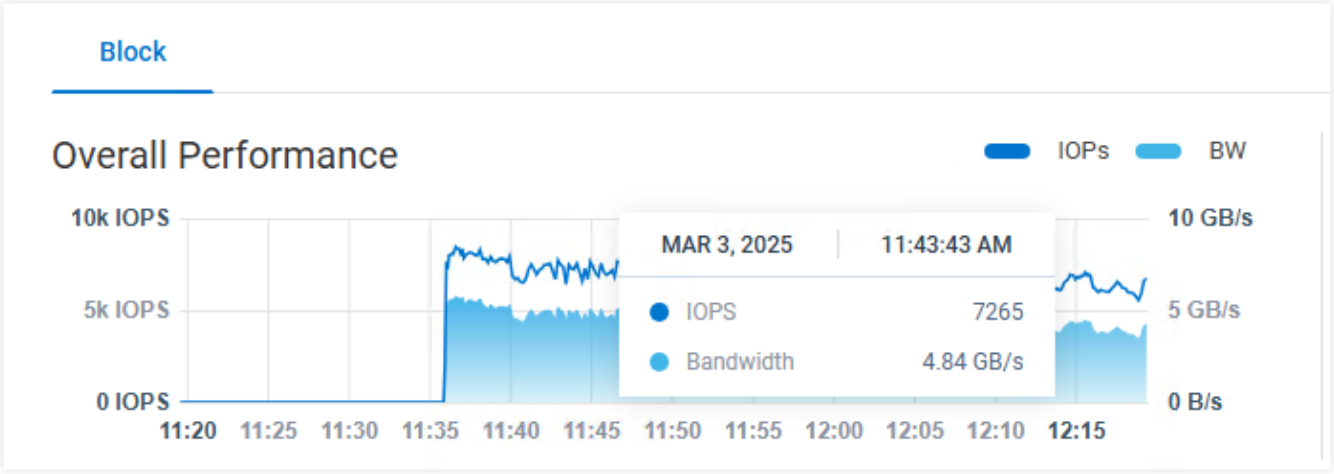


Figure 1. Overall performance

Interpreting latency

This section displays a graph that shows latency levels for a block configuration within the system.

Latency is the time that it takes for a single I/O operation to complete typically measured in milliseconds (ms). It represents the delay between when a request (reading or writing) for data is made and the data is delivered.

The relationship between latency, IOPS, and bandwidth can be summarized as follows:

- IOPS and latency: Higher IOPS means lower latency (faster response times) but only up to the limits of the system.

- Bandwidth and latency: High bandwidth does not necessarily mean low latency. A system with high bandwidth but high latency may still feel slow for small, frequent operations.

You can hover over the graph at any time to get the exact metrics for that point in time.

The following graph is an example of how latency for a block configuration is represented in the dashboard:

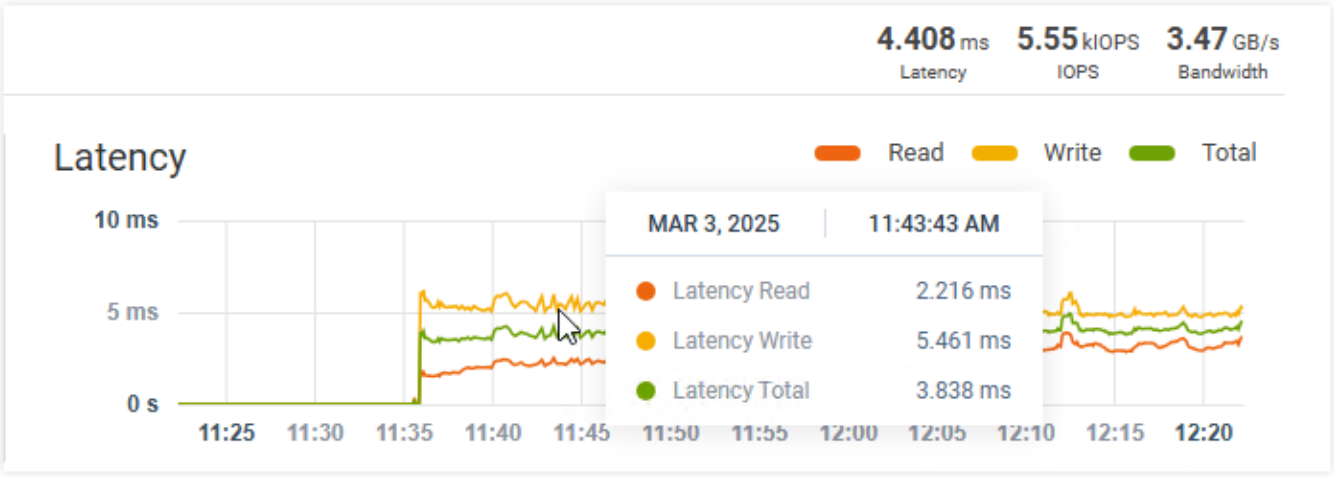


Figure 2. Latency

Interpreting usable capacity

This section displays a graph that shows the usable capacity within the system. Usable capacity is the physical capacity available for user data after applying a protection scheme and accounting for spare capacity.

For example, if you had 11 PowerFlex nodes with each PowerFlex node having 10 TB of capacity. Then, the raw capacity is 110 TB (11 x 10 TB = 110 TB). If you then allocate one PowerFlex node as a spare, the raw capacity reduces to 100 TB. If you apply the protection scheme (8 + 2), the total raw capacity reduces to 80 TB (80% of 100).

Used capacity is calculated as the amount of system capacity that is used as a percentage of the total system capacity.

The following diagram is an example of how usable capacity is represented in the dashboard:

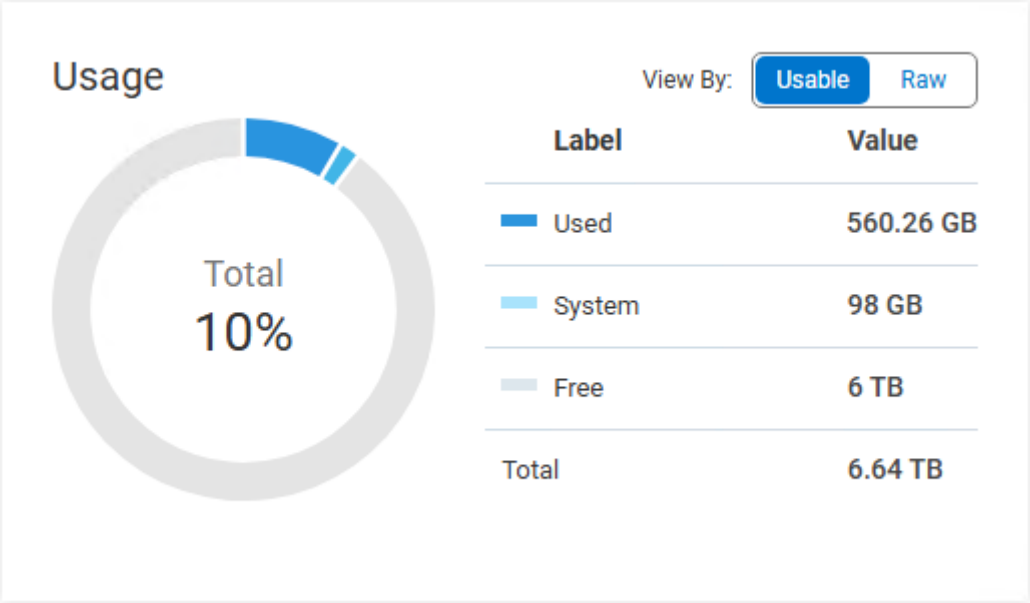


Figure 3. Usable capacity

Table 14. Usable capacity

Capacity	Description
Total	The total protected capacity that is available for user data and system (internal) usage.
Used	The physical capacity used by user data in the system.
System	The physical capacity in use by the system for internal use.
Free	The free physical capacity in the system.

Table 15. Raw capacity

Capacity	Description
Total raw	The total sum of devices in the system.
Raw used	The allocated raw capacity that is used by user data.
Raw system	The raw capacity that is in use by the system for internal use by the system.
Raw free	The free raw capacity that is available to allocate in the system.

Interpreting data savings

This section displays a graph that shows the data savings within the system. It provides details about the overall savings, thin and snapshot savings, and the data reduction ratio.

The following terms around data savings are used in the dashboard:

- Provisioned: The total size of volumes and snapshots in the system.
- Logical: The logical capacity in use by user data and snapshots in the system.
- Physical: The amount of storage that the user data and snapshots take after data reduction in the system. The physical resides on the storage pools.

The following graph is an example of how data savings are represented in the dashboard:

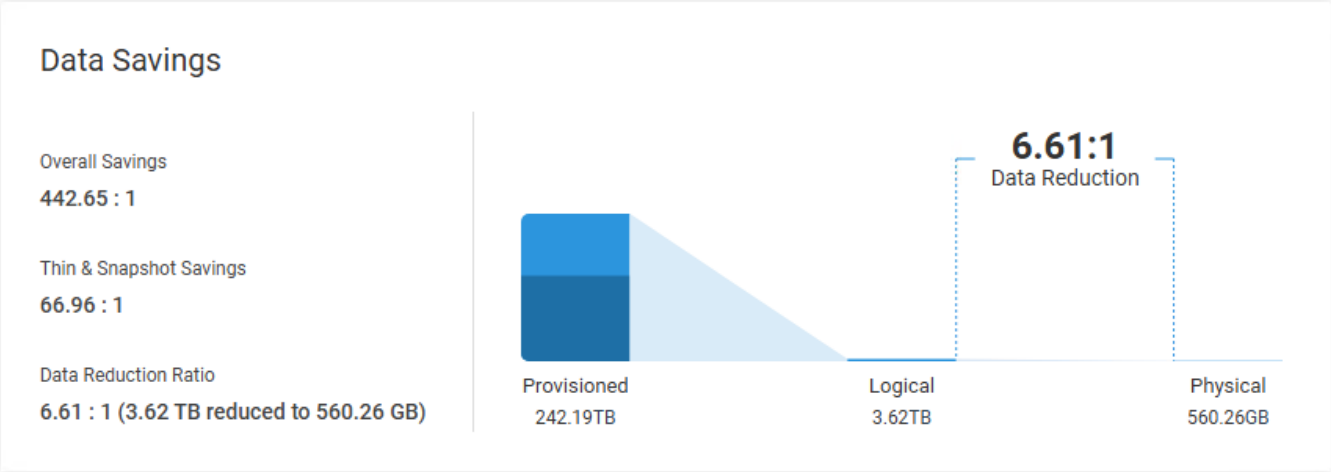


Figure 4. Data savings

Table 16. Data saving type

Savings	Description	Example
Overall savings	The ratio between provisioned and physical storage.	442.65% or 99.77% This is 247,611.36 GB / 560.26 GB = 441.95 GB.

Table 16. Data saving type (continued)

Savings	Description	Example
		Or 1/441.95 = 0.002262 (1-0.002262) x 100 = 99.77%
Thin and Snapshot Savings	The ratio between provisioned and logical used capacity.	For example, if 242.19 TB is provisioned to snapshots but only 3.62 TB is used on snapshots, the thin and snapshot savings ratio is 66.90% or 66%.
Data reduction ratio	The ratio of the logical used space to the data physical space after data reduction, for example, compression.	For example, if the original data size is 3706.88 GB and the reduced data size after compression is 560.26 GB, the data reduction ratio would be 6.61:1 (3706.88 GB/560.26 GB).

Resources/Inventory

This section displays the number of resources that are in the current inventory.

The list of resources include the following:

- VM managers
- Nodes
- Switches
- Protection domains
- Device groups
- Devices
- Storage pools
- Volumes
- Hosts

Resource groups

This section displays a graphical representation of the resource groups deployed based on status (for PowerFlex rack and PowerFlex appliance users only). The number next to each icon indicates the number of resource groups in a particular state.

The following diagram is an example of how the health and compliance of the resource groups are represented in the dashboard:

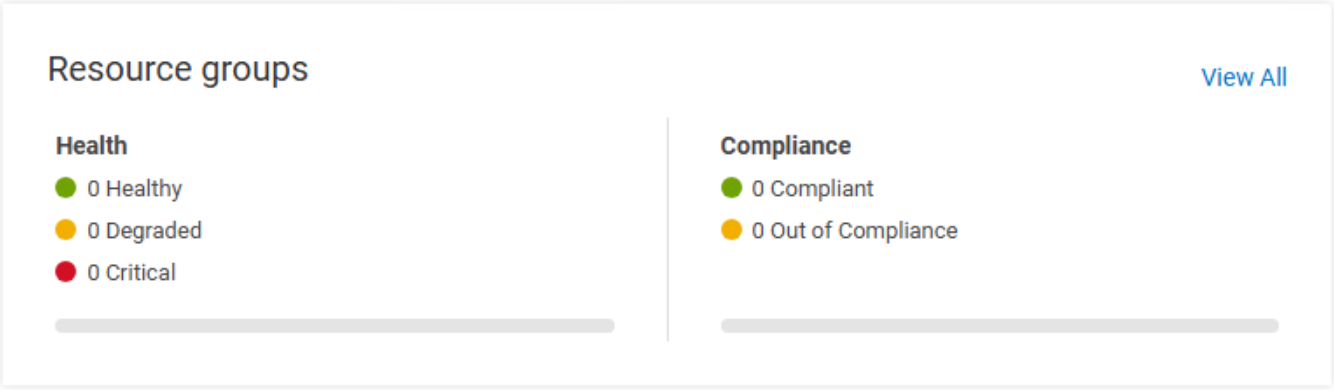


Figure 5. Resource group states

The resource groups are categorized into the following states:

Table 17. Resource group status

Tab	Status	Description
Health	**Healthy**	The green band on the graphic indicates that the resource group is successfully deployed.
	Degraded	The yellow band indicates that resources in a resource group require corrective action, but does not affect overall system health. For example, the firmware version that is installed on a resource in the resource group is not compliant.
	Critical	The red band indicates resource groups that have critical health problems.
	Service Mode	The yellow band indicates that the resource group has been placed in service mode.
Compliance	**Compliant**	The green band indicates that the resource group is compliant with the target compliance version.
	Out of compliance	The yellow band indicates that the resource group is non-compliant with the target compliance version.

You can monitor node health by viewing the status of the resource group on the **Resource Groups** page.

If a resource group is in a yellow (or warning) state, it means that one or more nodes is in a warning or failed state. If a resource group is in a red (or error) state, it indicates that the resource group has fewer than two nodes that are not in a failed state.

To view the status of a failed node component, hover the cursor on the image of the failed component in the resource group.

Alerts

This section lists the current alerts within the system that are categorized by the severity level.

The severity levels are:

- **Critical**: A critical level informs you that the system requires immediate attention. For example, the MDM certificate has expired.
- **Major**: A major level informs you that action is required soon. For example, the system is unable to receive MDM events.
- **Minor**: A minor level informs you that action is required but is not urgent. For example, PowerFlex is using a trial license.

The following diagram is an example of how the system alerts are represented in the dashboard:

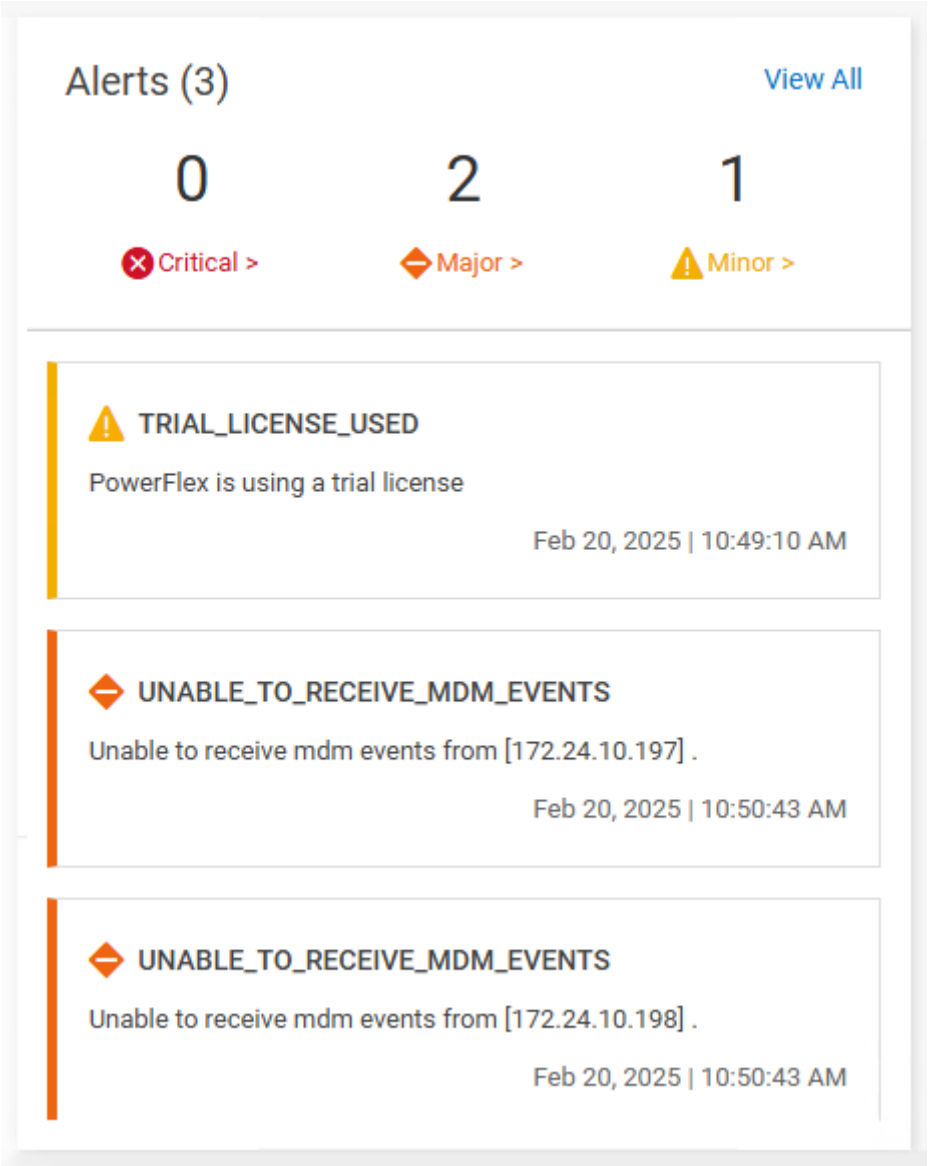


Figure 6. Alerts

For more detailed information about alerts, click **View All** to be redirected to the **Alerts** section under the **Monitoring** tab.

7

Managing block storage

Block storage management involves the configuration of the storage system components. The components includes the protection domain, device groups, devices, storage pools and NVMe targets. In addition, block storage management involves configuring volumes and hosts.

Related information

Configuring block storage

Storage system

A storage system consists of logical entities, such as devices and device groups, storage pools, and NVMe targets, which form the protection domain.

The devices are organized into a device group. A device group consists of devices that have the same drive type, either a solid-state drive (SSD) or persistent memory (PMEM), for example software-defined persistent memory (SDPM). It is from within the device group that storage pools are allocated.

A storage pool contains volumes and snapshots that hold user data.

Protection domains

A protection domain is a logical entity that contains a group of storage nodes and the devices in the storage nodes. The protection domain manages the storage of those storage nodes. It may also contain storage data targets (SDTs) and storage data clients (SDCs). These logical groups allow you to physically or logically isolate specific datasets and performance capabilities within specified protection domains and limit the effect of any specific device failure. You can add, modify, activate, inactivate, or remove a protection domain in the PowerFlex system.

Add a protection domain

A protection domain consists of storage nodes, with (optionally) SDTs and SDCs, configured in separate logical groups. You can add protection domains to a PowerFlex system.

1. Go to **Block** > **Storage System** > **Protection Domain**.
2. Click **Add Protection Domain**.
3. In the **Create Protection Domain** dialog box, enter the name of the protection domain and click **Create**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

You can now add storage nodes, device groups, devices, and storage pools to the protection domain.

Activate a protection domain

You must activate the protection domain to enable access to the data.

Before activating the protection domain, ensure that the storage nodes and the storage data targets (SDTs) in the protection domain are connected and have no alerts. Otherwise, an unnecessary rebuild may occur. If necessary, contact Dell Technologies Support for guidance.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Activate**.
3. In the **Activate Protection Domain** dialog box, click **Yes** for **Force activate** and then click **Activate** to enable access to the data on the protection domain.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

5. Repeat as necessary for each additional protection domain that you want to activate.

Inactivate a protection domain

Use this procedure for a graceful system shutdown.

NOTE: When you inactivate a protection domain, the data remains on the storage node. If you no longer need it, it is preferable to remove a protection domain.

While a protection domain is inactivated, the following activities can take place behind the scenes:

- Determine if there are any current rebuild/rebalance activities taking place. If so, the shutdown will be delayed (unless it is forced) until they are finished.
- Block future rebuild/rebalance activities.
- Temporarily disable application I/O and disable access to volumes.
- Move the DRL mode of the storage nodes to harden, in preparation for restarting the server.
- Reload of the storage nodes before re-enabling data access.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. In the list of protection domains, select the relevant protection domain, and click **More Actions** > **Inactivate**.
3. In the **Inactivate Protection Domain**, enter the user password, and click **Inactivate**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.
5. Repeat as necessary for each additional protection domain that you want to inactivate.

Configure I/O priorities and bandwidth use

PowerFlex includes advanced settings that control I/O priorities and bandwidth use and can be used to fine-tune system performance. The fine-tuning includes the ability to control secondary I/Os which affects the garbage collector, rebuild, and rebalance.

NOTE: These features affect system performance, and should only be configured by an advanced user. It is recommended to retain the default settings, unless you are an advanced user.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.
2. From the list of protection domains, select the relevant checkbox, and click **More Actions** > **Set I/O priority**.
3. In the **Set IO Priority for Protection Domain** dialog box, select one of the following secondary I/O priority policies that are described in the table below:

Table 18. Policies for the secondary I/O priority

Policy	Description
Unlimited	Secondary I/Os are not limited.
Favor Application	Limits the number of allowed concurrent I/Os to the values entered in the **Concurrent I/O limit** and **Bandwidth I/O limit** fields, regardless of user I/O. If you select the Favour Application IO policy, you must set the overall limit and the advanced bandwidth limits for each protection domain on each device. The value of the limit can range from **1** to **10**. The value of the advanced bandwidth can range from **1** to **1024**. The default value is **10**.

4. Click **Apply** and click **Dismiss**.

Remove a protection domain

Remove a protection domain from the PowerFlex system.

Ensure that the storage nodes and device group are removed from the protection domain before removing it from the system.

1. On the menu bar, click **Block** > **Storage System** > **Protection Domains**.

2. From the list of protection domains, select the relevant protection domain and click **More Actions** > **Delete**.
3. In the **Delete Protection Domain**, click **Delete**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Device group

A device group consists of devices that come from five or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same media type.

The capacity of a device group is derived from the capacity of the healthy storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group.

Add a device group

You should add one device group of type SSD per protection domain. There is an option to add additional persistent memory type device groups to a protection domain. The persistent memory device group is used for write cache.

NOTE: This feature is intended for advanced users only.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. Click **Add Device Group**.
3. In the **Create Device Group** dialog box, define the following settings:
   a. **Name**: Enter a name for the device group.
   b. **Protection Domain**: Select the protection domain associated with the device group.
   c. **Type**: Select the media type of the devices in the group: **SSD** or **PMEM**.
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a device group

You can change the name of a device group. You can only rename one device group at a time.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the check box of the device group that you want to rename, and click **More Actions** > **Rename**.
3. In the **Rename Device Group** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set spare capacity of a device group

You must set the spare capacity of a device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

Utilizing spare devices is a feature that is intended for advanced users and is not recommended for general use.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the check box of the device group that you want to set the capacity for, and click **More Actions** > **Spare Capacity**.
3. In the **Capacity Settings** dialog box, select the spare nodes for the device group.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Enable the device diagnostic check

You can enable the device awareness check to detect an issue with a device in advance. If there is an issue with a device, an alert is sent to the dashboard.

The device awareness check is enabled by default. If you disable this setting, it may cause potential device failures not to be detected early enough. To disable the diagnostic, turn **Enable Diagnostics Check** to off.

Device diagnostics is an advanced column and is not set to display by default in the PowerFlex Manager dashboard. You can add the device diagnostics setting to display through the table column arrangement icon at the upper right of the table.

1. On the menu bar, click **Block** > **Storage System**.
2. Click **Device Groups**.
3. Select the checkbox beside the device group to enable the device diagnostic check.
4. Click **More Actions** > **Advanced Settings**.
   The **Advanced Settings for Device Group** dialog box displays.
5. To enable the diagnostic, turn **Enable Diagnostics Check** to on and click **Apply**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a device group

You can remove a device group from a protection domain.

1. On the menu bar, click **Block** > **Storage System** > **Device Groups**.
2. From the list of device groups, select the checkbox of the device group that you want to remove, and select **More Actions** > **Delete**.
3. In the **Delete Device Group** dialog box, verify that you selected the wanted device group, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Devices

A device is a single storage medium, for example, a SSD, which is added to a device group in the storage system.

Activate a device

You must activate a device after it is added to the device group in order to start using it.

1. Go to **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the checkboxes of the required devices, and click **More Actions** > **Activate**.
3. In the **Activate Device** dialog box, click **Activate**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Update the device path

You can manually update the device path.

1. Go to **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the check boxes of the required devices, and click **More Actions** > **Update Device Path**.
3. In the **Update Device Path** dialog box, enter the new path and click **Update Path**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Clear a device error

If the system does not automatically clear the system, you can rectify the error by clearing it.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.

2. From the list of devices, select the check boxes of the relevant devices, and click **More Actions** > **Clear Errors**.
3. In the **Clear device errors** dialog box, click **Clear Errors**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a device

You can change the name of a device.

You can view the current device name by displaying the name column in the device list. The name column is hidden, by default. When no device name has been defined, the name is set by default to the path name of the device.

The device name must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the required device, and click **Modify** > **Rename**.
3. In the **Rename Device** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set device capacity limits

Prior to replacing a device with a device of a smaller capacity, set the capacity limit of the device being removed to the capacity of the new device. Capacity will be decreased, but the size of the disk remains unchanged.

NOTE: The capacity assigned to the storage device must be smaller than its actual physical size.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the relevant device, and click **Modify** > **Set Capacity Limit**.
3. In the **Set Capacity Limit** dialog box, enter the capacity and select a unit type (TB).
   This dialog box displays the maximum capacity available for the device.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify device LED settings

You can set a device's LED to blink or turn it off to help you physically identify a device in the system chassis.

1. On the menu bar, click **Block** > **Storage System** > **Devices**.
2. From the list of devices, select the relevant device, and click **Modify** > **LED Settings**.
3. In the **Set device LED settings** dialog box, do one of the following:
   - To turn on the device's LED, select the **On** check box.
   - To turn off the device's LED, clear the **On** check box.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Storage pools

A storage pool is a container of volumes that uses one device group and, if available, one PMEM device group. A storage pool consumes the capacity from the device group. A storage pool is a container of volumes that belongs to a single device group and consumes the capacity from the device group.

The capacity of a storage pool is based on the capacity of the device group. You can create one storage pool per device group and allocate the full capacity of the device group to that storage pool.

Add, modify, or remove a storage pool from the storage system. You can connect a storage pool to a persistent memory device group to enhance write performance by using the persistent memory as a write cache.

Add a storage pool

The maximum number of storage pools that can be created within a protection domain is one.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Click **+ Create Storage Pool**.
3. In the **Create Storage Pool** dialog box, define the following settings:
   a. Define the storage pool name according to the following rules:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   b. Select the relevant protection domain.
   c. If the system has a persistent memory device group, persistent memory is enabled by default. If there is no such device group, the persistent memory option is not available.
   d. Select one of the following protection schemes:
      - If there are fewer than 11 nodes in the protection domain, select **2+2** from the **Protection Scheme** dropdown menu.
      - If there are 11 or more nodes in the protection domain, select **8+2** from the **Protection Scheme** dropdown menu.

      By default, the storage pool capacity consumes the entire capacity of the device group. It is possible to set a size manually, but it is recommended for advanced users only.
   e. Select whether you want to enable compression. The default compression setting is enabled.
   f. Select whether you want over-provision control. The default over-provision control setting is disabled.
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a storage pool

You can change the name of a storage pool.

The storage pool name must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **Rename**.
3. In the **Rename Storage Pool** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Configure storage pool general settings

You can set the capacity alert thresholds that specify the amount of capacity which must be reserved as free space. The thresholds apply to both the capacity and the over-provisioning limits selected.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **General Settings**.
3. In the **Storage Pool Settings** dialog box, configure the following settings for the storage pool:

Option	Description
Enable compression	For storage pools, inline compression allows you to gain more effective capacity.
Enable Over-Provisioning Control	For storage pools, enabling over-provisioning enables you to limit the size of the thin clones that are created by percentages. For example, if the storage pool is 10 TB and the over-provisioning factor is 200%, you can create thin volumes up to 20 TB. The default is unlimited.

4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Configure storage pool capacity

You can set a storage pool capacity threshold which ensures that you are notified when consumption exceeds the threshold. If over-provisioning is enabled, the same threshold applies.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **Capacity**.
3. Set the capacity alert thresholds.
   - The default high alert threshold is 80%. If the used capacity exceeds 80% or if the over-provisioning limit exceeds 80%, a high alert is raised.
   - The default critical alert threshold is 90%. If the used capacity exceeds 90% or if the over-provisioning limit exceeds 90%, a critical alert is raised.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Expand storage pool capacity

You can choose to use all the available capacity, or you can choose to customize the capacity size of the storage pool. If you choose to use all the capacity, the system calculates the new size of the capacity.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool, and click **Modify** > **Expand**.
3. You can choose either to use the available capacity or customize the capacity.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a storage pool

You can remove the storage pool from a device group.

1. On the menu bar, click **Block** > **Storage System** > **Storage Pools**.
2. Select the storage pool that you want to remove.
3. Click **More Actions** > **Delete**.
4. In the **Delete Storage Pool** dialog box, verify that you have selected the wanted device, and click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

NVMe targets

NVMe targets (or SDT components) must be configured on the PowerFlex system side, in order to use NVMe over TCP technology.

The NVMe target (or SDT component) is a frontend component that translates NVMe over TCP protocol into internal PowerFlex protocols. The NVMe target provides I/O and discovery services to NVMe hosts configured on the PowerFlex system. A minimum of two NVMe targets must be assigned to a protection domain before it can serve NVMe hosts, to provide minimal path resiliency to hosts.

TCP ports, IP addresses, and IP address roles must be configured for each NVMe target (or SDT component). You can assign both storage and host roles to the same target IP addresses. Alternatively, assign the storage role to one target IP address, and add another target IP address for the host role. Both roles must be configured on each NVMe target.

- The host port listens for incoming connections from hosts, over the NVMe protocol.
- The storage port listens for connections from the MDM.

Once the NVMe targets have been configured, add hosts to PowerFlex, and then map volumes to the hosts. Connect hosts to NVMe targets, preferably using the discovery feature.

On the operating system of the compute nodes, NVMe initiators must be configured. Network connectivity is required between the NVMe targets and the NVMe initiators, and between NVMe targets (or SDT components) and storage nodes .

Add an NVMe target

Add an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. Click **+ Add NVMe Target**.
3. From the **Add NVMe Target** dialog box, configure the following settings:
   a. For **Target Name**, enter a name for the NVMe target (or SDT component).
   b. For **Protection Domain**, select a protection domain.
   c. Accept the default TCP ports, or modify them if necessary.
      - **Storage Port** listens for incoming connections from the MDM, and is only open on an IP address configured with the storage role. Default=12200
      - **I/O Port** listens for incoming connections for I/O, and is only open on an IP address configured with the host role. Default=4420 (which conforms to the NVMe over TCP standard)
      - **Discovery Port** listens for incoming discovery connections, and is only open on an IP address configured with the host role. Default=8009 (which conforms to the NVMe over TCP standard. Use zero to disable the discovery service.)
   d. For **Target IPs**, enter an IP address, and then select a role for the IP address from the **Target IPs Roles** menu. Click **Add Target IP**, and repeat this step until all IP addresses are configured.
4. Click **Add**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an NVMe target

Modify the configuration of an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Modify**.
3. In the **Modify NVMe Target** dialog box, modify the desired fields.
4. Optionally, add more target IP addresses, by clicking **Add Target IP**.
5. Click **Modify NVMe Target**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove an NVMe target

Remove an NVMe target (or SDT component).

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **Remove**.
3. In the **Remove NVMe Target** dialog box, verify that you are removing the wanted NVMe target, and click **Remove**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Enter an NVMe target into maintenance mode

You can enter multiple NVMe targets to maintenance mode at one time. You cannot enter an NVMe target that is already in maintenance mode. You must enter an NVMe target into maintenance mode before a storage node and exit the NVME target after a storage node.

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets, select the required NVMe target, and click **More Actions**.
3. In the **Enter NVMe Target to Maintenance Mode** dialog box, verify that you are selecting the wanted NVMe target, and click **Enter Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Exit an NVMe target from maintenance mode

You can exit an NVMe target from maintenance mode

1. On the menu bar, click **Block** > **Storage System** > **NVMe Targets**.
2. From the list of NVMe targets that are in maintenance mode, select the required NVMe target, and click **More Actions**.
3. In the **Exit NVMe Target from Maintenance Mode** dialog box, verify that you are selecting the wanted NVMe target, and click **Exit Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Volumes

Define, configure and manage volumes in the PowerFlex system.

Add a volume

A volume is distributed over the devices residing in the same storage pool. You can add one or multiple volumes at one time.

The system must have the following:

- At least five storage nodes
- Sufficient initial capacity is available to create thin volumes.

Dell Technologies highly recommends giving each volume that a meaningful name associated with its operational role.

1. On the menu bar, click **Block** > **Volumes**.
2. Click **+ Create Volume**.
3. In the **Create Volume** dialog box, configure the following items:
   a. Enter a name for the volume.
   b. Select a storage pool. You can search for the storage pool by name.
   c. Enter the number of volumes.
   d. Enter the volume size in GB (basic allocation granularity is 1 GB).
4. Click **Create**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

To use the created volume, you must map it to at least one host. If the restricted SDC mode is enabled for the system, you must approve SDCs prior to mapping volumes to them.

Map a volume to a host

Mapping exposes the volume to the specified host, creating a block device on the host. You can map a volume to one or more hosts.

Volumes can only be mapped to one type of host either an SDC or an NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select one or more volumes, and click **Provision** > **Map**.
3. In the **Map Volume to Hosts** dialog box, select one or more hosts to which you want to map to the volumes.
   - If a volume is already mapped to a host, only hosts of the same type, SDC or NVMe, are listed.
   - If the volume is not mapped to a host, click **SDC** or **NVMe** to set the type of hosts to be listed.
4. Click **Map**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Unmap a volume

You can unmap one or more volumes from a host.

1. On the menu bar, click **Block** > **Volumes**.

2. From the list of volumes, select the relevant volumes, and then click **Provision** > **Unmap**.
3. Select the host from which to remove mapping to the volumes.
4. Click **Unmap**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a volume snapshot

You can create instantaneous snapshots of one or more volumes.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volumes, and then click **Protect** > **Create Snapshot**.
3. In the **Create snapshot of volume** dialog box, enter the following details:
   a. The name of the snapshot. You can accept the default name, or give the snapshot a name according to the following rules:
      - Contains less than 32 characters
      - Contains only alphanumeric and punctuation characters
      - Is unique within the object type
   b. Enter an index value.
4. (Optional) If you want to prohibit the deletion of snapshots until a specific time, select **Use secure snapshots** option. The use secure snapshots option prohibits deletion of the snapshots until the defined expiration period has elapsed.
   a. To define the expiration period, enter the number in **Expiration Time** and select the units from the dropdown menu.
5. Click **Create**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

Restore volume data

The volume on the host must be taken offline to prevent data integrity issues.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Protect** > **Restore Data**.
3. In the **Restore Data of Volume** window, select the snapshot to restore from.
4. Click **Restore Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Assign a snapshot policy

You can assign a snapshot policy which is a set of rules that controls the creation, retention, and expiration of snapshots.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volumes or snapshots that you want to assign to the policy, and click **Protect** > **Assign Snapshot Policy**.
3. In the **Assign Snapshot Policy** dialog box, from the list of snapshot policies, select the relevant policy to assign and click **Assign Policy**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Unassign a volume from a snapshot policy

You can unassign a volume from a snapshot policy to detach the source volume from the policy and delete associated auto snapshots. This process helps manage snapshots and ensures that only relevant data is retained.

If the auto snapshots are locked, unassigning them from a policy only detaches them. The auto snapshots must be manually deleted.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to unassign from the policy, and click **Protect** > **Unassign Snapshot Policy**.
3. (Optional) If you want the snapshot to be removed, in the **Unassign Volumes from Snapshot Policy** dialog box, select the **Delete related snapshots** checkbox and click **Unassign Policy**.

4. In the **Unassign Volumes From all Snapshot Policies** dialog box, click **Yes**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Create a thin clone of a volume

Thin clones are writable copies of a volume or a snapshot.

A thin clone is a space efficient copy that shares data blocks with its parent object and not a full backup of the original resource.

Thin clones retain full read access to the original resource. You can modify the data within the thin clone while preserving the original data on the parent resource.

Using thin clones, you can establish hierarchical points in time to preserve data over different stages of data modifications. If the parent resource (volume or snapshot) is deleted, the thin clone is unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume that you want to create a thin clone from, and click **Repurpose** > **Create Thin Clone**.
3. In the **Create thin clone of volume** dialog box, enter a thin clone name, and click **Create**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Refresh volume data

Ensure that you have the most up-to-date data by refreshing.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **Repurpose** > **Refresh Data**.
3. In the **Refresh Data of Volume** window, select the volume, thin clone, or snapshot to refresh from.
4. Click **Refresh Volume**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Increase a volume size

You can increase, but not decrease, the logical size of one volume at any time, as long as there is enough capacity for the volume size to grow.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the volume, and click **Modify** > **Resize**.
3. In the **Resize Volume** dialog box, enter the new volume size, and select a unit type. You can set the volume to any size (1 GB granularity).
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a volume

You can use the default volume name or create a name.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and click **More Actions** > **Rename**.
3. In the **Rename Volume** dialog box, enter a new name and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete a volume

You can remove a volume from PowerFlex provided the volume is not mapped to a host and it is not assigned to a snapshot policy.

Before deleting a volume, ensure the following:

- If the volume is mapped to a host, unmap it before deleting the volume.

- If the volume is assigned to a snapshot policy, unassign it from the snapshot policy before deleting the volume.

To prevent causing a data unavailability scenario, avoid deleting volumes or snapshots while the MDM cluster is being upgraded.

CAUTION: All data is erased from a deleted volume. All direct decedent snapshots of the volume are deleted with it. Thin clones and snapshots of thin clones are unaffected.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the required volume, and click **More Actions** > **Delete**.
3. In the **Delete Volume** dialog box, verify the volumes to be removed, and click **Delete**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Set a volume bandwidth and IOPS limits

Setting bandwidth and IOPS limits for volumes lets you control the quality of service (QoS). Bandwidth and IOPS limits are set on a per-host basis.

Before you set these limits, ensure that the volumes are mapped .

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **More Actions** > **Set Limits**.
3. In the **Set IO limits for volume** dialog box, enter the required values for **Bandwidth Limits** and **IOPS Limits**, or select the corresponding **Unlimited** check box.
   - The number of IOPS must be larger than 10.
   - The volume network bandwidth is in MB/s.
   - The I/O limits are applied to every mapped SDC.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename a snapshot

You can rename a snapshot for clarity, version control, organization and categorization, and for audit purposes.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to rename.
5. Go to **Modify** > **Rename**.
6. In the **Rename Snapshot** dialog box, enter a new name and click **Apply**.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Delete a snapshot

You can delete a snapshot if it is no longer relevant, or if you want to free up storage space.

1. On the menu bar, click **Block** > **Volumes**.
2. From the list of volumes, select the relevant volume, and then click **View Details**.
3. Click **Snapshots**.
4. Select the checkbox beside the snapshot that you want to delete.
5. Go to **Modify** > **Delete**.
6. Verify that the operation is finished and is successful, and click **Dismiss**.

View volume details

You can view details about volumes including snapshots, host , and vTree details.

1. On the menu bar, click **Block** > **Volumes**.
2. Select the volume from the list that you want to view details about and click **View Details** in the lower right pane.

The following details display:

- Objects related to the volume that you are viewing.
- General information:
  - Type: Primary or thin clone
  - Size
  - Mapped: Whether the volume is mapped or not.
  - Created time: The time the volume was created.
  - Read-only: The volume is access mode.
  - VTree Logical written: The total sum of logical capacity used by the volumes, snapshots, and thin-clones that belong to the same family.
- Performance:
  - Latency: The time that it takes for a single I/O operation to complete and is typically measured in milliseconds (ms).
  - IOPs limit: The maximum number of read and write operations that the system can handle per second
  - Bandwidth: The maximum of data that is transferred per second in GB/s.

The following specific details are available for a host:

- Name: The name assigned during configuration.
- IP address: The IP address assigned during configuration.
- Storage access protocol: Either SDC or NVMe/TCP.
- Host access: The default access mode for newly mapped volumes.
- Version: The SDC software version, applicable for SCD only.
- Operating system(OS): The operating system of the host.
- BW limit: The maximum of data that is transferred per second in GB/s.
- IOPs limit: The maximum number of read and write operations that the system can handle per second.
- Latency: The time that it takes for a single I/O operation to complete and is typically measured in milliseconds (ms).

The following specific details are available for a snapshot:

- Name: The name assigned during configuration.
- Size: The size of the object is measured in GB.
- Creation time: The time the object was created,
- Parent ID: A unique identifier of the parent snapshot that is used to link snapshots together.
- VTree ID: The ID of the vTree the object belongs to.

The following specific details are available for a vTree:

- Name: The name assigned during configuration.
- Storage Pool: The storage pool to which the object belongs.
- Provisioned: The total provisioned size of each object.
- Logical: The logical capacity in use by each object.
- Type: The type of object.

Hosts

Hosts are entities that consume PowerFlex storage for application usage. There are two methods of consuming PowerFlex block storage: using the SDC kernel driver, or using NVMe over TCP connectivity. Therefore, a host is either an SDC or an NVMe host.

Once a host is configured, volumes may be mapped to it. In each case, hosts must be mapped to volumes.

Add an NVMe host

Add an NVMe host to PowerFlex.

Ensure the following:

- You know the NVMe Qualified Name (NQN) of the host. If you do not know the NQN, see the documentation for the host operating system.
- The host is connected to the Ethernet switch.
- The host is configured with the correct VLAN ID and routing rules.

1. On the menu bar, click **Block** > **Hosts**.

2. Click **+ Add Host**.
3. In the **Add NVMe Host** dialog box, enter a hostname in the **Host Name** field.
4. In the **Host NQN** field, enter the NQN string for the host.
5. In the **Number of Paths Per Volume** field, enter the maximum number of paths to be provided between the host and each volume (default is 4).
6. In the **Number of System Ports per Protection Domain** field, enter the maximum number of ports to be provided between the host and each protection domain (default is 10).
7. Click **Add**.
8. Verify that the operation is finished and is successful, and click **Dismiss**.
9. Configure the host operating system to connect to the cluster's NVMe discovery service on one or more of the NVMe targets.

Map volumes to hosts

Map hosts to volumes to ensure that the host can access the volumes.

Volumes can only be mapped to one type of host: either SDC or NVMe. Ensure that you know which type of hosts are being used for each volume, to avoid mixing host types.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Provision** > **Map**.
3. In the **Map Hosts to Volumes** dialog box, select the volumes to be mapped to the selected hosts, and click **Map**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Unmap volumes from hosts

Remove mapping between volumes and hosts.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Provision** > **Unmap**.
3. In the **Unmap** dialog box, ensure that the correct host is selected.
4. Click **Unmap**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove hosts

You can only remove hosts with unmapped volumes. For SDCs, the host must be disconnected from the PowerFlex cluster before it can be removed.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Remove**.
3. In the **Remove Host** dialog box, ensure that you have selected the correct host for removal.
4. Click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Configure or modify approved host IP addresses

When the restricted host (SDC) mode is set to `approved IP restriction`, configure the host IP addresses before mapping volumes to the hosts.

Ensure that the hosts are approved by GUID.

You can add up to four IP addresses.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Modify Approved IP Addresses**.
3. In the **Modify Approved IP addresses** dialog box, enter the IP address of the hosts, and click **Add IP Address**. Repeat this step for additional addresses.

4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Approve SDCs

When the restricted host (SDC) mode is set to GUID restriction, approve the SDCs before mapping the SDCs to volumes.

For more information about the restricted host feature, see the *Dell PowerFlex Technical Overview* or *Dell PowerFlex Family 5.x Security Configuration Guide*.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select one or more hosts and click **Modify** > **Approve**.
3. In the **Approve host** dialog box, verify that the hosts listed are the ones that you want to approve, and click **Approve**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Rename hosts

You can rename a host following the specific naming rules.

The hostname must adhere to the following rules:

- Contains less than 32 characters
- Contains only alphanumeric and punctuation characters
- Is unique within the object type

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Rename**.
3. In the **Rename Host** dialog box, enter the new name, and click **Apply**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Modify an SDC performance profile

SDC performance profiles are set by default to high and can be changed to compact. The compact setting may impact the system performance.

The default setting configures a predefined set of parameters for high-performance use cases.

NOTE: Performance tuning is case-specific. To prevent undesirable effects, Dell Technologies highly recommends that you thoroughly test all changes. For further assistance, contact Dell Technologies Support.

1. On the menu bar, click **Block** > **Hosts**.
2. From the list of hosts, select the relevant host, and click **Modify** > **Modify Performance Profile**.
3. In the **Modify Performance Profile Host** dialog box, select the wanted performance profile: **High** or **Compact**.
4. Click **Apply**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

8

Protecting a block storage environment

Use local features to protect your block storage.

Snapshot policies

In addition to being created manually, snapshots can be created automatically by the system according to a defined interval. Within PowerFlex, snapshot policies are used to achieve automatic data protection on storage resources (volumes and clones).

A snapshots policy is a group of user-defined rules that are used to establish local data protection on an assigned storage resource. In PowerFlex, you can assign a snapshot policy to a storage resource which defines the level of protection. Protection policies are also created at the cluster level, and not per individual protection domain. This means that any storage resource on any protection domain in the system can leverage a snapshot policy once it is created.

You can define a rule. For example:

Rule: Take snapshots every 60 minutes.

Retention Rules:

- 24 snapshots
- Seven snapshots
- Four snapshots

After defining the rules, select the storage resource to assign to the snapshot policy. You can assign multiple source volumes to a snapshot policy, but only a single policy per storage resource is allowed. Only one volume per vTree may be assigned to a policy (any policy).

Create a snapshot policy

Create snapshot policies with a maximum of six retention rules that each specify a retention period and number of snapshots to retain, and then associate storage resources to the policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. Click **Create Snapshot Policy**.
3. In the **Create Snapshot Policy** dialog box, assign the policy a name and the time interval between snapshots.
4. Create retention rules:
   a. In the **Retention** field, add the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention**, and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.
   Up to six retention rules can be created.
5. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
6. Click **Create** or **Create and Activate** to add the snapshot policy to PowerFlex.
   If created, but not activated, the snapshot policy shall be paused.
7. In the **Snapshot Policy was successfully added** dialog box, click **Assign volume to snapshot policy**.
8. In the **Select volumes you wish to assign to the policy** window, select one or more volumes, and click **Assign Policy**.
9. Verify that the operation has finished successfully, and click **Dismiss**.

Remove snapshot policy

Remove a snapshot policy to stop PowerFlex from automatically creating snapshots for the specified volumes. You cannot remove snapshot policies that are secured.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy.
3. Ensure that all storage resources are unassigned from the selected snapshot policy, and click **More Actions** > **Delete**.
4. In the **Remove Snapshot Policy** dialog box, click **Remove**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Modify snapshot policy

PowerFlex allows you to modify the parameters of snapshot policies, when required.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and then click **Modify** > **Modify Policy**.
3. In the **Modify Snapshot Policy** dialog box, change the desired parameters:
   a. In the **Retention** field, set the number of snapshots to retain.
   A retention period is created for the specified number of snapshots times the snapshot interval.
   b. Optionally, to create an additional retention rule, click **Add Retention** and repeat the previous step. Each rule is based on the specified number of snapshots times the previous retention period.
   Up to six retention rules can be created.
   c. Optionally, select the **Secured** checkbox to prevent the snapshots that are created by the policy from being modified or deleted before the end of the expiration date.
4. Click **Modify**.
5. Verify that the operation has finished successfully, and click **Dismiss**.

Rename snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Modify** > **Rename**.
3. In the **Rename Snapshot Policy** dialog box, enter a new name and click **Apply**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Activate snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **More Actions** > **Activate**.
3. In the **Activate Snapshot Policy** dialog box, click **Activate**.
4. Verify in the **State** column that the snapshot policy is active, and click **Dismiss**.

Pause snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **More Actions** > **Pause**.
3. In the **Pause Snapshot Policy** dialog box, click **Pause**.
4. Verify in the **State** column that the snapshot policy is in paused state, and click **Dismiss**.

Assign volumes to a snapshot policy

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Assign Volumes**.
3. In the **Assign Volumes to Snapshot Policy** dialog box, select the volumes that you want to assign to the policy, and click **Assign Policy**.
4. Verify that the operation has finished successfully, and click **Dismiss**.

Unassign a volume from a snapshot policy

When you unassign volumes from a snapshot policy, the storage resource is detached from the snapshot policy.

1. On the menu bar, click **Protection** > **Snapshot Policies**.
2. In the list of snapshot policies, select the relevant policy, and click **Protect** > **Unassign Volumes**.
3. In the **Unassign Volumes from Snapshot Policy** dialog box:
   a. To unassign the snapshot policy from a volume, select the checkbox of the relevant volume.
   b. Optionally, to delete the relevant snapshots of the selected volume, select the **Delete related snapshots** checkbox.
4. Click **Unassign Policy**.
5. Verify that the operation finished successfully, and click **Dismiss**.

9

Performing lifecycle operations for a resource group

This section provides tasks for deploying resource groups and building templates.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Deploying a resource configuration in a resource group

This section includes tasks for deploying and managing system resources in a resource group.

A resource group is a PowerFlex Manager object representing the infrastructure that is configured during a deployment. This object persists in PowerFlex Manager if the infrastructure is required. A resource group is viewed and interacted with to perform compliance, health, and life cycle management on the deployed infrastructure. PowerFlex Manager supports the deployment of two types of PowerFlex resource groups:

- Storage-only resource groups
- Compute-only resource groups

NOTE: Standard users see only the details of the resource groups that they created or for which they have permission.

The **Resource Groups** page displays the resource groups that are in the following states in both **Tile** and **List** view.

Table 19. Resource group states

State	Description
Healthy	The resource group is successfully deployed and is healthy.
Warning	One or more resources in the resource group requires corrective action.
Critical	The resource group is in a severely degraded or nonfunctional state and requires attention.
Pending	The deployment is scheduled for a later time or date.
In Progress	The resource group deployment is in progress, or has other actions currently in process, such as a node expansion or removal.
Cancelled	The resource group deployment has been stopped. You can update the resources or retry the deployment, if necessary.
Incomplete	The resource group is not fully functional because it has no volumes that are associated with it. Click **Add Resources** to add volumes.
Service Mode	The resource group is in service mode.
Lifecycle Mode	The resource group is in life-cycle mode. Resource groups in life cycle mode are enabled with health and compliance monitoring, and non-disruptive upgrade features only.
Managed Mode	The resource group is in managed mode. Resource groups in managed mode are enabled with health and compliance monitoring, non-disruptive update, automated resource addition, and automated resource replacement features.

On the **Resource Groups** page, you can:

- Click **Deploy New Resource Group** to deploy a new resource group.

NOTE: Standard users are allowed to deploy only those resource groups they have created or for which they have permissions.

- Click **Add Existing Resource Group** to add an existing resource group.

To switch views, click the **Tile View** icon or **List View** icon .

To view the resource groups based on a particular resource group state, select an option from the **Filter By** drop-down list. Alternately, in the **Graphical** view, click the graphic in a particular state.

In the **Tile** view, each square tile represents a resource group and has the status of the resource group at the bottom of the graphic. The state icon on the graphic indicates the state of the resource group. The components in blue indicate the component types that are in the deployment. The components that are in gray indicate the component types that are not in the resource group.

In the **List** view, the following information displays:

- **Status**—Status of the resource group.
- **Name**—Name of the resource group.
- **Deployed By**—Name of the user who deployed the resource group.
- **Deployed On**—Date and time when the resource group is deployed.
- **Components**—Components used in the resource group.

Click the resource group in the **List** view or **Tile** view to view the following information about the resource group in the right pane:

- Resource group name and description to identify the resource group.
- Name of the user who deployed the resource group.
- Date and time when the resource group is deployed.
- Name of the reference template that is used in the resource group.
- Number of resources that are in the resource group for deployment, based on component type (cluster or node).

Click **View Details** to view more details about the resource group. You can also generate troubleshooting bundles from the resource group details page.

Click the resource group name in the **List** view to open the **Resource Group Details** page.

Click **Update Resources** to update the firmware of all nodes in the resource group that are not compliant.

Related information

Configuring block storage
Deploying and provisioning

Basic tasks

This section provides basic tasks for resource group management.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows that are provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific life cycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
     ii. Select which users have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.

3. Click **Next**.
4. On the **Deployment Settings** page for **Device Group Spare Capacity** select **Custom** > **Nodes**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

   Utilizing spare devices is a feature that is intended for advanced users and is not recommended for general use.

5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file that is called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf` file from a previous release, it overrides these settings. If you must customize these settings, you can do it in `sysctl.conf`.

Related information

Lifecycle
Viewing resource group details
Adding components to a resource group
Build and publish a template
Component types
Removing a resource group

Adding an existing resource group

You can add an existing resource group to discover and import hardware resources that were not originally deployed with PowerFlex Manager.

Ensure that the following conditions are met before you add an existing resource group:

- The PowerFlex gateway, CloudLink Center, and hosts must be discovered in the resource list.
- You must have a resource group that includes the manager MDMs before importing other resource groups. The first resource group that you import must include the manager MDMs so that PowerFlex Manager has the manager MDM credentials.

1. On the menu bar, click **Lifecycle** > **Resource Groups** and then click **+ Add Existing Resource Group**.
2. On the **Add Existing Resource Group** page, enter a resource group name in the **Name** field.
3. Enter a description in the **Description** field.
4. Select the **Type** for the resource group.

   The choices are **Compute Only**, **Storage Only**, and **PowerFlex management cluster (KVM)**.

5. To specify the compliance version to use for compliance, select the version from the **Firmware and Software Compliance** list or select **Use PowerFlex Manager appliance default catalog**.

   You cannot specify a minimal compliance version when you add an existing resource group, since it only includes server firmware updates. The compliance version for an existing resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

   NOTE: Changing the compliance version might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

6. Specify the resource group permissions under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
7. Click **Next**.
8. Choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you choose **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.

9. On the **Cluster Information** page, enter a name for the cluster component in the **Component Name** field.
10. Select values for the cluster settings:

    For a compute-only resource group, select values for these cluster settings:
    a. **Target Virtual Machine Manager**—Select the vCenter name where the cluster is available.
    b. **Data Center Name**—Select the data center name where the cluster is available.

       NOTE: Ensure that the selected vCenter has unique names for clusters in case there are multiple clusters in the vCenter.

    c. **Cluster Name**—Select the name of the cluster that you want to discover.
    d. **OS Image**—Select the image or choose **Use Compliance File ESXi image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only ESXi images for a compute-only resource group.

    For a storage-only resource group, select values for these cluster settings:
    a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.

b. **Protection Domain**—Select the name of the protection domain in PowerFlex.
c. **OS Image**—Select the image or choose **Use Compliance File Linux image**. If you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for a storage-only resource group.

For the PowerFlex management cluster (KVM) resource group, select values for the following cluster setting:

a. **Target PowerFlex Gateway**—Select the gateway where the cluster is available.
b. **Protection Domain**—Select the name of the protection domain in PowerFlex.
c. **OS Image**—Select the image or choose **Use Compliance File Linux image** if you want to use the image provided with the target compliance version. PowerFlex Manager filters the operating system image choices to show only Linux images for the resource group.

11. Click **Next**.
12. On **OS Credentials** page, select the operating system credential that you want to use for each node and MVM.

    You can select one credential for all nodes or MVMs, or choose credentials for each item separately. You can create the operating system credentials on the **Credentials Management** page under **Settings**.

    PowerFlex Manager validates the credentials before it creates the resource group. This validation makes it possible for PowerFlex Manager to run a full inventory on all nodes and MVMs before creating the resource group.

    PowerFlex Manager runs the inventory on all nodes and MVMs for which the credentials are valid. The resource group uses any nodes and MVMs for which it has a successful inventory. For example, if you have four nodes, and one node has an invalid operating system password, PowerFlex Manager adds the three nodes for which the credentials are valid and ignores the one with an invalid password.

13. Click **Next**.

    The list of resources available in the cluster is displayed on the **Inventory Summary** page.
14. Review the inventory on the **Inventory Summary** page.

    The summary shows all nodes that are available. If a node is not available, it might be because this node does not match the **Type** you selected for the resource group (**Compute only**, **Storage only**).

    Depending on how the node is configured, the summary might show additional inventory information. For example, for a node that has NVDIMM compression, the summary shows additional information about the acceleration pool and compression settings.

    If the resources are discovered and in an available state, the **Available Inventory** displays the components as **Yes**. An unavailable PowerFlex gateway is shown as **No**.

    If the credentials are invalid for a node or if you have a network connectivity problem, PowerFlex Manager displays **No** in the **Available Inventory** column for the node, and displays an error message to notify you about the problem.

    PowerFlex Manager cannot update firmware and software versions for PowerFlex clusters that do not have available PowerFlex gateways. If expected PowerFlex gateways are not shown as available, you can discover the gateways and run the wizard again.

    NOTE: PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value can be displayed in PowerFlex Manager. Certain operating systems require extra packages that are installed for iDRAC to update the correct hostname.
15. Click **Next**.
16. On the **Network Mapping** page, review the networks that are mapped to port groups and make any required edits.

    PowerFlex Manager attempts to select the correct network based on the VLAN ID, subnet, or IP ranges entered in PowerFlex Manager. If PowerFlex Manager finds only one network for a given network type, it selects the network automatically. If it finds more than one, you must select the network from the **Network** drop-down list. The operating system Installation network does not get a VLAN ID.

    NOTE: If the operating system install VLAN is not already configured in your environment, add it. This network is required to perform node expansions. This network is typically added during PowerFlex Manager configuration.

    If there are any port groups for which you do not want PowerFlex Manager to manage access, leave those port groups cleared. If no network is selected for a particular port group, PowerFlex Manager leaves it out of the deployment data and does not add it to the nodes.

17. To import many general-purpose VLANs from vCenter, perform these steps:
    a. Click **Import Networks** on the **Network Mapping** page.

PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on the vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.

b. Optionally, search for a VLAN name or VLAN ID.
PowerFlex Manager filters the list of available networks to include only those networks that match your search.

c. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.

d. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
PowerFlex Manager updates the **Selected Networks** to show the ones that you have chosen.

e. Click **Save**.

18. Click **Next**.

19. Review the **Summary** page and click **Finish** when you are ready to add the resource group.

PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

The process of adding an existing resource group causes no disruption to the underlying hardware resources. It does not shut down any of the nodes or the vCenter.

For an existing resource group, the **Reference Template** field shows **Generated Existing Resource Group Template** on the **Resource Group Details** page. You can distinguish existing resource groups from new resource groups that were deployed with PowerFlex Manager.

When PowerFlex Manager must put a resource group in life cycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for life cycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

When you add an existing resource group, PowerFlex Manager matches the hosts, vCenter, and other items it finds with discovered resources in the resource list. If you missed a component initially, you can change your resource inventory, and update the resource group to reflect these changes. Go back to the resources list, select the component, and mark it as **Managed** by selecting **Change resource state to Managed**. Then, perform an **Update Resource Group Details** operation on the resource group to pull in the missing component.

When you deploy an existing resource group, PowerFlex Manager reserves any IP addresses from vCenter or the PowerFlex gateway that it needs. If you later tear down the resource group, it releases those IP addresses so that they can be reused.

If you add an existing resource group that supports NSX-T or NSX-V, PowerFlex Manager displays a banner indicating that the resource group supports a limited set of actions. Most resource group actions are disabled for an NSX-T or NSX-V configuration, except the ability to update the firmware and software components, remove resources (or the resource group as a whole), and update resource group details.

PowerFlex Manager creates a configuration file called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf`file from a previous release, it will override these settings. If you need to customize these settings, you can do it in `sysctl.conf`.

Related information

Getting started
Lifecycle
Support for full and partial network automation
Defining a network

Entering and exiting service mode

PowerFlex Manager enables you to put a node in service mode when you must perform maintenance operations on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

PowerFlex Manager detects when a node is in instant maintenance mode. It automatically places the node in service mode and also ensures that the resource group itself goes into service mode.

If the node has a VMware NSX-T or NSX-V configuration, PowerFlex Manager does not enable you to enter service mode. PowerFlex Manager also does not enable you to enter service mode if the PowerFlex gateway used in the resource group is being updated on the **Resources** page.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.

3. Click **Enter Service Mode** under **More Actions**.
4. Select the node on the **Node Lists** page and click **Next**.
5. Click **Instant Maintenance Mode**.
6. Click **Finish**.

   PowerFlex Manager displays a yellow warning banner at the top of the Resource Groups page. The **Service Mode** icon displays for the **Deployment State** and **Overall Resource Group Health**, and for the **Resource Health** for the selected nodes.
7. When you are ready to leave service mode, click **More Actions** > **Exit Service Mode**.

Replacing a drive

You can replace a failed drive in a deployed node. PowerFlex Manager supports the replacement of SSD and NVMe drives for PowerFlex storage-only nodes.

Ensure that you have a replacement drive and can access the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

PowerFlex Manager supports drive replacement for:

- Nodes that have NVDIMM/SDPM enabled.
- Nodes that have NVMe drives for HBA SAS controllers
- **NOTE:** RAID controllers are not supported.
- CloudLink-enabled PowerFlex nodes with self-encrypting drives (SEDs) or software encryption.

If the node has an NSX-T or NSX-V configuration, PowerFlex Manager does not enable you to replace a drive.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. Scroll down to the **Physical Nodes** section of the page.
4. Under **Physical Nodes**, click **Component ReplacementDrive Replacement**.
   PowerFlex Manager displays the **Node List** panel in the **Drive Replacement** wizard.
5. Click the node for which you want to perform a drive replacement.
6. Click **Next**.
   PowerFlex Manager displays the **Select Drive** panel. Any empty slots are shown in gray. The slots that have drives are shown in black.
7. Select the drive that you want to replace by clicking the drive slot in the hardware image, or select the drive from the table.

   Be sure to click the correct drive, because the drive replacement process is irreversible.

   The color for the selected drive changes to blue and the table below the hardware image is selected. The table shows details about the selected drive. To help you pick the correct drive, PowerFlex Manager provides the iDRAC name for the drive, the PowerFlex drive name, and the serial number.
8. Optionally, click **Launch iDRAC GUI** to see iDRAC details about the selected drive before proceeding.

   When you launch iDRAC, the iDRAC user interface opens in a different tab. Log in and go to the drive details.
9. Optionally, click **Blink LED** to cause the selected drive to blink.
   PowerFlex Manager turns on the blinking LED on the selected drive and keeps it blinking until you pull the disk out. A new disk is not in a blinking state. PowerFlex Manager is not able to determine whether the drive is blinking. If the drive is already blinking, you can click **Unblink LED** to turn off the blinking LED.
10. Click **Next** when you are ready to delete the drive.
    PowerFlex Manager displays the **Confirmation** page. PowerFlex Manager provides details about the drive and shows a hardware image again with the drive that is selected in blue.
11. Review the information carefully under **Confirm Drive Replacement** to be sure that you have selected the correct drive. If you are certain that you want to proceed, type **`REMOVE DRIVE`**.
12. Click **Confirm**.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a drive removal operation is started. After several minutes, if the node ejection is successful, PowerFlex Manager places the resource group and the node in service mode and deletes the drive from the storage pool. The resource group and the node remain in service mode until the drive is replaced in the node, and the drive replacement process has completed. During this

period, PowerFlex Manager does not enable you to enter service mode for the node, or perform another drive replacement within the same resource group.

PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that a drive replacement is in process for the selected drive.

For NVMe disks, PowerFlex Manager initiates a job to instruct iDRAC to delete the NVMe disk safely from the operating system.

If the drive ejection process fails, PowerFlex Manager displays a banner at the top of the **Resource Group Details** page to notify you that the initial step to eject the drive failed. When the delete drive step fails in the drive replacement process, the banner displays a **Retry Drive Replacement** option to help you to restart the drive replacement process.

If there is a failure, the resource group and the node do not remain in service mode. Instead, they return to the previous state that the resource group was in before you attempted the drive replacement.

13. Pull the disk out of the slot, and insert the new disk.
14. On the disk replacement banner at the top of the **Resource Group Details** page, click **Component Replacement**.

    PowerFlex Manager opens the **Complete Drive Replacement** wizard and runs inventory on the node. It takes a couple of minutes to get the updated inventory for the node.

    After getting the updated inventory, PowerFlex Manager displays the **Found New Drive** page or the **Continue Without Drive** page. If PowerFlex Manager finds the new drive, it displays the drive that is selected in green on the **Found New Drive** page.

    If PowerFlex Manager does not find the new drive, it displays the **Continue Without Drive** page. If you are certain that you want to proceed without a new drive, type **`CONTINUE WITHOUT DRIVE`**. If you proceed, the node is taken out of service mode with fewer drives, which might impact the capacity and performance of the resource group. Click **Confirm** to finish the process without adding a drive.
15. Click **Complete** to finish the disk replacement process.
    PowerFlex Manager puts the resource group into the **In Progress** state and updates the **Recent Activity** to show that a complete drive replacement operation is started. Then, PowerFlex Manager replaces the drive in the resource group, and takes the resource group and the node out of service mode.

    When the drive addition is complete, PowerFlex Manager displays a banner at the top of the resource group to notify you that the drive replacement is complete. PowerFlex Manager enables the **Reconfigure MDM Roles** and **Drive Replacement** options again. The resource group returns to the previous state after the drive replacement successfully completes.

Reconfiguring MDM roles

PowerFlex Manager enables you to change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.

NOTE: This is not applicable for PowerFlex in a PowerFlex management cluster (KVM), as there are only three nodes.

You can launch the wizard for reconfiguring MDM roles from the **Resource Groups** page or from the **Resources** page. The nodes that are listed and the operations available are the same regardless of where you launch the wizard.

1. To access the wizard from the **Resource Groups** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group that has the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
      The **Resource Group Details** page is displayed.
   d. On the **Resource Group Details** page, click **Reconfigure MDM Roles** under **More Actions**.
      The **MDM Reconfiguration** page is displayed.
2. To access the wizard from the **Resources** page:
   a. On the menu bar, click **Lifecycle** > **Resources**.
   b. Select the PowerFlex system for which you want to reconfigure MDM roles.
   c. In the right pane, click **View Details**.
      The **Details** page is displayed for the selected PowerFlex gateway.
   d. On the **Details** page, click **Reconfigure MDM Roles**.
      The **MDM Reconfiguration** page is displayed.
3. Review the current MDM configuration for the cluster.
4. Find each MDM role that you want to reassign and choose the new Host Name or IP address for the role in the **Select New Node for MDM Role** list.

You can reassign multiple roles at one time.

5. Click **Next**.
   The **Summary** page is displayed.
6. Type **CHANGE MDM ROLES** to confirm your changes.
7. Click **Finish**.

Related information

Viewing PowerFlex system details

Adding components to a resource group

You can add one or more node, volume, or network components to a new resource group that PowerFlex Manager deploys, or to an existing resource group not deployed by PowerFlex Manager.

You can add components to a resource group for which deployment was successful or to a failed resource group deployment. You do not need to go back to the original template from which the resource group was originally deployed. Standard users can add components only to a resource group for which they have permission.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, click one of the following options from the **Add Resources** menu:
   - **Add Nodes** —Add one or more nodes to the resource group
   - **Add Volumes**—Add one or more volumes to the resource group
   - **Add Network**—Add one or more networks to the resource group

   If a node has an NSX-T or NSX-V configuration, PowerFlex Manager removes the **Add Resources** button under **Resource Actions**.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager also removes the **Add Resources** button.

   If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not allow you to add a node.

Related information

Component types
Deploying a resource group

Adding nodes to a resource group

You can add a node to a resource group. To add a node, you must define the PowerFlex, operating system, and hardware settings.

1. On the **Resource Groups** page, select the check box beside the resource group that want to add a node to and click **View Details**.
2. On the **Resource Groups** page, click **Add Resources** and choose **Add Nodes**.
   The **Duplicate Node** wizard opens.
3. From the **Duplicate Node** page, complete the following steps:
   a. From **Resource to Duplicate**, select a node.
      Select a node that is of the same type as the other nodes within the resource group.
   b. From **Number of Instances**, enter the number of node instances that you want to add to the resource group.
   c. Click **Next**.

   The **Additional Settings** window opens.
4. Under **PowerFlex Settings**, enter the **Protection Domain Name**, **Device Group Name**, **Device Group Spare Capacity**, and **Custom Spare Node Count**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

Utilizing spare devices is a feature that is intended for advanced users and is not recommended for general use.

5. Under **OS Settings**, set the **Host Name Selection** to **Auto-Generate**, **Specify at Deployment Time**, or **Reverse DNS Lookup**.
   - If you choose **Specify at Deployment**, provide a name for the host in the **Host Name** field.
   - If you choose **Auto-Generate**, specify a template for the name in the **Host Name Template** field.

   For an existing resource group that PowerFlex Manager did not deploy, the **Host Name Selection** option is automatically set to **Specify at Deployment Time** and you must type the hostname.

   In the **IP Source** box, provide an IP address. For an existing resource group that PowerFlex Manager did not deploy, the default choice is **User Entered IP** and the IP settings for each network default to **Manual Entry**. However, you can change the setting to **PowerFlex Manager Selected IP**.
   - If the **Multi-Network Selection** option is selected and the IP source is **PowerFlex Manager Selected IP**, the IPs are assigned sequentially from all the networks of that type.
   - If an IP cannot be assigned from any of the multiple networks of the selected type, an error is displayed.
   - If the IP source is **User Entered IP**, a generic name is displayed for the **IP Source** setting. For example, **PowerFlex Management IP Source**.
   - If multi networking is not enabled and the IP source is **User Entered IP**, the name of the network is displayed as on the **Network Settings** page. If the selected IP Source setting is of **Manual Entry** type, the IP you entered is validated against all the networks of the selected type. An error is displayed when the IP is not validated in one of the networks.

   Under **Hardware Settings**, the **Target Boot Device** option is automatically set to **Use Local Flash Storage for Dell PowerFlex** for an existing compute-only resource group that PowerFlex Manager did not deploy.
6. Under **Hardware Settings**, in the **Node Source** box, select **Node Pool** or **Manual Entry**.

   For an existing resource group that PowerFlex Manager did not deploy, the node source defaults to **Manual Entry**, but you can change it to **Node Pool**.

   In the **Node Pool** box, select the node pool. Alternatively, if you chose **Manual Entry**, select the specific node in the **Choose Node** box.

   You can view all user-defined node pools and the global pool. Standard users can see only the pools for which they have permission.

   For an existing resource group that PowerFlex Manager did not deploy, the **Node Pool** defaults to **Global**. This is valid for compute-only and storage-only node deployments.
7. Click **Next**.
8. Click **Save**.
9. Review the **Summary** page and click **Finish**.

Related information

Add a storage node

Adding volumes to a resource group

You can create volumes for a storage-only resource group. You can add existing volumes to a compute-only resource group.

PowerFlex Manager allows you to create up to 50 new volumes at a time for a storage-only resource group.

Initially, there are no volumes for a new resource group. The resource group is in an incomplete state until you add volumes. Add at least one volume to a resource group to get it out of the incomplete state.

A compute-only resource group consumes volumes that the storage-only resource group provides.

After you create the volumes in a storage-only resource group, they are added to PowerFlex and are mapped to the storage node . When you add the volumes to a compute-only resource group, PowerFlex maps the volumes and creates the datastores.

PowerFlex Manager allows you to enable compression when adding volumes to a storage-only resource group.

1. On the **Resource Groups** page, select check box beside the resource group that you want to add a volume to and click **View Details**.
   The **Add Volume** wizard opens.
2. Ensure the **Create New Volumes** button is selected and click **Next**.
3. Click **+ Add New Volume** to add a volume section. PowerFlex Manager adds a volume section to the **Add Volume** wizard.
4. From the **Volume Name** drop-down, choose how you want the volume name to be created:

- To enter a specific name for a volume.
- To automatically create multiple volumes that share a common naming pattern, select **Auto Generate Name**.

5. If you choose to create a volume with a specific name, complete the following steps:
   a. From the **Volume Name** field, select **Create New Volume** from the drop-down.
   b. To name the volume, enter a name in the **New Volume Name** field.
   c. To choose the storage pool where the volume will reside, select the location from the **Storage Pool** drop-down,
   d. To specify the size of the volume, enter the size in GB in the **Volume Size (GB)** field. The minimum size is 8 GB and the value that you specify must be divisible by eight.
   e. Click **Next**.
   f. Review the summary and click **Finish**.
6. If you choose to automatically create multiple volumes that share a common naming pattern, complete the following steps:
   a. From the **Volume Name** field, select **Auto Generate Name**.

      NOTE: The template used to generate volume names must contain a variable that will produce a unique volume name. The variables that are allowed are: ${num} , ${num_5d}(an auto generated unique number.

   b. In the **Volume Name Template** field, enter a template name that includes a variable which enables PowerFlex Manager to produce a unique volume name.
   c. In the **How Many Volumes** field, select the number of volumes you want to create.

      The number must be less than 50.
   d. To choose the storage pool where the volume will reside, select the location from the **Storage Pool** drop-down.
   e. To specify the size of the volume, enter the size in GB in the **Volume Size (GB)** field. The minimum size is 8 GB and the value that you specify must be divisible by eight.
   f. Click **Next**.
   g. Review the volume details on the summary page and click **Finish**.

The resource group moves to the **In Progress** state and the new volume icons appear on the **Resource Group Details** page. You may see multiple volume components while the `add` operation is still in progress. One the operation is complete, you see one volume component with the count updated.

After the deployment completes successfully, you can click **View Volumes** in the **Storage** list on the **Resource Group Details** page to search for volumes that are part of the resource group.

The new volumes are displayed under the storage pool.

For a storage-only resource group, the volumes are created and mapped to a storage node

For a compute-only resource group, the volumes are mapped to SDCs. In the vSphere client, you can see the volumes in the storage section and also see the hosts that are mapped to the volumes, once the mappings are in place.

Related information

Resize a volume
Viewing and selecting volumes

Adding networks to a resource group

If you have PowerFlex compute-only nodes, you can update the workload network by adding a general purpose LAN to a resource group. Alternatively, you can add a static route to allow nodes to communicate across different networks.

1. On the **Resource Group Details** page, click **Add Resources** and choose **Add Network**.
   The **Add Network** window is displayed. All used resources and networks are displayed under **Resource Name** and **Networks**.
2. Click **Add Additional Network** to add an additional network:
   a. From the **Available Networks** list, select the network, and click **Add**.
   The selected network is displayed under **Network Name**. You can define a new network by clicking **Define a New Network**.
   b. Select **Port Group** from the **Select Port Group** list.
   c. Click **Save**.

      NOTE: You cannot remove an added network using PowerFlex Manager.

3. Click **Add Additional Static Route** to add an additional static route:

a. Click **Add New Static Route**.
b. Select a **Source Network**.

   The source network must be a PowerFlex data network.

c. Select a **Destination Network**.

   The destination network must be a PowerFlex data network.

d. Type the IP address for the **Gateway**.
e. Click **Save**.

Resize a volume

After adding volumes to a resource group, you can resize the volumes.

For a storage-only resource group, you can increase the volume size. For a VMware ESXi compute-only resource group, you can increase the size of the datastore that is associated with the volume.

If you resize a volume in a storage-only resource group, you must update the datastore size in the corresponding VMware ESXi compute-only resource group. The datastore size cannot exceed the size of the volume.

1. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.
2. Choose the volume that you want to resize:
   a. Click **Select Volume**.
   b. Enter a volume or datastore name search string in the **Search Text** box.
   c. Optionally, apply additional search criteria by specifying values for the **Size**, **Type**, **Compression**, and **Storage** filters.
   d. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   e. Select the row for the volume you want to resize.
   f. Click **Apply**.
3. Update the sizing information:

   If you are resizing a volume for a storage-only resource group, enter a value in the **New Volume Size (GB)** field. Specify a value that is greater than the current volume size. Values must be in multiples of eight, or an error occurs.

   If you are resizing a volume for a compute-only resource group, review the **Volume Size (GB)** field to see if the volume size is greater than **Current Datastore Size (GB)**. If it is, PowerFlex Manager expands the datastore size.

4. Click **Save**.

Related information

Adding volumes to a resource group

Removing resources from a resource group

You can remove resources that were previously added to a resource group.

If you are removing a node that is fulfilling a Meta Data Manager (MDM) role, verify that you will have at least two MDMs and one tiebreaker in the cluster after the removal of the node. When you remove a node, PowerFlex Manager puts a standby node in place of the one that you removed. For a storage-only resource group, the platform minimum is four nodes in the cluster. This enables you to scale down to three nodes for a component replacement.

If the node has an NSX-T or NSX-V configuration, you can remove the node information that PowerFlex Manager displays for the resource group, but not delete the node from the resource group entirely. If the PowerFlex gateway used in the resource group is being updated on the **Resources** page, PowerFlex Manager does not enable you to remove resources from the resource group.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group from which you want to delete resources.
3. In the right pane, click **View Details**.
4. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Remove Resource**.

5. In the **Remove Resource** dialog box, select the node that you want to remove and click **Next**.
6. Select the **Resource removal type**:
   - **Delete Resource** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components, and returns the components to the available inventory.
   - **Remove Resource** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex, and returns the components to the available inventory.
7. If you choose **Remove Resource**, perform the following steps:
   a. To keep the node in the inventory, select **Leave resource in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the node, select **Remove resource from the PowerFlex Manager inventory**.
   c. Click **Remove**.
8. If you choose **Delete Resource**, perform the following steps:
   a. If you are certain that you want to proceed, type **`DELETE RESOURCE`**.
   b. Click **Delete**.

When you delete a node from a resource group, PowerFlex Manager performs the following steps to clean up the underlying components associated with the node:

1. Reconfigures the MDM cluster to ensure that the cluster is in good working order. This step is necessary only if the node is part of an MDM cluster.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a scale-down operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.
2. If the node is on the list of storage nodes, it removes the node from the list of storage nodes within PowerFlex. The node might only be removed from the storage node list if the spare capacity is sufficient, and the environment provides enough space.
3. Removes the operating system on the node.
4. Removes the node from the SDC list within PowerFlex.

If you remove a node that is CloudLink enabled from a storage-only resource group, PowerFlex Manager also performs these cleanup tasks:

- Unencrypts the drives on the server.
- Removes the machine from the machine group.
- Removes IP addresses from approved networks.

PowerFlex Manager does not remove the machine groups or keystores. The machine groups and keystores can be reused for subsequent deployments.

After a node is deleted from a resource group, the **Resource Group Details** page shows the cluster without the removed node, and updates the MDM roles to reflect the new configuration.

Adding NVMe/TCP storage access

PowerFlex Manager enables you to add NVMe/TCP storage access to nodes in a resource group. When you add storage access, it installs the SDT (storage data target) package on all nodes in the resource group.

NVMe storage access is only supported for a resource group that is on PowerFlex 4.x. In addition, the option to add NVMe storage access is only available if the resource group was deployed with the **SDC Only** option selected for **Client Storage Access** in the template.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Add NVMe Storage Access**.
4. Review the list of nodes.
5. Type **`ADD SDT TO NODES`** to confirm your changes.
6. Click **Confirm**.

Additional resource group management tasks

This section provides additional tasks for resource group management.

Redeploying a resource group

You can redeploy a resource group that failed to deploy because of errors. Standard users can redeploy only a failed resource group that they have deployed.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, select a resource group in an error state and click **View Details** in the right pane.
3. On the **Resource Group Details** page, in the right pane, under **More Actions**, click **Retry**.
4. Click **Yes** when a confirmation message appears.

   If necessary, PowerFlex Manager reconfigures the MDM role for each node to be a manager or tiebreaker when you perform a retry operation. PowerFlex Manager rebalances the roles as needed for a new resource group or for an existing resource group. PowerFlex Manager applies the required role changes for nodes that are within the same resource group or in a different resource group.

Edit resource group information

You can edit information about a resource group such as the resource group name, description, and who has access to the resource group information.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. On the **Resource Groups** page, click the resource group.
3. On the **Resource Group Details** page, in the right pane click **Modify**.
4. In the **Modify Resource Group Information** dialog box, modify the **Resource Group Name** and **Resource Group Description** that identifies the resource group.
   - To update the firmware and software running on the nodes when you deploy a resource group that uses this template, select **Firmware and Software Compliance**.

     If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in lifecycle mode and restricts the actions that can be performed. In lifecycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

     NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains firmware on shared devices.

5. Set the **OS Image** for the resource group.

   You can optionally select a custom operating system image that will be used for future add node operations on the resource group. The custom image must be added to the **OS Image Repositories** tab on the **Settings** > **Compliance and OS Repositories** page.

   PowerFlex Manager filters the list of images available for selection to include only those images that are suitable for the current resource group.

6. Specify the permissions for this resource group under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriverReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
7. Click **Save**.

Viewing resource group details

You can view the state of a resource group at the component level. You can view the component topology and connections in a selected resource group template. You can view component logs, display the port view, place a component in service mode, and replace a disk.

If the resource group state is incomplete, a banner at the top of the page indicates that you must add volumes to make it fully functional. In some cases, you may only need to run **Update Resource Group Details** to ensure that the resource group is complete.

To view the details of a resource group component, scroll down on the **Resource Group Details** page. The following information is displayed based on the resource types in the resource group:

Table 20. Resource group details

Resource type	Description
Clusters	View the following information about the clusters in the VMware vCenter environment: • **Health** • **Data Center Name** • **Cluster Name** • **Asset/Service Tag** • **Management IP**
PowerFlex Gateways	View the following information about the gateways that are part of the resource group: • **System Name** • **Primary MDM IP** • **Management IP** • **Virtual IP** • **Total Device Groups** Provides the number of device groups in the resource group. A device group consists of devices that come from four or more nodes that are contained in a single protection domain. The nodes that belong to a device group have the same drive type or media type. The capacity of a device group is derived from the capacity of the storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group. The device group details include the following information for each device group: ○ **Name** ○ **Device Group Type** ○ **Size** ○ **Total Storage Pools** The storage pool details include the name of each storage pool, size, device type, and RAID set. • **Protection Domain** A protection domain is a logical entity that consists of and manages a group of storage nodes each of which contains devices. A storage node contains a PowerFlex Data server (PDS) and a Device Gateway Target (DGWT). These services enable the storage and processing of data to be separated. Data is logically isolated into specific datasets and performance capabilities within a specified protection domain. The effect of any specific device failure is limited.
CloudLink	View the following information about the CloudLink Centers participating in the resource group: • **Health** • **Hostname** • **Management IP** • **Machine Group** • **Keystore**

Table 20. Resource group details (continued)

Resource type	Description
	When a resource group is deployed, if CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers on the **Resource Group Details** page. PowerFlex Manager ensures that the resource group deployment succeeds as long as at least one CloudLink Center in the cluster is working.
Storage	View details about the storage volumes added for the resource group. Click **View Volumes** to search for volumes and see the following information about the volumes: • **Name** • **Size** • **Type** • **Compression** • **Storage Pool** • **Datastore**
Physical Nodes	View the following information about the nodes that are part of the resource group: • **Health** • **Asset/Service Tag** • **iDRAC IP/Hostname** • **PowerFlex Mode** The mode for each node is one of the following: ○ **Storage Only** includes only the storage node component. ○ **Compute Only** includes only the SDC component. • **MVM Hostname** • **Associated IPs** • **MDM Role** The MDM role is the metadata manager role. The MDM role applies only to those nodes that are part of a PowerFlex cluster. The MDM role is one of the following: ○ **Primary**: The MDM in the cluster that controls the storage nodes and SDCs. The primary MDM contains and updates the MDM repository, the database that stores the storage node configuration, and how data is distributed between the storage nodes. This repository is constantly replicated to the secondary MDMs, so they can take over with no delay. Every PowerFlex cluster has one primary MDM. ○ **Secondary**: An MDM in the cluster that is ready to take over the primary MDM role if necessary. ○ **Tie Breaker**: An MDM whose sole role is to help determine which MDM is the primary. ○ **Standby MDM**: A standby MDM can be called on to assume the position of a manager MDM when it is promoted to be a cluster member. ○ **Standby Tie Breaker**: A standby node that is prepared to take over as a tiebreaker.

View all settings

View All Settings allows you to view the settings configured on the resources in a resource group for deployment.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component. By displaying the details on separate tabs, PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

For a PowerFlex resource group, the **VMware Cluster** section of the **Settings** page shows the **PowerFlex Settings** section, and the **Target PowerFlex Gateway** should specify the gateway that is included with the resource group.

For a PowerFlex resource group, the **Use Local Storage For Dell PowerFlex** property is set to **true** within the **OS Settings** section for any node in an existing PowerFlex resource group deployment, and for the nodes in a new PowerFlex resource group deployment.

Generate troubleshooting bundle

Generate Troubleshooting Bundle allows you to generate a compressed file of logs to use for troubleshooting. The following logs are included:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes

View compliance report

View Compliance Report allows you to view the software or firmware compliance report.

Add resources

Add Resources allows you to add nodes, volumes, and networks to a resource group.

More actions

Under **More Actions**, you can perform the following tasks:

Table 21. More actions

Click...	To...
Update Resource Group Details	Update resource group definition.
Enter Service Mode	Put nodes in service mode so that maintenance operations can be performed.
Exit Service Mode	Take nodes out-of-service mode.
Reconfigure MDM Roles	Change the MDM role for a node in a PowerFlex cluster. For example, if you add a node to the cluster, you might want to switch the MDM role from an existing node to the new node.
Remove Resource Group	Remove a resource group that is no longer required. PowerFlex Manager supports two types of removal: • Delete the entire resource group, which deletes the deployment information and makes any required configuration changes for components that are associated with the resource group. This includes making destructive changes such as unconfiguring the nodes in the resource group and powering them down. • Delete the deployment information for the resource group without making any configuration changes to the deployed components.
Remove Resource	Removes resources from a resource group.
Retry	Redeploy a failed or canceled resource group. This button is only visible when a resource group is in a critical state.

Table 21. More actions (continued)

Click...	To...
	If you see a resource group fail for any reason, try clicking **Retry** to correct the problem. For example, a server deployment might fail with a `ChangeBootOrderByInstanceID failed` message. You can correct this failure by retrying the resource group.
Cancel	Cancel the resource group.

Resource actions

Under **Resource Actions**, you can perform the following tasks:

Table 22. Resource actions

Click...	To...
Add Resources	Select the type of the resources that you want to add to the resource group. If you must input data for the template that is used to create the running resource group, click **Confirm Resource Group Settings**. In the **Update Resource Group Component** window, enter values for all displayed fields. Click **Save**.
Remove Resource	Remove resources from a resource group.

Resource group information

In the **Resource Group Information** box on the right, you can view the following information:

- **Overall Resource Group Health**—Displays the health of the resource group. The following information determines the overall resource group health:
  - **Resource Health** —Displays the health monitoring of resources.
  - **Compliance**—Displays if the resources are firmware or software-compliant. This option is applicable only if you have enabled firmware or software update on the resource group.
  - **Deployment State**—Indicates if the deployment of the resource group completed successfully.
- **Deployed By**—Displays the name of the user who deployed the resource group.
- **Deployed On**—Displays the date and time when the resource group is deployed.
- **Reference Template**—Displays the name of the reference template that is used in the resource group.

  NOTE: For existing resource groups, the name displays as **User Generated Template** and not a template name from the inventory.
- **User Permissions**—Displays one of the following:
  - **Enabled**—Indicates that the permission is granted for one or more standard users to deploy this resource group.
  - **Disabled**—Indicates that the permission is not granted for the standard users to deploy this resource group.

Recent activity

Recent Activity displays component deployment status and information about the current deployed resource group.

You can also click the **Port View** tab to display port view details.

If you want to see the current firmware or software repository that is in use, look at **Target Version**. To change the compliance version, click **Change Target**.

If you select a minimal compliance version for a resource group, PowerFlex Manager puts the resource group in lifecycle mode and restricts the actions that can be performed. In lifecycle mode, the resource group supports monitoring, service mode, and compliance upgrade operations only. All other resource group operations are blocked.

Related information

Deploying a resource group

Viewing all settings

The **View All Settings** option lets you display the settings that are used to configure the resources in the deployment of a resource group.

When you click **View All Settings**, the **Resource Group Deployment Settings** page shows each top-level component on a separate tab. Click on a tab for a component to see the details for the component. By displaying the details on separate tabs, PowerFlex Manager greatly improves the performance when you are working with a resource group that includes a large number of nodes.

You can see all settings that are defined for the clusters and nodes, as well as storage. To search for volumes within the resource group, you can click **View Volumes** under **Storage**.

During volume operations, the **Resource Group Deployment Settings** page displays additional information about the volume operations that are in progress. If you create a new volume or add an existing volume, or resize a volume, the page provides details that can be helpful for troubleshooting. For example, if you specified an invalid name template string variable, the **Resource Group Deployment Settings** page indicates that this is the cause of a volume not being added.

Viewing and selecting volumes

Use this procedure to search for volumes in a resource group, and optionally select volumes for an add or resize operation.

You can use the volume search dialog to search for volumes using various types of search criteria. You can search by volume name or datastore name, as well as by volume type, compression setting, and storage pool name.

You can launch the volume search dialog from various places with the PowerFlex Manager user interface. You can launch the dialog from the following locations:

- **Storage** section of the **Resource Group Details** page
- **View All Settings** page
- **Add Volumes** wizard
- **Resize Volume** wizard

When you perform a search, PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the results would include more than 50 volumes, you must refine the search criteria to return only 50 volumes.

After performing a search in the **Add Volumes** wizard, you can select one or more existing volumes to add to a resource group. In the **Resize Volume** wizard, you can select the volume that you want to resize.

Identify the naming patterns that are used for the volumes and datastores within the resource group, so you can easily enter a search pattern.

1. To access the **Volumes** dialog from the **Storage** section of the **Resource Group Details** page:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Group Details** page, click **View Volumes** in the Storage section.

      You can also click **View Volumes** in the Storage section of the **View All Settings** page, which is also available from the **Resource Group Details** page.

      The **Volumes** dialog is displayed.
2. To access the **Volumes** dialog from the **Add Volumes** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the **Add Resources** button and choose **Add Volumes**.
   d. When PowerFlex Manager displays the **Add Volume** wizard, click **Add Existing volumes**.
   e. Click **Select Volumes**.
      The **Select Existing Volumes** dialog is displayed.
3. To access the **Volumes** dialog from the **Resize Volume** wizard:
   a. On the menu bar, click **Lifecycle** > **Resource Groups**.
   b. Select a resource group.
   c. On the **Resource Groups** page, click the volume component and choose **Volume Actions** > **Resize**.

d. Click **Select Volume**.
   The **Select Volume** dialog is displayed.

4. Provide the search criteria needed to filter the results to show 50 volumes or fewer:
   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, **Thick** or **Thin** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
      PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.
   g. If you are viewing volumes, click **Close** when you are done looking at the search results.
5. If you are adding existing volumes, you can select one or more volumes to add:
   a. Select each volume you want to add.
   b. Click **>>** to add the volumes.

      To remove a selected volume, click the checkbox for the volume to remove and click **<<**.

   c. When you are ready to add the selected volumes, click **Add**.
6. If you are selecting a volume for a resize operation, you can select the volume that you want to resize and click **Apply**.

Related information

Adding volumes to a resource group

Updating the details for a resource group

When new resources are added or removed outside of PowerFlex Manager, you can update the resource group details to pull in those changes. If nodes are removed externally, updating the resource group details does not pull in the changes.

You can update the details for a new resource group that was deployed with PowerFlex Manager, or for an existing resource group that includes resources that were not originally deployed with PowerFlex Manager.

If the PowerFlex gateway used in the resource group is currently being updated on the **Resources** page, PowerFlex Manager does not enable you to update the resource group details.

Whenever you update the resource group details, PowerFlex Manager pulls in a representation of the components from the VMware vCenter cluster, PowerFlex gateway, or CloudLink Center that was specified for the resource group. These components might include new nodes, volumes, SVMs, or management virtual machines. When a new component is added to the definition of a resource group, PowerFlex Manager does not modify the component.

If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Before you update the details for a resource group, ensure that you run inventory for VMware vCenter and the PowerFlex gateway.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Update Resource Group Details**.
4. Review the **OS Credentials** page, and click **Next**.

   PowerFlex Manager shows all nodes, SVMs, and management virtual machines, along with their credentials. This enables you to update the username and password if it has changed for any of these items.

5. Review the **Inventory Summary** page, and click **Next**.

   PowerFlex Manager does a rediscovery of the resource group, and tracks whether items are new to the resource group.

   In the **Inventory Update** column, PowerFlex Manager shows which items are added, removed, or kept as is within the resource group:

   - **Adding** indicates that an item was added externally and will now be added to the resource group.
   - **Removing** indicates that an item was removed externally and will now be removed from the resource group.
   - **No Change** indicates that an item was not modified externally and will be kept in the resource group.

6. Review the **Summary** page, and click **Finish**.
   PowerFlex Manager checks the configuration and inventory of selected resources to ensure that everything is configured properly. The resource group is moved to the **In Progress** state, and any new component types are displayed on the **Resource Group Details** page. **Recent Activity** shows that a new deployment started.

Confirming resource group settings

If a PowerFlex Manager upgrade added new required fields to components within the template from which a resource group was deployed, the **Confirm Resource Group Settings** button is displayed on the **Resource Groups** page. Although confirming resource group settings is not mandatory, certain resource group or resource functions are not available until this step is complete.

Click **Confirm Resource Group Settings** to launch the **Upgrade Resource Group Components** window. Fields in this window vary depending on which components contain newly required settings. Complete all the displayed fields, and click **Save**.

Removing a resource group

You can remove a resource group that is no longer required. PowerFlex Manager supports two types of removal:

- Delete the entire resource group, which deletes the deployment information and also makes any required configuration changes for components that are associated with the resource group.
- Remove just the deployment information for the resource group without making any configuration changes to the deployed components.

If the node has an NSX-T or NSX-V configuration, you can remove the deployment information for a resource group, but not delete the resource group entirely. PowerFlex Manager also does not allow you to delete a resource group if the PowerFlex gateway used in the resource group is currently being updated on the **Resources** page.

Standard users can delete only the resource groups that they have deployed.

To remove a resource group, perform the following steps:

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select the resource group.
3. On the **Resource Group Details** page, under **More Actions**, click **Remove Resource Group**.
4. In the **Remove Resource Group** dialog box, select the **Resource group removal type**:
   - **Delete Resource Group** makes configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex to unconfigure those components. Also, it returns the components to the available inventory.
   - **Remove Resource Group** removes deployment information, but does not make any configuration changes to the nodes, switch ports, virtual machine managers, and PowerFlex. Also, it returns the components to the available inventory.
5. If you choose **Remove Resource Group**, perform the following steps:
   a. To keep the nodes in the inventory, select **Leave nodes in PowerFlex Manager inventory and set state to** and select the state:
      - **Managed**
      - **Unmanaged**
      - **Reserved**
   b. To remove the nodes, select **Remove nodes from the PowerFlex Manager inventory**.
   c. Click **Remove**.
6. If you choose **Delete Resource Group**, perform the following steps:
   a. Select **Delete Clusters(s) and Remove from vCenter** to delete and remove the clusters from VMware vCenter.
   b. Select **Remove Protection Domain and Storage Pools from PowerFlex** to remove the protection domain and storage pools that are created during the resource group deployment.

      If you select this option, you must select the target PowerFlex gateway. The PowerFlex gateway is not removed. PowerFlex Manager removes only the protection domain and storage pools that are part of the resource group. If multiple resource groups are sharing a protection domain, you might not want to delete the protection domain.

      For a compression enabled resource group, PowerFlex Manager deletes the acceleration pool and the DAX devices when you delete the resource group.

   c. Select **Delete Machine Group and remove from CloudLink Center** to clean up the related components in CloudLink Center.

CloudLink Center cleanup includes the deletion of the machine group, keystore, and approved network that is related to the resource group being deleted. These components are removed from CloudLink Center only if all the machines that are associated to the machine group are deleted first. If all the machines that are related to this machine group are not removed, the cleanup does not succeed as there are machines associated with the machine group.

d. If you are certain that you want to proceed, type **DELETE RESOURCE GROUP**.
e. Click **Delete**.

Related information

Deploying a resource group

Canceling a resource group

Canceling a resource group stops a scheduled/pending, or in-process resource group deployment. After canceling a resource group, you can update the content and retry the deployment later.

1. On the menu bar, click **Lifecycle** > **Resource Groups**.
2. Select a resource group that is in **In Progress** or **Pending state**.
3. On the **Resource Group Details** page, click **Cancel** under **More Actions**.
4. On the **Cancel Deployment** page, click **Cancel Deployment**.

   You cannot cancel existing resource groups or resource groups that are used for managing the firmware.

   When you cancel a deployment, a banner displays at the top of the screen that displays a warning.

Preparing a template for a resource group deployment

This section includes tasks for preparing a template that will be used to deploy a set of infrastructure resources in a resource group.

A template is an object in PowerFlex Manager that represents resource types and their required configuration and topology. A template specifies requirements for the deployment of a set of infrastructure resources through PowerFlex Manager's automation workflows.

On the **Templates** page, you can access the default sample templates or create templates that meet your specific requirements. For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

PowerFlex Manager allows you to create templates that support storage-only and compute-only resource group deployments.

After you create a template, PowerFlex Manager saves the template in a draft state. You must publish the template before deploying it.

Lifecycle administrators can view and use only those templates for which they have been granted permission by a super user.

Related information

Deploying and provisioning

Basic tasks

This section provides basic tasks for template management.

Clone a template

The **Clone** feature allows you to copy an existing template into a new template. A cloned template contains the components that existed in the original template. You can edit it to add additional components or modify the cloned components.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment. If you already created a template using the **Add a template** option, skip this task.

To clone an existing template:

1. On the menu bar, click **Templates**.
2. Click **Sample Template** > **Select** and click a template from the list.

   You can also click **Clone** on the **My Templates** page if you want to clone one of your own templates, rather than one of the sample templates.
3. In the **Clone Template** dialog box, enter a template name in the **Template Name** box.
4. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
5. In the **Template Description** box, enter a description for the template.
6. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
7. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
8. Click **Next**.
9. On the **Additional Settings** page, provide new values for the **Network Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
10. Click **Finish**.
11. To modify the cluster components: Node (software only) and PowerFlex cluster, see the *Edit a Template* topic in the related information,

Related information

Configuring block storage
Edit a template

Sample templates

This topic lists the PowerFlex compute-only node and PowerFlex storage-only node sample templates that are provided with PowerFlex Manager.

To search for a sample template, click the filter icon and search for the template in the **Filter Templates** field.

PowerFlex compute-only node sample templates require that PowerFlex storage-only nodes are deployed first.

Table 23. Sample templates

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
Compute ESXi 100Gb - 2 Data - LACP	An ESXi PowerFlex compute-only node 100 GB networking deployment with full network automation.	3	2
Compute ESXi 25Gb - 2 Data - LACP	A standard ESXi PowerFlex compute-only node deployment with full network automation.	3	2
Compute ESXi 25Gb - 2 Data - LACP - Partial Network Automation	A standard ESXi PowerFlex compute-only node deployment with partial network automation.	3	2

Table 23. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
Compute ESXi 25Gb - 2 Data - Trunked	An ESXi PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port2) ● Two networks (data 1 and data 2) as SDC-storage node traffic on the other two ports ● 25 GB NIC	3	2
Compute Linux 100Gb - 4 Data - Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) ● Two networks (data 2 and data 3) on interface1 port2 ● Two networks (data 1 and data 4) on interface2 port2 ● 100 GB NIC	3	4
Compute Linux 25Gb - 2 Data - LACP - Partial Network Automation	A Linux PowerFlex compute-only node deployment with partial network automation.	3	2
Compute-Linux-25Gb-2 Data-Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port2) ● Two networks (data 1 and data 2) as SDC-storage node traffic on the other two ports ● 25 GB NIC	3	2
Compute-Linux-25Gb-4 Data-Trunked	A Linux PowerFlex compute-only node deployment with full network automation. ● Management and customer networks on two ports in mode1 bond (interface1 port1 and interface2 port1) ● Two networks (data 2 and data 3) on interface1 port2 ● Two networks (data 1 and data 4) on interface2 port2 ● 25 GB NIC	3	4
Storage 100 Gb - 2 Data - LACP	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 100 Gb - 2 Host Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. ● Management network in mode1 bond on two ports ● Two networks (data 1 and data 2) as SDC-storage node traffic ● Two networks (data 3 and data 4) as storage node-storage node traffic ● 100 GB NIC ● interface1 port1: management, data1 SDC-storage node traffic	5	2

Table 23. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
	• interface1 port2: data 3 storage node-storage node traffic • interface2 port1: management, data 2 SDC-storage node traffic • interface2 port2: data 4 storage node-storage node traffic		
Storage 100 Gb - 4 Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. • Management network in mode1 bond on the four ports • Four data networks, each on a different port • 100 GB NIC • interface1 port1: management, data 1 • interface1 port2: management, data 2 • interface2 port1: management, data 3 • interface2 port2: management, data 4	5	4
Storage 25 Gb - 2 Data - LACP - Compression Enabled with persistent memory	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - Compression Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. Requires nodes with NVDIMM.	5	2
Storage 25 Gb - 2 Data - LACP - Encryption Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - NVMe/TCP Enabled	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	2
Storage 25 Gb - 2 Data - LACP - Partial Network Automation	A Linux PowerFlex storage-only node deployment using erasure coding with partial network automation.	5	2
Storage 25 Gb - 2 Host Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation. • Management network in mode1 bond on two ports • Two networks (data 1 and data 2) as SDC-storage node traffic • Two networks (data 3 and data 4) as storage node-storage node traffic • 25 GB NIC • interface1 port1: management, data 1 SDC-storage node traffic • interface1 port2: data 3 storage node-storage node traffic • interface2 port1: management, data 2 SDC-storage node traffic • interface2 port2: data 4 storage node-storage node traffic	5	2
Storage 25 Gb - 4 Data - Trunked	A Linux PowerFlex storage-only node deployment using erasure coding with full network automation.	5	4

Table 23. Sample templates (continued)

Template name	Description	Minimum number of nodes required	Minimum number of data networks required
	• Management network in mode1 bond on the four ports • Four data networks, each not in a bond • 25 GB NIC • interface1 port1: management, data 1 • interface1 port2: management, data 2 • interface2 port1: management, data 3 • interface2 port2: management, data 4 This template was called **Storage-NoBonding** in previous releases.		
Management - CloudLink Center	A two-node CloudLink Center cluster deployment, for use in templates with encryption.	2(VMs)	N/A

Add a template

The **Create** feature allows you to create a template, clone the components of an existing template into a new template, or import a pre-existing template.

For most environments, you can clone one of the sample templates that are provided with PowerFlex Manager and edit as needed. Choose the sample template that is most appropriate for your environment.

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click **Create**.
3. In the **Create** dialog box, select one of the following options:
   - **Clone an existing PowerFlex Manager template**
   - **Upload External Template**
   - **Create a new template**

   If you select **Clone an existing PowerFlex Manager template**, select the **Category** and the **Template to be Cloned**. The components of the selected template are in the new template.
4. Enter a **Template Name**.
5. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
6. Enter a **Template Description** (optional).
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?** by performing one of the following actions:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific life-cycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the displayed list.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the check box next to the users to grant or block access.
   - To grant access to super users and all life-cycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Next**.
10. In **Additional Settings** provide the new values for the following settings:
    - **Network**
    - **PowerFlex Gateway**

- **Node Pool**

11. Click **Finish**.

Build and publish a template

After creating a template using the **Create** feature, you use the template builder page to build and publish the customized template. Publishing a template indicates that a template is ready for deployment.

1. Click **Modify Template**.
2. To add a component type to the template, click **Add Node**, **Add Cluster**, or **Add VM** at the top of the template builder.

   The corresponding **<component type> component** dialog box appears.
3. If you are adding a node, choose one of the following network automation types:
   - **Full Network Automation**
   - **Partial Network Automation**

   When you choose **Partial Network Automation**, PowerFlex Manager skips the switch configuration step, which is normally performed for a resource group with **Full Network Automation**. Partial network automation allows you to work with unsupported switches. However, it also requires more manual configuration before deployments can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches.

   In the **Number of Instances** box, provide the number of component instances that you want to include in the template.
4. If you are adding a cluster, in the **Select a Component** box, choose the **PowerFlex Cluster** type.
5. If you are adding a VM, in the **Select a Component** box, choose one of the following cluster types:
   - **CloudLink Center**
   - **PowerFlex Gateway**
6. Under **Related Components**, perform one of the following actions:
   - To associate the component with all existing components, click **Associate All**.
   - To associate the component with only selected components, click **Associate Selected** and then select the components to associate.

   Based on the component type, specific settings and properties appear automatically that are required and can be edited.
7. Click **Save** to add the component to the template builder.
8. Repeat steps 1 through 6 to add additional components.
9. After you finish adding components to your template, click **Publish Template**.

   A template must be published to be deployed. It remains in draft state until published.

   After publishing a template, you can use the template to deploy a resource group on the **Resource Groups** page.

Related information

Support for full and partial network automation
Deploying a resource group

Edit a template

You can edit an existing template to change its draft state to "published" for deployment or to modify its components and their properties.

1. On the menu bar, click **Templates**.
2. Open a template, and click **Modify Template**.
3. Make changes as required to the settings for components within the template.

   Based on the component type, required settings and properties are displayed automatically. You can edit these settings by performing these steps:

   NOTE: If an RCM associated with the template is modified, a wrench icon labelled **Modified** is displayed. However, if the update file is moved or deleted, the wrench icon with the text, Needs Attention, is displayed.

a. To edit PowerFlex cluster settings, select the **PowerFlex Cluster** component and click **Modify**. Make the necessary changes and click **Save**.
b. To edit node settings, select the **Node** component and click **Modify**. Make the necessary changes, and click **Save**.

4. Optionally, click **Publish Template** to make the template ready for deployment.

Related information

Component types

(Optional) Edit template information

Optionally, edit the information for a template, with the following procedure:

1. On the menu bar, click **Templates**.
2. On the **Templates** page, click the template that you want to edit and click **Modify Template** in the right pane.
3. On the template builder page, in the right pane, click **Modify**.
4. In the **Modify Template Information** dialog box, enter a template name in the **Template Name** field.
5. Select a template category from the **Template Category** list. To create a template category, select **Create New Category**.
6. In the **Template Description** box, enter a description for the template.
7. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
8. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriverReplacer users to the displayed list.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
9. Click **Save**.

Deploying a resource group

Deployment is the automated process of selecting and configuring specific resource requirements that are outlined in a template using the integrated automation workflows that are provided with PowerFlex Manager. You cannot use a template that is in a draft state to deploy a resource group. Publish the template before using it to deploy a resource group.

1. On the menu bar, click one of the following:
   - **Lifecycle** > **Resource Groups**. Then, click **Deploy New Resource Group**.
   - **Lifecycle** > **Templates**. Then, click **Deploy**.

   The **Deploy Resource Group** wizard opens.
2. On the **Deploy Resource Group** page, perform the following steps:
   a. From the **Select Published Template** list, select the template to deploy a resource group.
   b. Enter the **Resource Group Name** (required) and **Resource Group Description** (optional) that identifies the resource group.
   c. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

      You cannot select a minimal compliance version when you deploy a new resource group, since it only includes server firmware updates. The compliance version for a new resource group must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Firmware and Software Compliance** list.

      PowerFlex Manager checks the VMware vCenter version to determine if it matches the VMware ESXi version for the selected compliance version. If the ESXi version is greater than the vCenter version, PowerFlex Manager blocks the

resource group deployment and displays an error. PowerFlex Manager instructs you to upgrade vCenter first, or use a different compliance version that is compatible with the installed vCenter version.

NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. The global default firmware repository maintains the firmware on the shared devices.

d. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific life cycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     i. Click **Add User(s)** to add one or more LifecycleAdmin or DriveReplacer users to the list displayed.
     ii. Select which users have access to this resource group.
     iii. To delete a user from the list, select the user and click **Remove User(s)**.
     iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all life cycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.

3. Click **Next**.
4. On the **Deployment Settings** page for **Device Group Spare Capacity** select **Custom** > **Nodes**.

   You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.

   Utilizing spare devices is a feature that is intended for advanced users and is not recommended for general use.

5. Click **Next**.
6. On the **Schedule Deployment** page, select one of the following options and click **Next**:
   - **Deploy Now**—Select this option to deploy the resource group immediately.
   - **Deploy Later**—Select this option and enter the date and time to deploy the resource group.
7. Review the **Summary** page.
   The **Summary** page gives you a preview of what the resource group will look like after the deployment.
8. Click **Finish** when you are ready to begin the deployment. If you want to edit the resource group, click **Back**.

   PowerFlex Manager displays a message indicating that a new job is being created for this resource group.

   PowerFlex Manager creates a configuration file that is called `/etc/sysctl.d/powerflex-so.conf` for a storage-only resource group. PowerFlex Manager maintains default values in a conf file for resource group deployment settings. If you have an old `sysctl.conf` file from a previous release, it overrides these settings. If you must customize these settings, you can do it in `sysctl.conf`.

Related information

Lifecycle
Viewing resource group details
Adding components to a resource group
Build and publish a template
Component types
Removing a resource group

Additional template management tasks

This section provides additional tasks for template management.

Add cluster component settings to a template

To deploy a PowerFlex compute-only node , you must add VMware and PowerFlex cluster components settings to a template. For a PowerFlex storage-only node deployment, you must add cluster settings for the PowerFlex cluster component only.

1. On the template builder page, click **Add Cluster**.
2. In the **Cluster Component** dialog box, from the **Select a Component** list, select **VMware cluster**, or **PowerFlex Cluster**.
3. Under **Related Components**, select the components that you want to map to the selected cluster instance.

4. Click **Continue**.
5. Under **Cluster Settings**, specify the settings that you want to configure on the cluster components and click **Add**.

Related information

Cluster component settings

Add a new interface

Use the **Add New Interface** feature to create a network interface to match to a network card on a node when deploying a template.

PowerFlex Manager supports a network layout for compute-only, and storage-only deployments. This network layout allows you to use trunk ports and port channels for data networks, instead of access ports. In this type of configuration, both data networks are on both NICs, teamed or bonded together in the operating system.

- For a compute-only deployment, the first port on both interfaces is for trunk traffic, whereas the second port is for data 1 and data 2.
- For a storage-only deployment, the first port is for trunk traffic and data 1, whereas the second port is for data 2. PowerFlex Manager still supports the legacy network layout, but the sample templates use the new configuration.

1. On the node component page, under **Network Settings**, click **Add New Interface**.
2. Enter the following information for the new interface:
   - **Port Layout**—Select the NIC type from the list.

     For a VMware ESXi deployment with an NVIDIA card, you must select the two port, 25-gigabit NIC type.
3. Enter the network VLANs for each port.
   a. Click **Choose Networks** for a port.
   b. To add one or more networks to the port, select **Add Networks to this Port**, then click the check box for each network you want to add from the **Available Networks** list. Alternatively, click the check box in the upper left corner next to the **Name** label to select all the available networks.

      If you want to filter the list by network type, select a **Network Type**, then enter a name or VLAN ID to search.

      Click **>>** to move the selected items to the **Selected Networks** list on the right.
   c. To mirror network settings from another port for which you have already chosen the network VLANs, select **Mirror this Port with Another Port**. Then, select the other interface and port from which you want to mirror this port.
   d. Click **Save**.
4. To view the list of nodes that match the network configuration parameters, click **Validate Settings**.

   The list of nodes is filtered according to the target boot device and NIC type settings specified.

   When you enable PowerFlex settings for the node, the **Validate Settings** page filters the list of nodes according to the supported storage types (NVMe, All flash, and HDD). Within the section for each storage type, the nodes are also sorted by health, with the healthy (green) nodes displayed first and the critical (red) nodes displayed last.

NOTE: If you select the same network on multiple interface ports or partitions, PowerFlex Manager creates a team or bond on systems with the VMware ESXi operating system. This configuration enables redundancy.

Add a new static route

A static route allows nodes to communicate across different networks. Use the add a new static route feature to create a static route in a template.

1. On the node component page, under **Network Settings**, click **Enabled** under **Static Routes**.
2. Click **Add New Static Route**.
3. Enter the following information for the static route:
   - **Source Network**—Select the PowerFlex data network that is the source.

     If you add or remove a network for a port, the **Source Network** list still shows the old networks. In order to see the changes, you must save the node settings and edit the node again.
   - **Destination Network**—Select the PowerFlex data network that is the destination for the static route.

- **Gateway**—Enter the IP address for the gateway.

View template details

View more details about the topology of the components in the template builder.

1. On the **Templates** page, select a template.
2. In the right pane, click **View Details**.
3. To view all the component settings, click **More Actions** > **View All Settings** in the right pane.
   The **Template Settings** page shows each top-level component on a separate tab. Click a tab for a component to see the details for the component.
   NOTE: If an RCM is mapped to the template and is modified, a wrench icon with the text, "Modified", is displayed. However, if the update file is moved or deleted, the wrench icon with the text, "Needs Attention", is displayed.

Related information

Component types

Export a template

To export a template:

1. On the menu bar, click **Templates**.
2. Select the template that you want to export.
3. Click **Export**.
4. In the **Export Template to ZIP File** window, enter values as follows:
   a. Enter a name for the template file in the **File Name**.
   b. If you have set an encryption password, select **Use Encryption Password from Backup Setting** to use that password. To set an encryption password, clear this option.
   c. If you clear **Use Encryption Password from Backup Setting**, two additional fields display. Enter a new password in the **Set File Encryption Password** field and enter the password again to confirm it.
5. Click **Export** to download the file. Select a location to save the file and click **OK**.

Import a template

The **Import Template** feature allows you to import the components of an existing template and its component configurations into a template. For example, you can create a template that defines a specific cluster and node topology and import this template definition into another template. After importing, you can modify the component properties of the imported components.

Editing an imported template does not affect the original template.

As an alternative to this procedure, you can also start from a new template and import an existing template as you create it. This is a better approach to use. To do this, you select **Create**, then choose **Clone an existing VxFlex Manager template** in the wizard.

To import a template, perform the following steps:

1. On the menu bar, click **Templates**.
2. On the **Templates** page, select the template into you want to import an existing template and click **Edit** in the right pane.
3. On the **Template Builder** page, in the right pane, click **Import Template**.
4. In the **Import Template** dialog box, select a specific template from the **Select a template** list and click **Import**.

Upload an external template

To upload a template, perform the following steps:

1. On the **Templates** page, click **Create**.
2. In the **Create** dialog box, select **Upload External Template**.

The **Upload External Template** page appears.

3. Click **Browse** to select the exported file from your directory.
4. Select the **Use Encryption Password from Backup Settings** checkbox, if you have set the encryption password in **Backup and Restore**.

   Clear the **Use Encryption Password from Backup Settings** checkbox if you have not set the encryption password in **Backup and Restore**.

   If you clear the **Use Encryption Password from Backup Settings**, you must enter the encryption password in the **Encryption Password** field.
5. Enter the template name in the **Template Name** field.
6. Select a template category from the **Template Category** list.

   If you want to create a template category, select **Create New Category**.
7. In the **Template Description** box, enter a description for the template.
8. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.

   NOTE: Changing the firmware repository might update the firmware level on nodes for this resource group. Firmware on shared devices is maintained by the global default firmware repository.
9. Indicate **Who should have access to the resource group deployed from this template** by selecting one of the following options:
   - To restrict access to super users, select **Only PowerFlex SuperUser**.
   - To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
     a. Click **Add User(s)** to add one or more LifecycleAdmin or DriverReplacer users to the list displayed.
     b. Select which users will have access to this resource group.
     c. To delete a user from the list, select the user and click **Remove User(s)**.
     d. After adding the users, select or clear the checkbox next to the users to grant or block access.
   - To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.
10. Click **Upload and Continue**.
    The **Additional Settings** page appears.
11. On the **Additional Settings** page, provide new values for the **Network Settings**, **OS Settings**, **Cluster Settings**, **PowerFlex Gateway Settings**, and **Node Pool Settings**.
12. Click **Finish**.

Deploy a CloudLink Center

Use this task to deploy a CloudLink Center. Although PowerFlex Manager supports up to three instances of CloudLink Center, two are recommended.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Ensure the following:

- Hypervisor management or PowerFlex management networks are added to PowerFlex Manager.
- A VMware vCenter with a valid data center, cluster, network, and datastore is discovered.

1. On the menu bar, click **Templates** > **Sample Templates**.
2. On the **Sample Templates** page, click **Management - CloudLink Center**.
3. Click **More Actions** > **Clone**.
4. In the **Clone Template** wizard, do the following:
   a. Enter a template name.
   b. From the **Template Category** list, select a template category. To create a category, select **Create New Category** from the list.
   c. Enter a **Template Description** (optional).
   d. To specify the version to use for compliance, select the version from the **Firmware and Software Compliance** list or choose **Use PowerFlex Manager appliance default catalog**.
   e. Specify the resource group permissions for this template under **Who should have access to the resource group deployed from this template?**:

- To restrict access to super users, select **Only PowerFlex SuperUser**.
- To grant access to super users and some specific lifecycle administrators and drive replacers, select the **PowerFlex SuperUser and specific LifecycleAdmin and DriveReplacer** option, and perform the following steps:
  i. Click **Add User(s)** to add one or more LifecycleAdmin or DriverReplacer users to the list displayed.
  ii. Select which users will have access to this resource group.
  iii. To delete a user from the list, select the user and click **Remove User(s)**.
  iv. After adding the users, select or clear the checkbox next to the users to grant or block access.
- To grant access to super users and all lifecycle administrators and drive replacers, select **PowerFlex SuperUser and all LifecycleAdmin and DriveReplacer**.

f. Click **Next**.

5. On the **Additional Settings** page, do the following:
   a. Under **Network Settings**, select **Hypervisor Network** (PowerFlex management network).
   b. Under **OS Settings**, select **CLC** credential or click **+** to create a credential with root or CloudLink user.
   c. Under **CloudLink Settings**, do the following:
      i. Select the secadmin credential from the list or click **+** to create a secadmin credential and do the following:
         i. Enter **Credential Name**.
         ii. Enter **User Name** as **`secadmin`**.
         iii. Leave the **Domain** empty.
         iv. Enter the password for **`secadmin`** in **Password** and **Confirm Password**.
         v. Select **V2** in **SNMP Type**.
         vi. Click **Save**.
      ii. Select a license file from the list based on the types of drives or click **+** to upload a license through the **Add Software License** page.

      NOTE: For SSD/NVMe drives, upload a capacity-based license. For SED drives, upload an SED-based license.

   d. Under **Cluster Settings**, select **Management vCenter**.
   e. Click **Finish**.
6. Select the **VMware Cluster** and click **Edit** > **Continue**.
   a. Under **Cluster Settings**, select **Datacenter Name**, and then select **Cluster Name** from the drop-down list.
   b. Under **vSphere Network Settings**, select the hypervisor management port group or PowerFlex management port group.
   c. Click **Save**.

   NOTE: To deploy a CloudLink Center from PowerFlex Manager, you need a Management vCenter, Datacenter, and Cluster along with DvSwitch port groups for PowerFlex management or hypervisor management.

7. Select the **VM** and click **Edit** > **Continue** (by default, the number of CloudLink instances is two and PowerFlex Manager supports up to three instances).
   a. Under **VM Settings** select the **Datastore** and **Network** from the drop-down list.
   b. Under **CloudLink Settings** select the following:
      i. For **Host Name Selection**, either select **Specify At Deployment Time** to manually enter at deployment time or **Auto Generate** to have PowerFlex Manager generate the name.
      ii. Enter the vault passwords.

      NOTE: Other details such as operating system credentials, NTP, and secadmin credentials are auto populated.

8. Under **Additional CloudLink Settings**, you can choose either or both of the following settings:
   - **Configure Syslog Forwarding**
     a. Select the checkbox to configure syslog forwarding.
     b. For **Syslog Facility**, select the **syslog** remote server from the list.
   - **Configure Email Notifications**
     a. Select the checkbox to configure email alerts.
     b. Specify the IP address of the email server.
     c. Specify the port number for the email server. The default port is 25. Enter the port numbers in a comma-separated list, with values between 1-65535.
     d. Specify the email address for the sender.
     e. Optionally, specify the username and password.

9. Click **Save**.
10. Click **Publish Template** and click **Yes** to confirm.
11. In the **Deploy Resource Group** wizard, do the following:
    a. Select the published template from the drop-down list, and enter **Resource Group Name** and description.
    b. Select who should have the access to the resource group and click **Next**.
    c. Provide a **Hostname** and click **Next**.
    d. Select **Deploy Now** or **Schedule deployment** and click **Next**.
    e. Review the details in the **Summary** page and click **Finish**.

Delete a template

The **Delete Template** option allows you to delete a template from PowerFlex Manager.

1. On the menu bar, click **Templates**.
2. Select the template that you want to delete. Click **More Actions** > **Delete Template** in the right pane.
3. Click **Yes** to confirm the deletion.

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Managing resources

This section includes tasks for discovering and managing resources in the PowerFlex Manager inventory.

A resource is a physical and virtual data center object that PowerFlex Manager interacts with, including but not limited to nodes, network switches, VM managers (for example, VMware vCenter), and element managers (for example, CloudLink Center).

The **Resources** page displays detailed information about all the resources and node pools that PowerFlex Manager has discovered and inventoried. You can perform various operations from the **All Resources** and **Node Pools** tabs.

NOTE: Depending upon the number of resources, it might take few minutes to display the discovered resources every time you run the inventory.

The **All Resources** tab displays the following information about the resources that are discovered and managed in PowerFlex Manager:

Table 24. All Resources tab

Field name	Description
Health	Indicates the resource health.
Compliance	Indicates if the resource firmware and software compliance state is **Compliant**, **Non-compliant**, **Update required**, or **Update failed**. Compliance is determined by the firmware/software version for the selected resource, based on the default compliance version. Click the compliance status to view the compliance report.
OS Hostname	Indicates the resource hostname. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only. **NOTE:** PowerFlex Manager retrieves the hostname value from iDRAC and not the operating system. If the hostname field is not updated in iDRAC, an incorrect value might display in PowerFlex Manager. Certain operating systems require additional packages to be installed for iDRAC to update the correct hostname.
Resource Name	Indicates the name of the resource.
Management IP	Indicates the resource IP address. Click the IP address to open the **Element Manager**.
Asset/Service Tag	Indicates the resource service tag. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.
Deployment Status	Indicates if the resource deployment status is **In Use**, **Not in use**, **Available**, **Updating resource**, or **Pending Updates**. Click the deployment status to view resource group details.
Managed State	Indicates the resource state. If you did not upload a license file in the **Initial Configuration Wizard**, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

Table 24. All Resources tab (continued)

Field name	Description
Model	Indicates the resource manufacturer name and model number. This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To filter the resources that display, click the toggle filters icon on the **Resources** page.

Table 25. Resources page filters

To filter based on...	Select a value...
Resource Type	• **Element Manager** • **Node (Hardware/Software Management)** • **Switches** • **VM Manager** • **PowerFlex Gateway** • **Node (Software Management)**
Health	• **Healthy** • **Warning** • **Critical** • **Unknown** • **Service Mode**
Resource Groups	Allows you to choose one or more resource groups.
Resource State	**Managed**, **Unmanaged**, or **Reserved**
Node Pool	Lets you choose **Global** or one or more node pools.

On the **Resources** page, you can also perform the following tasks:

Table 26. Resources page tasks

Task	How to perform
Discover new resources	Click **Discover Resources**.
Manually run an inventory operation on a resource and update PowerFlex Manager with the latest resource details	Select a resource and click **Run Inventory**. After running the inventory, you can perform an **Update Resource Group Details** operation on any resource group that requires the updated resource data.
Remove a resource from PowerFlex Manager	Click **Remove**.
Update the firmware or software of a resource	Select one or more resources and click **Update Resources**.
Export a compliance details report for all resources to a CSV or PDF file	Click **Export Report** and select a report format.
Modify the current state of a resource	Select a resource and click **Change resource state to**. You can change the state to **Managed**, **Unmanaged** or **Reserved**.
View detailed information about a resource	Select the resource. In the right pane, click **View Details**.
View a firmware and software compliance report for a resource	Select the resource. In the right pane, click the link corresponding to **Compliance** field.
Update password for a resource	Select the resource and click **Update Password**.

Basic tasks

This section provides basic tasks for resource management.

Discover a resource

A resource must be discovered in PowerFlex Manager in order for PowerFlex Manager to manage it.

Before you start discovering a resource, complete the following:

- Gather the IP addresses and credentials that are associated with the resources.
- Ensure that both the resources and PowerFlex Manager are connected to the network.

A resource in PowerFlex Manager is categorized into one of the following groups: Element Manager, Node (Hardware/Software Management), Switch, VM manager, PowerFlex gateway, Node (Software Management), or PowerFlex System.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources**.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select a resource that you want to discover.
      - **Element Manager**, for example, a CloudLink Center
      - **Node (Hardware / Software Management)**
      - **Switch**
      - **VM Manager**
      - **PowerFlex Gateway**
      - **Node (Software Management)**
      - **PowerFlex System**

      If you want to discover a PowerFlex software node or management VM (MVM) node with a credential that uses SSH key pairs, you must add the public keys to the nodes manually before performing the discovery. If you do not add the public keys first, the discovery fails.

      For a PowerFlex 5.0 system, the PowerFlex instance that is used is a management cluster.

   b. Enter the management IP address (or hostname) of the resources that you want to discover in the **IP/Hostname Range** field.

      To discover one or more nodes by IP address, select **IP Address** and provide a starting and ending IP address.

      To discover one or more nodes by hostname, select **Hostname** and identify the nodes to discover in one of the following ways:

      - Enter the fully qualified domain name (FQDN) with a domain suffix.
      - Enter the FQDN without a domain suffix.
      - Enter a hostname search string that includes one of the following variables:

      Table 27. Hostname search strings

Variable	Description
${num}	Produces an automatically generated unique number.
${num_2d}	Produces an automatically generated unique number that has two digits.
${num_3d}	Produces an automatically generated unique number that has three digits.

      If you use a variable, you must provide a start number and end number for the hostname search.

   c. Select one of the following options from the **Resource State** list:

      Table 28. Resource state

Option	Description
Managed	Select this option to monitor the firmware version compliance, upgrade firmware, and deploy resource groups on the discovered resources. A managed state is the default option for the switch, vCenter, element manager, and PowerFlex gateway resource types.

Table 28. Resource state (continued)

Option	Description
	Resource state must be set to **Managed** for PowerFlex Manager to send alerts to Dell Technologies connectivity services.
Unmanaged	Select this option to monitor the health status of a device and the firmware version compliance only. The discovered resources are not available for a firmware upgrade or deploying resource groups by PowerFlex Manager. This option is the default for the node resource type. If you have yet not uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, **Unmanaged** is the only option available.
Reserved	Select this option to monitor firmware version compliance and upgrade firmware. The discovered resources are not available for deploying resource groups by PowerFlex Manager.

d. To discover resources into a selected node pool instead of the global pool (default), select an existing or create a node pool from the **Discover into Node Pool** list. To create a node pool, click the **+** sign to the right of the **Discover into Node Pool** box.

e. Select an existing or create a credential from the **Credentials** list to discover resource types. To create a credential, click the **+** sign to the right of the **Credentials** box. PowerFlex Manager maps the credential type to the type of resource that you are discovering. The credential types are as follows:

- Element Manager
- Node (Hardware/Software Management)
- Switch
- VM Manager
- PowerFlex Gateway
- Node (Software Management)
- PowerFlex System

The default node (Hardware/Software Management) credential type is **Dell PowerEdge iDRAC Default**.

f. If you want PowerFlex Manager to automatically reconfigure the iDRAC nodes it finds, select the **Reconfigure discovered nodes with new management IP and credentials** checkbox. This option is not selected by default, because it is faster to discover the nodes if you bypass the reconfiguration.

g. To have PowerFlex Manager automatically configure iDRAC nodes to send alerts to PowerFlex Manager, select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox.

4. Click **Next**.

   You might have to wait while PowerFlex Manager locates and displays all the resources that are connected to the managed networks.

   To discover multiple resources with different IP address ranges, repeat steps 2 and 3.

5. On the **Discovered Resources** page, select the resources from which you want to collect inventory data and click **Finish**. The discovered resources are listed on the **Resources** page.

Related information

Getting started
Configuring block storage
Resource health status
Compliance status

Discover a PowerFlex system

Perform this task to discover a PowerFlex system.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

Before you start discovering a PowerFlex system, complete the following:

- Gather the IP addresses and credentials that are associated with the resource.
- Ensure that both the resource and PowerFlex Manager are connected to the network.

1. Access the **Discovery Wizard** by performing either of the following actions:
   - On the **Getting Started** page, click **Discover Resources**.
   - On the menu bar, click **Resources**. On the **Resources** page, click **Discover Resources** on the **All Resources** tab.
2. On the **Welcome** page of the **Discovery Wizard**, read the instructions, and click **Next**.
3. On the **Identify Resources** page, click **Add Resource Type**, and perform the following steps:
   a. From the **Resource Type** list, select **PowerFlex System**.
   b. Enter the management IP addresses of the LIA nodes in the **MDM Cluster IP Address** field. You must provide the IP addresses for all the nodes in a comma-separated list. The list should include a minimum of three nodes and a maximum of five nodes.

      If you forget to add a node, the node will not be reachable after discovery. To fix this problem, you can rerun the discovery later to provide the missing node. You can enter just the one missing node, or all the nodes again. If you enter IP addresses for any nodes that were previously discovered, these nodes are ignored on the second run.

   c. Select an existing credential or create a new one from the **Credentials** list. The credential must be a **PowerFlex Management System** credential. Be sure to provide the LIA password that was used for the original setup. The LIA password is required for the mTLS configuration.
4. Click **Next**.
5. On the **Summary** page, click **Finish**.
   After you complete the discovery, you should see a **PowerFlex System** resource on the **Resources** page.
6. SSH to the two SVMs associated with the **`powerflex-mds`** system. Run `scli --add_certificate --certificate_file /opt/emc/scaleio/mdm/cfg/mgmt_CA.pem`.

Viewing resource details

Standard users can view details only for resources that are part of node pools for which they have permissions.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a resource for which you want to view details.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the resource and associated components.

   For the PowerFlex rack Intelligent Physical Infrastructure (IPI) cabinet, you can view information including system name and location, model number, IP address, cabinet temperature, and humidity details.

   For PowerEdge servers, you can also see performance details, including system usage, CPU usage, memory usage, and I/O usage. Performance usage values are updated every five minutes.

   For a PowerFlex gateway, you can view MDM cluster, storage, and node details.

   For supported switches, you can view packet count by port for all ports or a specific port. The packet count for all ports shows the average packet count for all the ports in that switch.

   For storage only, you can view details of the NIC interfaces, port connections, firmware/software revisions, CPUs, memory, and local storage.

Viewing node details

From the **Node Details** page, you can:

- Open the remote console of the node's Integrated Dell Remote Access Controller (iDRAC).
- View recent activities performed on the node.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select a node from the resources list to view its details.
3. In the **Details** pane, click **View Details**.
   The **View Details** page displays the detailed information about the node on the following tabs:
   - **Network Interfaces**
   - **Port View**

- **Firmware/Software Revisions**
- **CPUs**
- **Memory**
- **Local Storage**

  NOTE: You can select one of the following options from the **Filter By** list.

  - **Logical Disks**
  - **Physical Disks**

    To see which disks are self-encrypting drives (SED), look at the **Security Status**. The value **Encryption Capable** indicates that the disk is an SED. The value **Not Capable** indicates that the disk is not an SED.

    On the **Resources** page for the node, filter by "Physical Disks" to view the details of the Vault Optimized Storage Solution (VOSS) under the PCIe Extender section.

Viewing VMware vCenter details

You can view details for VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click VMware vCenter from the resource list to view the details.
   In the right pane, you can see the basic information about the VMware vCenter, such as power state, management IP address, data centers, clusters, hosts, and virtual machines.
3. In the **Details** pane, under **vCenter Details**, expand **vCenter** > **Datacenter** > **Cluster** to view the lists of nodes and applications.

Opening the iDRAC remote console

To simplify routine node maintenance, you can open a remote console to the node's Integrated Dell Remote Access Controller (iDRAC) directly from PowerFlex Manager.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a node.
3. In the **Details** pane, click **View Details**.
4. Click **Launch iDRAC GUI**.

Viewing PowerFlex system details

You can view performance metrics and storage details for a PowerFlex system.

PowerFlex Manager stores up to 15 GB of PowerFlex gateway PowerFlex system metrics. Once this threshold is exceeded, PowerFlex Manager automatically purges the oldest data to free up space.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click a PowerFlex system resource from the list to view its details.
   The right pane displays basic information about the resource, such as the unused and spare space for PowerFlex. It also shows the number of protection domains, volumes, storage data clients (SDC), and storage nodes.
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the PowerFlex System resource on the following tabs:
   - **MDM Cluster**: Displays details about the MDM roles within the cluster.
   - **Storage**: Displays the protection domain, device groups, storage pools, volumes, and raw capacity (TB).
   - **Nodes**: Displays the protection domain, type (storage node or SDC), connection, and the name, IP address for the storage node or SDC.

   The **Details** page also allows you to launch a wizard to see the status of a recent upgrade or reconfigure the MDM roles in the PowerFlex cluster.

4. Click the **MDM Cluster** tab to see information about the MDM roles within the cluster.
5. Click the **Storage** tab. For each volume within a storage pool, PowerFlex Manager shows the **Compression** and **Type** settings.

   For volumes, you may see a different number in the **Volumes** column on the **Resources** tab than you see for the total number of reported volumes on the **Volumes** page on the **Block** tab. The count of volumes that are listed on the PowerFlex system details view is the "vTree" count (volume + snapshots). The count of volumes that are listed on the **Block** tab is just the count of volumes.

   To search for volumes associated with the gateway:

   a. Enter a volume or datastore name search string in the **Search Text** box.
   b. Optionally, select a **Size** range.
   c. Optionally, select **Thick** for the **Type**.
   d. Optionally, select **Enabled** or **Disabled** for the **Compression** setting.
   e. Optionally, select a specific **Storage Pool**.
   f. Click **Search**.
   PowerFlex Manager updates the results to show only those volumes that satisfy the search criteria. If the search returns more than 50 volumes, you must refine the search criteria to return only 50 volumes.

Related information

Reconfiguring MDM roles

Viewing PowerFlex rack IPI cabinet details

The IPI cabinet contains an appliance that is configured to provide information about power, thermals, alerts, and all components in the cabinet's physical infrastructure. PowerFlex Manager uses SNMPv2 to poll the IPI appliance status and send results about temperature and humidity to be viewed in PowerFlex Manager.

This capability is available on the PowerFlex rack offering only.

- If your cluster is running on several nodes, you must add the IP address of each node to the SNMP network management system (NMS) section in the Panduit Appliance UI.
- Ensure that PowerFlex Manager discovers the IPI cabinet. You can verify by clicking the list of resources and checking that the IPI cabinet is present.

You can view additional cabinet details in the IPI appliance management application.

To view the details for the IPI cabinet, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the IPI cabinet.
3. In the right pane, click **View Details**. The **Details** page displays detailed information about the IPI cabinet, including system name and location, model number, and IP address. It also displays cabinet temperature and humidity details.
4. To view additional details, click **Launch IPI Cabinet Manager** to launch the IPI appliance management application.

Viewing CloudLink Center details

You can view details for CloudLink Center.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

To view the details for CloudLink Center, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing the CloudLink Center.
3. In the right pane, click **View Details**.
   The **Details** page displays detailed information about the CloudLink Center resource on the following tabs:
   - **Summary**: Displays cluster details, system performance, alarms, pending machines, and security events. If CloudLink has clustered CloudLink Centers, PowerFlex Manager shows all CloudLink Centers.
   - **Machines**: Displays machine groups and machines.
   - **Approved Networks**: Displays a list of the names and IP addresses for all approved networks going in and out of the CloudLink Center.

The **Details** page displays information that is similar to information provided in the CloudLink application. All the data is based on the last inventory, as indicated by the **Last Inventory** timestamp shown on the **Details** page under **Resource Information**. Whenever the inventory is updated, either automatically or manually, any changes that are made in the CloudLink Center are reflected in the **Details** page.

4. To view additional details, click **Launch Console** to launch the CloudLink application.

Viewing NVDIMM details

To view NVDIMM details for a node, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the row containing a node that has NVDIMM compression.
3. In the right pane, click **View Details**.

   The **Details** page displays detailed information about the node.
4. Click the **Memory** tab.

   The **Memory** tab shows the NVDIMM modules and all DIMMs in the server. The DIMMs with rank 1 are the NVDIMM devices. The NVDIMMs shown in the top section of the **Memory** tab provide the name of each **NVDIMM Module**, as well as the **Battery Enabled State**, **Battery Health State**, **Battery Operational Status**, **Remaining Rated Read Write Endurance**, and **Size** for the module.

   All of the data is based on the last inventory.

Update resource inventory

You can run inventory to incorporate changes that are made to resource data outside of PowerFlex Manager. After running the inventory to incorporate these changes into the **Resources** page, you can update the details on any resource group that needs to include the new resource data.

Superusers can run the inventory on any resources. Lifecycle administrators can run the inventory only on resources that are part of a node pool for which they have permission.

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, click a resource, and in the **Details** pane, click **Run Inventory**.

   The resource state changes to **Pending**. When the inventory is complete, the resource state changes to **Available**. See PowerFlex Manager logs to view the start time and end time of the resource inventory operation.

Viewing a compliance report for a resource

Perform the following steps to view a compliance report for a single resource:

1. On the menu bar, click **Resources**.
2. Select the resource for which you want to view the compliance report.
3. In the right pane, click the link corresponding to the **Compliance** option.
   The **Compliance Report** page displays.

   NOTE: If an RCM is mapped to the resource group and is modified, a wrench icon with the text, "Modified", is displayed. However, if the update file is moved or deleted, the wrench icon with the text, "Needs Attention", is displayed.
4. Select the **Firmware Components** option to view the details of the firmware available on the selected resource. For example, a switch.
5. Select the **Software Components** option to view the software components in the compliance report or to update the software components for a resource. If the resource is in the **In Use** state, it redirects you to the **Resource Groups** page.
6. To update the non-compliant resources, click **Update Resource**.

   NOTE: If there is no compliance version attached to the resource, you cannot perform the firmware update or software update.

Removing resources

Only super users can remove resources from PowerFlex Manager.

To remove a resource from PowerFlex Manager, perform the following steps:

1. On the menu bar, click **Resources**.
2. On the **Resources** page, click the **All Resources** tab.
3. From the list of resources, select one or more resources, and click **Remove**.
4. Click **OK** when the confirmation message appears.

If you remove a node, the node state changes to **Pending** and it powers off.

NOTE: Before you add new nodes using an existing iDRAC IP address in inventory, ensure that you remove the old nodes from PowerFlex Manager before discovering the new nodes.

Related information

Resource health status
Compliance status

Additional resource management tasks

This section provides additional tasks for resource management.

Creating a node pool

You can create a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, click **Create**.
   The **Create Node Pool** wizard starts.
3. On the **Welcome** page, read the instructions, and click **Next**.
4. On the **Node Pool Information** page, enter a name and description for the node pool. Click **Next**.
5. On the **Add Nodes** page, select the nodes that you want to add to the node pool. Click **Next**.
6. On the **Assign Users** page, select the users that you want to grant access rights to the node pool. Click **Next**.
7. On the **Summary** page, review the node pool configuration and click **Finish**.

Modifying a node pool

You can modify the details for a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, click **Modify**.
   The **Modify Node Pool** wizard starts.
3. To change the name and description of the node pool, in the left pane, click **Node Pool Information**. Make the updates and click **Save**.
4. To add or remove nodes from the node pool, in the left pane, click **Add Nodes**. Make the updates and click **Save**.
5. To add or remove the access rights to the node pool, in the left pane, click **Assign Users**. Make the updates and click **Save**.

Removing a node pool

You can remove a node pool on the **Resources** page.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

1. On the menu bar, click **Resources**. Click the **Node Pools** tab.
2. On the **Node Pools** page, select one or more node pools to delete. Click **Remove**.
3. Click **OK** when the confirmation message appears.

Viewing the port view details

Port view is a PowerFlex Manager feature that is associated with a resource group deployment. It enables you to view the physical network configuration characteristics and topology from the network interface cards to upstream switches, highlighting such details as switch ports, VLANs, and network identifiers such as MAC addresses.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

You can access port view from both the **Resources** and **Resource Groups** pages. Port view provides connectivity information for both new and existing resource groups.

The **Port View** page displays the following information:

- Topology information for all networks and VLANs deployed in a resource group
- Network connections between the devices
- Health of the resources
- Connection details section with information about the network devices

Port View is only available for PowerFlex nodes and is not available for management VMs, such as CloudLink Center.

PowerFlex Manager does not show switch connectivity for deployments with partial network automation.

To view the port view details:

1. On the **Resource Groups** page, view the details for a resource group and click **Port View**. If you are accessing the port view from the **Resources** page, select a node that is in a **Deployed** state and click **View Details** on the right pane first.

   If you select a node that is not in a deployed state, only the interface card information is displayed.

2. To view device details such as hostname, model name, and management IP address, or information about associated devices, click the specific ports or devices.
3. To view information about intermediate devices in port view, ensure that the devices are discovered and available in the inventory. Sometimes, connectivity cannot be determined for an existing resource group because the switches have not yet been discovered. In this case, you see only the node in port view, but you do not see connectivity information. You can correct this by going back and discovering the switches, and updating the resource group again.

   PowerFlex Manager cannot discover interface cards that do not have integrated LLDP support (such as Intel X520).

4. To filter information based on the connectivity, select an option from the **Display Connections** list. **Show All Connections** is the default option.

Exporting a configuration report for all resources and resource groups

You can download a PDF report that lists configuration details for all resources and resource groups.

The configuration report shows the result of various configuration checks that PowerFlex Manager performs against the system. You can use the report to troubleshoot your resources and resource groups.

The report shows the following kinds of information:

Table 29. Configuration report

Column name	Description
Resource Name	Provides the name of the resource for which the checks have been run. This column corresponds to the **Resource Name** shown on the **Resources** page.

Table 29. Configuration report (continued)

Column name	Description
Asset/Service Tag	Provides the asset or service tag of the resource for which the checks have been run. This column corresponds to the **Asset/Service Tag** shown on the **Resources** page.
Management IP	Specifies the **Management IP** of the resource for which the check is run. For a PowerFlex cluster, the management IP address is the MDM cluster IP address. For a PowerFlex gateway, the management IP address is the PowerFlex gateway IP address.
Result	Shows the result of the check (`PASS` or `FAIL`).
Severity	Indicates the severity of the check. The severity is based on the result. The severity levels are INFO, WARNING, HIGH, or CRITICAL. If the result is PASS, the severity is INFO. If the result is FAIL, the severity depends on the type of check. PowerFlex Manager supports only CRITICAL checks.
Details	Provides a description of the check that was run.
Affected Resources	Gives a list of the IP addresses or unique identifiers of resources that are impacted by the check. The list of affected resources helps with troubleshooting.

1. On the menu bar, click **Resources**.
2. Click **Export Report**.
3. Select **Export Configuration PDF Report** from the drop-down list.

   The selected report downloads.

Related information

Configuration checks

Importing networks

You can import many general-purpose VLANs from VMware vCenter.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

After importing networks, you can add them to templates or resource groups. You can also import networks when you add an existing resource group.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, click the VMware vCenter from which you want to import networks.

   NOTE: Adding a standard vSwitch network is not supported and leads to unsupported scenarios.

   For compute-only or hyperconverged add one of the supported networks:

   - Hypervisor management
   - PowerFlex management
   - PowerFlex data
   - vMotion
   - Out-of-Band management

   Importing networks from VMware vCenter is not supported for PowerFlex software nodes.

3. In the right pane, click **Import Networks**.
   PowerFlex Manager displays the **Import Networks** wizard. In the **Import Networks** wizard, PowerFlex Manager lists the port groups that are defined on the VMware vCenter as **Available Networks**. You can see the port groups and the VLAN IDs.
4. Optionally, search for a VLAN name or VLAN ID.
   PowerFlex Manager filters the list of available networks to include only those networks that match your search.

5. Click each network that you want to add under **Available Networks**. If you want to add all the available networks, click the checkbox to the left of the **Name** column.
6. Click the double arrow (>>) to move the networks you chose to **Selected Networks**.
   PowerFlex Manager updates the **Selected Networks** to show the ones you have chosen.
7. Click **Save**.

PowerFlex Manager creates a job to add the networks. The job may finish quickly, so you may not see on the **Jobs** page.

Go to **Settings** > **Networks** to see the imported networks.

Updating passwords for system components

You can update the passwords for some system components from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more resources of the same type for which you want to change passwords.
   For example, you could select one or more iDRAC nodes or you could select one or more PowerFlex gateway components.
3. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, select one or more components for which you want to update a password and click **Next**.
   The component choices vary depending on which resource type you initially selected on the **Resources** page.
5. On the **Select Credentials** page, create a credential or change to a different credential having the same username.
6. Click **Finish** and click **Yes** to confirm the changes.

Updating passwords for nodes

You can update the passwords for one or more nodes from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more nodes for which you want to change the passwords.
3. Click **Update Password**.
   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, specify which passwords you want to update for the selected nodes by clicking one or more of the following check boxes.
   - **iDRAC Password**
   - **Node Operating System Password**
   - **MVM Operating System Password**

   The option to update passwords for SVMs and MVMs will only appear if a node is in a resource group.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **iDRAC**, **OS Password**, or **MVM Password** object under the **Type** column to see credential details for each node you selected on the **Resources** page.
      The **MVM Password** sections do not appear if there is nothing to show for SVMs or MVMs.
   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.
      Specify the **Credential Name** and the **User Name** for which you want to change the password. Enter the new password in the **Password** and **Confirm Password** fields.
   c. To modify the credential, click the pencil icon for the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.

   You must perform the same steps for the node operating system and SVM operating system password changes. For a node operating system credential, only the OS admin credential type is updated.
7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. The node operating system, SVM, and MVM operating components are updated only if PowerFlex Manager is managing a cluster with these components. If PowerFlex Manager is not managing a cluster with these components, these components are not displayed and their credentials are not updated. Credential updates for iDRAC are allowed for managed and reserved nodes only. Unmanaged nodes do not provide the option to update credentials.

Updating passwords for PowerFlex gateway components

You can update the passwords for one or more PowerFlex gateway components from PowerFlex Manager.

1. On the menu bar, click **Resources**.
2. On the **All Resources** tab, select one or more PowerFlex gateway components for which you want to change the passwords.
3. Click **Update Password**.

   PowerFlex Manager displays the **Update Password** wizard.
4. On the **Select Components** page, select **PowerFlex Password**.
5. Click **Next**.
6. On the **Select Credentials** page, create a credential with a new password or change to a different credential.
   a. Open the **PowerFlex** object under the **Type** column to see details about each gateway you selected on the **Resources** page.
   b. To create a credential that has the new password, click the plus sign (**+**) under the **Credentials** column.

      Specify the **Credential Name**, as well as the **Gateway Admin User Name** and **Gateway OS User Name** for which you want to change passwords. Enter the new passwords for both users and confirm these passwords.

   c. To modify the credential, click the pencil icon for one of the nodes under the **Credentials** column and select a different credential.
   d. Click **Save**.
7. Click **Finish**.
8. Click **Yes** to confirm.

PowerFlex Manager starts a new job for the password update operation, and a separate job for the device inventory. If PowerFlex Manager is managing a cluster for any of the selected PowerFlex gateway components, it updates the credentials for the **Gateway Admin User** and **Gateway OS User**, as well as any related credentials, such as the LIA and lockbox credentials. If PowerFlex Manager is not managing the cluster, it only updates the credentials for the **Gateway Admin User** and **Gateway OS User**.

11

Monitoring events and alerts

This section includes tasks for monitoring events and alerts from resources that are installed or automatically discovered.

Event and alert monitoring provides information regarding occurrences in your system.

An event is a notification that something happened in the system. Events monitor configuration changes, activities of the system, faults, and errors in the system and send notifications accordingly to inform the administrator.

An event that requires attention generates an alert as well. Other events can update or clear an alert when the system detects a change in the condition that needs attention.

An alert is a state in the system which is usually on or off. Alerts monitor serious events that require user attention or action.

Although the events may be interesting for troubleshooting purposes, it is not necessary to monitor events. Whereas an alert is a summation of one or more events that need or needed attention.

The **Monitoring** menu displays the events, alerts, and jobs on your system.

Events

An event is a notification that something happened in the system. An event happens at a single point in time and has a single timestamp. An event may be unique or be part of a series.

Each event message is associated with a severity level. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.

PowerFlex Manager stores up to 3 million events or events that are up to 13 months old. Once this threshold is exceeded, PowerFlex Manager automatically purges the events to free up space. The threshold is reviewed daily.

NOTE: An event is published each day that lists the events which have been removed that day.

Events severity level

The severity levels are as follows:

Table 30. Events severity level

Severity level	Explanation	Example
Information	Informs you of events that you should be aware of, but that do not put the system at risk (no urgency).	`The volume is created.`
Minor	Indicates a failure that may result from an acceptable condition (for example user error), but can also indicate a possible failure.	`The Device media type is mismatched.`
Major	An error alarm raised by the system. This error requires your attention. The system is stable but could be degraded.	`The device failed.`
Critical	A major error alarm raised by system. The system requires immediate attention. Some data may be unavailable.	`The storage pool is running out of capacity.`

Event default and advanced fields

The default and advanced fields are as follows:

Table 31. Event default and advanced fields

Field	Description	Example
Timestamp	The time the event occurred in the following format: `YYYY-MM-DD hh:mm:ss.sss`	`2022-04-03T13:49:18.709Z`
Severity	A predefined severity level that is associated with an event.	`INFORMATION`
Event ID	A unique identifier specific to the event instance that is always 16 digits.	`B46b66c8c79e4a2a`
Code	A unique event code that is always eight digits and identifies a specific event.	`80120001`
Event Name	An alphanumeric descriptive name for the event that is unique.	`MISCELLANEOUS`
Description	A free text description of the event.	`Successfully initialized the Discovery Manager`
Sub system	A resource domain that identifies the area that the alert relates to. The resource domains are: • Management • Block • File • Compute • Network • Security	`MANAGEMENT`
Resource Type	The type of resource that is involved in the event.	`basic-system-config`
Resource Name	The name of the resource that is involved in the event.	`asm manager`
Resource ID	The system specific ID of the resource involved in the event.	`asmcore`
Category	There are six categories of events that help you to filter: • Security • Audit • Software • Hardware • Maintenance • State changed	`STATE_CHANGED`
Service Name	The name of the service that generated the event.	`mdm_gw`
Details	Extra details that are relevant to the operation, for example, user command URL or a list of command parameters.	`[cmd: login, user: admin]`
Originator Application Name	The name of the application that generated the event. This field can be null.	`REST`
Message	A message that describes the event.	`Successfully initialized the Discovery Manager`

Related information

Enabling Dell Technologies connectivity services

Viewing an event

You can view an event in PowerFlex Manager.

1. Go to **Monitoring** > **Events**.
   The **Events** page opens which displays a list of system events. You can add one or more filters to control the list of events displayed.
2. Select the event that you want to view.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

Related information

Enabling Dell Technologies connectivity services

Alerts

An alert is a state in the system which is usually on or off. Alerts monitor serious events that require user attention or action.

When an alert is no longer relevant or is resolved, the system automatically clears the alerts with no user intervention. This action ensures that cleared alerts are hidden from the default view so that only relevant issues are displayed to administrators. Cleared alerts can be optionally displayed through table filtering options. Alerts can also be acknowledged which removes the alert from default view. Acknowledging an alert does not indicate that the issue is resolved. You can view acknowledged alerts through the table-filtering options.

Alert severity level

The severity levels are as follows:

Table 32. Alert severity levels

Severity level	Explanation	Example
Information	An information level informs you of things that happen in the system that do not require action.	NDU
Minor	A minor level informs you that action is required but is not urgent.	This Storage Pool contains devices that are not being accelerated.
Major	A major level informs you that action is required soon.	The MDM certificate is about to expire.
Critical	A critical level informs you that the system requires immediate attention.	The MDM certificate has expired.

Alert default and advanced fields

The default and advanced fields are as follows:

Table 33. Alert default and advanced fields

Field	Description	Example
Description	A message that describes the alert.	The MDM cluster is degraded, and data is not protected.
System impact	A free text, description of the system impact of the alert.	Risk of cluster unavailability
Repair Flow	Free text, actions that you can take in order to repair the issue.	Check that all MDM cluster nodes are functioning correctly, and fix and

Table 33. Alert default and advanced fields (continued)

Field	Description	Example
		`replace faulty nodes, if necessary, to return to full protection`
Alert Details	Any extra details that are relevant to the object and or incident involved.	`Percentage of SP capacity usage`
Associated Events	A list of events that modified the life cycle of the alert.	`[object Object]`
Resource ID	The ID of the resource that is associated with the alert.	`86fb0000000000`
Resource Name	The name of the resource that is associated with the alert. This is usually the resource that is involved with raising the event.	`storagenode212`
Resource Type	The type of resource that is associated with the alert. this is usually the resource that is involved with raising the event.	`storagenodes`
Last Updated	The UTC date and time that the alert was last updated.	`2022-04-28T07:58:46.16Z`
Timestamp	The UTC date and time that the alert was initially raised.	`2022-04-28T07:58:46.16Z`
Acknowledged status	The acknowledged status indicates whether the alert is acknowledged or unacknowledged.	`Y`

The total number of alerts that can be sent to secure remote services gateway from the PowerFlex Manager dispatcher service is restricted to 200 per day. The threshold for the number of alerts for a particular event is set to three per hour. After the first three alerts are sent, if fourth alert is generated for the same event, it is not sent to the secure remote services. However, if the alerts are sent from two different systems running iDRAC, the alerts are sent to the secure remote services gateway.

A new threshold alert is triggered from PowerFlex Manager when the threshold of 200 alerts per 24 hours is crossed per day and is automatically sent to secure remote services gateway. Similarly, a new alert is triggered from PowerFlex Manager and sent to secure remote services gateway when threshold of three alerts per hour for same alert type, symptom code, or resource is reached.

Related information

Enabling Dell Technologies connectivity services

Viewing and acknowledging an alert

You can view and acknowledge an alert in PowerFlex Manager. When you acknowledge an alert, PowerFlex Manager removes the alert from the default view. Acknowledging an alert only indicates that you have seen the alert. It does not indicate that the issue is resolved. You can view acknowledged alerts through the table filtering options.

1. Go to **Monitoring** > **Alerts**.
   The **Alerts** page opens and displays a list of system alerts. You can add one or more filters to control the list of alerts displayed.
2. Select the alert that you want to view or acknowledge.

   NOTE: You can reset the default and advanced field columns by clicking the **Reset Columns** icon.

3. You can:
   - Click **Acknowledge** to acknowledge an alert.
   - Click **Unacknowledge** to remove an alert acknowledgment.

Jobs

In PowerFlex Manager, you can view the details of discovery, firmware update, inventory, and resource group deployment jobs. Only a user with an administrator role can view jobs.

The **Jobs** page displays the following information about the jobs that are scheduled or running in PowerFlex Manager:

- **State** — Displays one of the following states that is based on the job status:
  - **Error** —Job has completed with errors (job is complete but failed on one or more resources).
  - **Scheduled**—Job is scheduled to run at a specific time. It can be scheduled to run at a single time or at several times as a recurring job.
  - **In progress**—Job is running.
- **Job Name**—Identifies the name of the job.
- **Started By**—Displays the name of the user who started the job.
- **Start Time**—Displays the date and time when the job is scheduled to run.
- **Time Elapsed**—Displays the time that has elapsed from the start time to the end time of a job instance.
- **Details**—Displays the detail of a job instance.

If a job scheduled is a one-time job, after execution, it is not listed in the **Jobs** page.

To cancel the job, select the job that you want to cancel, and click **Cancel**.

To filter for a job, click the filter icon.

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Configuring system settings

This section provides information about configuring settings for PowerFlex Manager.

Only a user with the administrator role can configure certain settings.

Go to **Settings** to complete the following tasks:

- Manage local users, LDAP users, and directory services
- Upgrade a single firrmware or software component
- Manage compliance versions, OS images, and compliance management versions
- Define networks and specify the port numbers that PowerFlex Manager should use to verify IP addresses
- Configure settings for events and alerts, including notification policies, inputs, destinations, and alert severities
- Upload a production license for PowerFlex, as well as other software licenses
- Configure security settings, including SSL trusted certificates, appliance SSL certificates, and resource credentials
- Perform serviceability operations, such as generating a trouble-shooting bundle and performing a backup
- Upgrade the management software

User management

The **User Management** page allows you to manage local users, LDAP users, and directory services.

Under **Settings** > **User Management**, you can find three pages:

- Local Users
- LDAP Users
- Directory Services

User roles

User roles control the activities that different types of users can perform when using PowerFlex Manager.

Ensure that you configure the active directory before assigning roles. The roles that can be assigned to local users and LDAP users are identical. Each user can only be assigned one role. If an LDAP user is assigned directly to a user role and also to a group role, the LDAP user is provided with permissions of both roles.

The following table summarizes the activities that can be performed for each user role:

Table 34. User roles and activities

Role	Description	Activities
SuperUser	A SuperUser can perform all system operations.	• Manage storage resources • Manage lifecycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Manage users, certificates • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
SystemAdmin	A SystemAdmin can perform all operations, except for user management and security ones.	• Manage storage resources

Table 34. User roles and activities (continued)

Role	Description	Activities
		• Manage lifecycle operations, resource groups, templates, deployment, backend operations • Manage snapshots, snapshot policies • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations • Update system settings
StorageAdmin	A StorageAdmin can perform all storage-related front-end operations including block systems, such as create volume. **NOTE:** Operations such as create storage pool cannot be performed by the StorageAdmin role, but can be performed by the LifecycleAdmin role.	• Manage storage resources • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
LifecycleAdmin	A LifecycleAdmin can manage the life cycle of hardware and PowerFlex systems.	• Manage lifecycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View resource groups and templates • System monitoring (events, alerts)
SnapshotManager	SnapshotManager is a subset of StorageAdmin, working only on existing systems. This role includes all operations required to set up and manage snapshots.	• View storage configurations, resource details • System monitoring (events, alerts)
SecurityAdmin	The SecurityAdmin manages PowerFlex role-based access control (RBAC), and LDAP user federation. It includes all security aspects of the system.	• Manage users, certificates • System monitoring (events, alerts)
Technician	The Technician can perform all hardware FRU operations on the system, including entering a node into maintenance mode.	• Replace drives • Hardware operations • System monitoring (events, alerts) • Perform serviceability operations
DriveReplacer	This is a subset of the Technician role. The DriveReplacer can perform only operations required for a drive replacement.	• Replace drives • System monitoring (events, alerts)
Monitor	The Monitor role has read-only access to the system, including topology, alerts, events, and metrics.	• View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts)
Support	The Support role is a special kind of SystemAdmin (all activities except for user/security management operations) to be used only by Dell Technologies support staff and developers. This user role has access to undocumented, special operations and options for common operations, required only for support purposes. **NOTE:** This special role should be used only by Dell Technologies support. It opens special,	• Manage storage resources • Manage lifecycle operations, resource groups, templates, deployment, backend operations • Replace drives • Hardware operations • View storage configurations, resource details • View platform configuration, resource details • System monitoring (events, alerts) • Perform serviceability operations

Table 34. User roles and activities (continued)

Role	Description	Activities
	often dangerous, commands for advanced troubleshooting.	• Special Dell Technologies Support operations

LDAP users and security permissions

LDAP security permissions are managed through group memberships, allowing permissions to be assigned to groups instead of individual users. This streamlines administration and improves security by ensuring only authorized users can access specific resources.

You cannot assign multiple roles to a group. All roles in a group have a proximate hierarchy.

Table 35. LDAP hierarchy

Ranking	PowerFlex group role without security officer permissions	PowerFlex group role with security officer permissions
1	N/A (local user)	N/A (local user)
2	Admin	Super-User
3	Admin	Super-User
4	Admin	Super-User
5	StorageAdmin or LifecycleAdmin	Super-User
6	Monitor	SecurityAdmin
Not ranked	SecurityAdmin	SecurityAdmin

Local users

You can create and manage local users within PowerFlex Manager.

Creating a user

Perform this task to create a local user and assign a role to that user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, click **Create**.
4. Enter a unique **User Name** to identify the user account.
5. Enter the **First Name** and **Last Name** of the user.
6. Enter the **Email** address.
7. Enter a **New Password** that a user enters to access PowerFlex Manager. Confirm the password in the **Verify Password** field.

   The password must be at least 8 characters long and contain one lowercase letter, one uppercase letter, one number, and one special character. Passwords cannot contain a username or email address.
8. In the **User Role** box, select a user role. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician

- Monitor
- Support

9. Select **Enable User** to create the account with an `Enabled` status, or clear this option to create the account with a `Disabled` status.
10. Click **Submit** and click **Dismiss**.

PowerFlex Manager creates the new user with the specified password. The first time you login with the new user and password, PowerFlex Manager asks you to change the password.

Modifying a user

Perform this task to edit a PowerFlex Manager user profile.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select the user account that you want to edit.
4. Click **Modify**. For security purpose, confirm your password before editing the user.
5. You can modify the following user account information from this window:
   - **First Name**
   - **Last Name**
   - **Username**
   - **User Role**
   - **Email**
   - **Enable User**

   If you select the **Enable user** checkbox, the user can log in to PowerFlex Manager. If you disable the checkbox, the user cannot log in.
6. Click **Submit** and click **Dismiss**.

Deleting a user

Perform this procedure to remove an existing local user.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one or more user accounts to delete.
4. Click **Delete**.

   Click **Apply** in the warning message to delete the user. Click **Dismiss**.

Resetting the password for a user

You can reset the password for a local user in PowerFlex Manager.

1. On the menu bar, click **Settings** and click **User Management**.
2. Click **Local Users**.
3. On the **Local Users** page, select one user account to reset the password.
4. Click **Reset Password**.
5. Enter **New Password** and **Verify Password**.
6. If you wish to set the password as a temporary one, check **Temporary Password**.
7. Click **Apply** and click **Dismiss**.

Remote users and groups

You can add and modify remote users and groups in PowerFlex Manager. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

For LDAP configurations, the LDAP realm name must not exceed the 155 character limit defined by Microsoft.

Add remote users or groups

Add remote users or groups to PowerFlex, and assign roles to them. Remote users are users tied to an LDAP directory service or to an SSO identity provider.

When you add users and groups, you can also assign roles to the users and groups. The roles control access permissions for the corresponding LDAP user or group. The users and groups must also be configured in the LDAP directory service or SSO identity provider. On the PowerFlex side, these users and groups are mapped to PowerFlex roles.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Remote Users/Groups**.
3. Click **Add**.
4. In the **Add Remote User/Group** dialog box, select a **Type** option.
   - `User`—a user definition will be configured for an individual user.
   - `Group`—a group definition will be configured for a specific group.
5. Select the LDAP directory service or SSO identity provider in the **Provider** drop-down.
6. In the **User Name** box, enter the user or group name.
7. In the **User Role** box, select the role to be assigned to the user or group. Options include:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
8. Click **Apply**.

Modify a remote user or group

You can modify the user role of a user or group tied to an LDAP directory service or an SSO identity provider. User roles control access permissions to the corresponding user or group.

1. On the menu bar, click **Settings** > **Remote Users**.
2. Click **Modify**.
3. In the **Modify Remote User/Group** dialog box, change the user role by selecting one of the **User Role** options:
   - SuperUser
   - SystemAdmin
   - StorageAdmin
   - LifecycleAdmin
   - SnapshotManager
   - SecurityAdmin
   - DriveReplacer
   - Technician
   - Monitor
   - Support
4. Click **Apply**.

Directory services

You can create a directory service that PowerFlex Manager can access to authenticate users.

An Active Directory or Open LDAP user is authenticated against the specific directory domain to which a user belongs.

The **Directory Services** page displays the following information about PowerFlex Manager active directories:

- LDAP configuration

- User search settings
- Group search settings

From this page, you can:

- Add a directory service (only available when no service is defined in the system)
- Modify a directory service
- Remove a directory service

Add a directory service

Add a directory service for user authentication.

The hostname character limitation is 15 characters.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Add**.
4. For **LDAP Configuration**, configure the following:
   a. In the **Address** box, enter the address of the authentication server.
      The address must be specified in URL-like format:
      - Enter **`ldap://HOSTNAME or IP ADDRESS`** for a plaintext LDAP connection.
      - Enter **`ldaps://HOSTNAME or IP ADDRESS`** for a secure LDAP connection.

      For example: **`ldap://100.68.68.1`**
   b. In the **Bind DN** box, enter the bind distinguished name attributes.
      The Bind Distinguished Name (DN) uniquely identifies an entry and its position in the hierarchy of entries contained in a directory server.
      The user account entered here is the user that PowerFlex Manager uses to connect to your directory service.
      For example, for the administrator account in the **Users** folder of **`asm.delllabs.com`**, enter the following: **`CN=administrator,CN=Users,DC=asm,DC=delllabs,DC=net`**.
   c. In the **Bind DN Password** box, enter the Bind DN password.
      This is the password used to access the LDAP server.
   d. In the **Timeout** box, enter a value in milliseconds.
      For example: **`1000`**
5. For **User Search Settings**, configure the following:
   a. In the **Username LDAP Attribute** box, enter the name of an LDAP attribute that is mapped as the username. For many LDAP servers, it can be **`uid`**. For Active Directory, it can be **`sAMAccountName`** or **`cn`**. The attribute should be filled in for all LDAP users you want to import from LDAP to PowerFlex.
      NOTE: If you are using Microsoft Windows Active Directory (AD) as an LDAP server, the hostname cannot exceed 15 characters. A hostname that exceeds the 15-character limit may cause issues during deployment and integration.
      For example: **`sAMAccountName`**
   b. In the **ID Attribute** box, enter the ID attribute for users.
      For example: **`sAMAccountName`**
   c. In the **Object Class** box, enter an object class.
      For example: **`top,person,organizationalPerson,user`**
   d. In the **Search Path** box, enter the search path.
      The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria.
      For example: **`CN=Users,DC=asm,DC=delllabs,DC=net`**
6. For **Group Search Settings**, configure the following:
   a. In the **Group Member Attribute** box, enter a group member name.
      For example: **`member`**
   b. In the **Group ID Attribute** box, enter the group ID.
      For example: **`name`**
   c. In the **Group Object Class** box, enter the group object class.
      For example: **`group`**
   d. In the **Group Search Path** box, enter the group search path.

The search path is used to identify and retrieve entries in the directory information tree that match a set of criteria for groups.

For example: **`CN=Users,DC=asm,DC=delllabs,DC=net`**

7. Click **Test Connection**.

   If the test is successful, the **Submit** button will become active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

8. When you have finished making your changes, click **Submit**.

Modify a directory service

The **Modify** option allows you to edit the existing directory service settings.

Perform the following procedure to edit the settings:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Modify**.
4. In the **LDAP Settings** dialog box, edit the desired fields.

   Note that the **Bind DN Password** must be reentered.

5. When you have finished making your changes, click **Test Connection**.

   If the test is successful, the **Submit** button becomes active. If the test fails, you will not be able to proceed until you fix the connectivity issue.

6. Click **Submit**.

Remove a directory service

The **Remove** option allows you to remove the directory service configuration from PowerFlex.

Perform the following procedure to remove a directory service:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **Directory services**.
3. Click **Remove**.
4. In the warning dialog box, click **Submit**.

Single sign-on identity provider configuration

You can configure single sign-on (SSO) authentication through an identity provider (IdP) using the SAML 2.0 and OIDC protocols. This configuration makes it possible for users to take advantage of SSO capabilities.

You can add up to three IdPs for SSO authentication through PowerFlex Manager.

The supported IdPs and protocols are described in the following table:

Table 36. Supported IdPs and protocols

IdP	SAML	OIDC
Microsoft Active Directory Federation Services	Yes	No
Microsoft Azure	Yes	Yes
Keycloak	No	Yes
Okta	Yes	Yes
PingOne	Yes	No

Add an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.
6. Configure PowerFlex as the service provider within the IdP.

   NOTE: To configure PowerFlex as the service provider within the IdP, you must log in to the portal of the IdP.

   To configure PowerFlex as the service provider within Active Directory Federation Services, see the *Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services* section in the *PowerFlex and PowerFlex Manager Software Administration Guide*.

   To configure PowerFlex as the service provider within Microsoft Azure, see the *Set up PowerFlex single sign-on with Microsoft Azure* topic in the *PowerFlex and PowerFlex Manager Software Administration Guide*.
7. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
8. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
   a. To upload the metadata as a file, select **Upload File** and specify the file location.
   b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://hostname/FederationMetadata/2007-06/FederationMetadata.xml`

      The URL is always the same, except for the hostname, which you must specify for your environment.
   c. Click **Next**.
9. Under **Claims Mapping**, review the attribute mappings that are imported from the identity provider.
   a. Check the attribute mappings to ensure that they are correct.
   b. Click **Next**.
10. Under **Summary**, review the details and click **Finish**.

After you add an identity provider, PowerFlex Manager adds it to the list of identity providers on the **SSO Identity Provider (IdP) Configuration** page.

In addition, PowerFlex Manager updates the login page to show a login button with the new identity provider. You can see this button the next time you log in PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Active Directory Federation Services

To set up PowerFlex single sign-on with Microsoft Active Directory Federation Services (ADFS), you must first add ADFS as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the ADFS portal, configure PowerFlex as the service provider, and choose an access control policy. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Add Active Directory Federation Services as an identity provider

Add an SSO Identity Provider (IdP) for PowerFlex Manager to enable users to take advantage of single sign-on (SSO) capabilities through other applications.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.

3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Active Directory Federation Services

To configure PowerFlex as a service provider for Active Directory Federation Services, upload the root CA certificate and log in with appropriate permissions. Then, establish trust by adding a relying party, configuring claim rules, importing metadata, and setting access control policies.

You must upload the root CA certificate for ADFS.

1. Log in to the Active Directory Federation Services portal with the security or administrator role.
2. Under **ADFS**, go to the **Relying Party Trusts** folder.
3. Under **Actions**, click **Add Relying Party Trust...**.
4. On the **Welcome** screen, select **Claims aware** and click **Start**.
5. On the **Select Data Source** screen, select **Import data about the relying party from a file**.
6. In the **Federation metadata file location** field, specify the location of the downloaded service provider metadata file and click **Next**.

   NOTE: If you copied the metadata instead of downloading the file, select **Enter data about the relying party manually** and input the copied metadata.

7. On the **Specify Display Name** screen, enter the name that you would like to use for the service provider and click **Next**.
8. On the **Choose Access Control Policy** screen, choose a policy and click **Next**.
9. On the **Ready to Add Trust** screen, click **Next**.
10. On the **Finish** screen, select **Configure claims insure policy for this application** and click **Close**.
11. On the **Relying Party Trusts** screen, under **Actions** select the display name for the newly created service provider, and click **Edit Claim Insurance Policy...**.
12. Click **Add Rule...** to add the following LDAP attribute rule:
    a. For the **Claim rule template**, select **Send LDAP Attributes as Claims**.
    b. For the **Claim rule name**, enter `LDAP attributes`.
    c. For the **Attribute store**, select **Active Directory**.
    d. For the Mapping of **LDAP attributes to outgoing claim types**, select the following attributes:

Table 37. LDAP attributes and outgoing claim types

LDAP attribute	Outgoing claim type
Email-Addresses	Email Address
SAM-Account-Name	Name ID
Surname	Surname
Given-Name	Given Name

    e. Click **Finish**.
13. Click **Add Rule...** to add the following custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter `Get groups`.
    c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.microsoft.com/ws/2008/06/identity/claims/
windowsaccountname", Issuer == "AD AUTHORITY"]  => add(store = "Active Directory",
types = ("http://schemas.xmlsoap.org/claims/Group"), query = ";tokenGroups;{0}",
param = c.Value);
```

d. Click **Finish**.

14. Click **Add Rule...** to add another custom rule:
    a. For the **Claim rule template**, select **Send Claims Using a Custom Rule**.
    b. For the **Claim rule name**, enter **`Claim of groups membership`**.
    c. For the **Custom rule**, paste in the following string:

```
c:[Type == "http://schemas.xmlsoap.org/claims/Group"]  => issue(Type = "Group",
Issuer = c.Issuer, OriginalIssuer = c.OriginalIssuer, Value = c.Value, ValueType =
c.ValueType);
```

    d. Click **Finish**.
15. Click **OK**.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, upload the identity provider metadata so that PowerFlex can establish a connection to the identity provider.
    a. To upload the metadata as a file, select **Upload File** and specify the file location.
    b. To retrieve the metadata from a URL, select **URL** and specify the following URL: `https://hostname/FederationMetadata/2007-06/FederationMetadata.xml`

       The URL is always the same, except for the hostname, which you must specify for your environment.
    c. Click **Next**.
4. On the **Settings** page, review the attribute mappings are imported from the identity provider.
    a. Check the attribute mappings to be sure they are correct.
    b. Click **Next**.
    c. Review the **Summary** page and click **Finish**.

After you add ADFS as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with ADFS. You can see this button the next time you log in PowerFlex Manager.

Set up PowerFlex single sign-on with Microsoft Azure

To set up PowerFlex single sign-on with Microsoft Azure, you must first add Microsoft Azure as an IdP through PowerFlex Manager.

You must then download the service provider metadata for PowerFlex. After you complete this step, log in to the Microsoft Azure portal and configure PowerFlex as the service provider. You must then return to PowerFlex Manager to complete the setup of single sign-on.

Upload the PowerFlex metadata file

You must upload the metadata file through PowerFlex Manager to ensure that PowerFlex can establish a connection to the IdP.

1. In PowerFlex Manager, go back to the **Add Identity Provider (IdP)** window.
2. Under **Service Provider**, select the **I have configured PowerFlex as a SP in my IdP using the metadata above** checkbox and click **Next**.
3. Under **IdP Setup**, select **Upload File**.
4. Browse and select the `Federated Metadata XML` file and click **Next**.

You must now leave PowerFlex Manager. Go back to the Microsoft Azure portal to create a group claim.

Add Microsoft Azure as an identity provider

You must add Microsoft Azure as an IdP and download the service data for PowerFlex through PowerFlex Manager.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Add**.
   The **Add Identity Provider (IdP)** window opens.
4. Under **Name**, provide a name for the identity provider and click **Next**.
5. Under **Service Provider**, choose a method to download the service provider metadata for PowerFlex:
   - To download the metadata as a file, select **Download File** and click **Download SP Metadata (XML)**.
   - To copy the metadata, click **Manual Copy** and click **Copy** for each piece of metadata.

Configure PowerFlex as a service provider for Microsoft Azure

You must export the PowerFlex metadata file and edit it to include additional attributes.

You have 15 minutes from when the SP Metadata (XML) file is downloaded from PowerFlex until the `Federation Metadate XML` file is uploaded to complete the procedure, or a timeout may occur. If there is a timeout, see *Troubleshooting a timeout error*.

1. Using an editor, open the `SP Metadata XML` file, replace any occurrence of the string `keycloak-http` with the PowerFlex Manager IP address, and save the file.
2. Log in to the Microsoft Azure Portal.
3. Go to **App registrations**.
4. Under **Owned applications**, select the Microsoft Azure application that PowerFlex uses.
5. Click the link beside **Managed Application in local directory**.
6. Go to **Manage** > **Single sign-on**.
7. Select **SAML**.
8. From the menu, click **Upload metadata file**.
9. To import the metadata file, browse and select the file, and then click **Add**.
10. Check that the URL contains the PowerFlex system IP and click **Save**.
11. Exit the import screen.
12. In the **SAML-based Sign-On screen** window, go to the **SAML Certificates** section, and download the `Federation Metadate XML` file by clicking **Download.**.
    The file name that is created contains the application name with an `.xml` extension. The file is downloaded to the `Downloads` directory. In this case, the `Federation Metadate XML` directory.
13. Go to the `Federation Metadate XML` file, open it in a text editor and edit it to include the additional attributes as follows:
    a. Before `</IDPSSODescriptor>`, add the following lines:

```
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/emailaddress"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="E-Mail
Address"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/givenname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri" FriendlyName="Given
Name"/>
<Attribute xmlns="urn:oasis:names:tc:SAML:2.0:assertion"
Name="http://schemas.xmlsoap.org/ws/2005/05/identity/claims/surname"
NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:uri"
FriendlyName="Surname"/>
```

    b. Click **Save**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and upload the `Federation Metadate XML` file to PowerFlex.

Create a group claim

You must create a group claim to manage user authorization to the application.

1. In the Microsoft Azure portal, go back to the **SAML-based Sign on** window.

2. From the left navigation, click **Single sign-on**.
   The **Set up Single Sign On with SAML** window opens.
3. From the **Attributes & Claims** window, click **Edit**.
4. Select **+ Add a group claim**.
5. From the **Group Claims** window, complete the following steps:
   a. For **Which groups associated with the user should be returned to the claim?**, select **Groups assigned to the application**.
   b. For **Source attribute**, select **Group ID**.
   c. Expand **Advanced options**.
   d. Select the **Customize the name of the group claim** checkbox.
   e. For **Name (required)**, enter `Group`.
   f. Click **Save**.

Assign a user or group to the application

The user and or group you assign is given access to authenticate to PowerFlex through PowerFlex Manager.

1. In the Microsoft Azure portal, from the **SAML-based Sign on** window, go to the left side navigation, and click **Users and groups**.
2. In the **Add Assignment** window, click **Users and groups**.
3. Select the checkbox beside the users and groups that should be assigned.
4. Click **Assign**.

You must now leave the Microsoft Azure portal. Go back to PowerFlex Manager and provide permissions to the remote users and groups you created.

Provide permissions to remote users and groups

You can provide permissions to users and groups that are based on their role. The roles control access permissions for the corresponding group.

You must not connect to SSO from PowerFlex Manager before you assign permissions.

1. Log in to PowerFlex Manager using the local administrator rights.
2. Go to Settings > **User Management** > **Remote users/Groups**.
3. Click **Add**.
   a. Select the type.

      If you select the **Group** type for the **Group Name**, enter the group-id that is taken from Microsoft Azure (not the Group Name).

      If you select the **User** type for the **User Name**, enter the username that is defined in Microsoft Azure.
   b. Select the provider.
   c. Select a role.
   d. Click **Apply**.

After you add Microsoft Azure as an IdP, it is added to the list of IdPs on the **SSO Identity Provider (IdP) Configuration** page in PowerFlex Manager.

In addition, PowerFlex Manager updates the login page to show a login button with Microsoft Azure. You can see this button the next time you log in PowerFlex Manager.

Troubleshooting a timeout error

If a timeout error occurs, you can log in to PowerFlex Manager and create the IdP again.

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings**.
3. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
4. Select the IdP and click **Delete**.
5. Add the IdP.
6. Export the `Federation Metadate XML` file.
7. Import the `Federation Metadate XML` file that you downloaded from Microsoft Azure to PowerFlex.
8. Import the exported `Federation Metadate XML` file that you created in step 7 to Microsoft Azure.

In the Microsoft Azure portal, update the entity-id by reimporting the PowerFlex metadata file through the application.

Modify an identity provider

The **Modify** option allows you to edit the existing identity provider settings.

Perform the following procedure to edit the settings:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Select an identity provider.
4. Optionally, change the name for the identity provider.
5. Optionally, enable or disable the identity provider.
6. Click **Apply**.

Enable or disable an identity provider

You can disable an identity provider after the initial configuration. If needed, you can also reenable the provider after it has been disabled.

Perform the following procedure to enable or disable an identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Enable** or **Disable**.

Delete an identity provider

You can delete an identity provider if you no longer need it.

Perform the following procedure to delete an existing identity provider:

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Click **Delete**.

Update the certificate for an identity provider

You can update the certificate for an identity provider if it is about to expire.

1. On the menu bar, click **Settings**.
2. In the left pane, click **User Management**, then in the right pane, click **SSO Identity Provider (IdP) Configuration**.
3. Check the **Status** column to see if any of the identity providers are about to expire.
4. If an identity provider is about to expire, select the identity provider and click **Update Certificate** in the details pane to the right of the list of providers.

Repositories

On the **Repositories** page, you can load compliance files, operating system images, and compatibility management files.

You can load compliance files and specify a default version for compliance checking on the **Compliance Versions** tab. You can load multiple compliance files into PowerFlex Manager with different operating system images included. The compliance file includes operating system images for VMware ESXi and PowerFlex. When you add a compliance file, these images are automatically added to the **OS Images** tab. If necessary, you can also create your own operating system image repositories. You can also load the compatibility management file on the **Compatibility Management** tab to allow PowerFlex Manager to determine which upgrade paths are valid and which are not.

On the **Compliance Versions** tab, click **Add** to load a compliance version. Click **Change** to change the default compliance version to a new file.

You cannot make a minimal compliance version the default version for compliance checking, since it includes only server firmware updates. The default version must include the full set of compliance update capabilities. PowerFlex Manager does not show any minimal compliance versions in the **Default Version** drop-down list.

The **Compliance Versions** tab displays the following information:

Table 38. Compliance version fields

Field	Description
State	Displays an icon indicating one of the following states: ● Pending—The compliance file download process is in progress. ● Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. ● Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. ● Synchronizing—The compliance file is being synchronized with the management software after unpacking. ● Available—The compliance file is downloaded and copied successfully. ● Error—There is an issue downloading the compliance file. ● Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or **Full** if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option for a compliance file that is in an **Error** state: ● Delete the compliance version ● Resynchronize a compliance version
	Select an option for a compliance file that is in a **Needs Approval** state: ● Allow unsigned file ● Delete

Use the **OS Image Repositories** tab to create operating system image repositories and view the following information:

Table 39. Operating system image repositories

Field	Description
State	Displays the following states: ● Available—The operating system image repository is downloaded and copied successfully. ● Pending—The operating system image repository download process is in progress. ● Error—There is an issue downloading the operating system image repository.
Repository	Displays the repository name.
Image Type	Displays the operating system type.
Source Path	Displays the share path of the repository in a file share.
In Use	Displays **True** (the operating system image repository is in use) or **False** (the operating system image repository is not in use).
Available Actions	Select an option to delete or resynchronize.

You cannot perform any actions on repositories that are in use. However, you can delete repositories that are in an **Available** state, but not in use and not set as a default version.

All the options are available only for repositories in an **Error** state. The **Resynchronize** option appears only when you must perform a backup and restore of a previous image.

Use the **Compatibility Management** tab to load the compatibility management file. To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not. When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions.

Related information

Configuring block storage
Deploying and provisioning

Compliance versions

PowerFlex Manager displays the following information about the compliance versions:

Table 40. Compliance versions

Field	Description
State	Displays an icon indicating one of the following states: ● Pending—The compliance file download process is in progress. ● Downloading—The compliance file is being downloaded and PowerFlex Manager provides the percentage complete for the download operation. ● Unpacking—The compliance file is being unpacked and provides the percentage complete for the unpacking operation. ● Synchronizing—The compliance file is being synchronized with the management software after unpacking. ● Available—The compliance file is downloaded and copied successfully. ● Error—There is an issue downloading the compliance file. ● Needs Approval—The compliance file is unsigned and needs approval for use.
Version	Displays the compliance version. If one or many components in the RCM are modified, a wrench icon and the text, **Modified** is displayed next to the name of the RCM. But, if the file is deleted from the file share after a component is updated, the wrench ion with the text, "Modified | Needs Attention", is displayed.
Source	Displays the share path of the compliance version in a file share.
Signature	Displays whether the digital signature on the compliance file is signed or unsigned.
File Size	Displays the size of the compliance file in GB.
Type	Displays **Minimal** if the compliance file only contains firmware updates, or Full if it contains firmware and software updates.
View bundles	Displays details about any bundles added for the compliance version.
Available Actions	Select an option if the compliance file is in an **Error** or **Needs Approval** state: ● Delete the compliance version ● Resynchronize a compliance version ● Allow unsigned file

Select the compliance version to view the following information about the firmware package:

- **Bundles**—Displays the number of bundles available in the firmware package.
- **Components**—Displays the number of software components available in the firmware package.
- **Created On**—Displays the date when the compliance version was created.
- **Last Updated**—Displays the date when the compliance version was last updated.
- **Resource Groups Affected**—Displays the resource groups in which the firmware is used.
- **Additional Components**:
  - **VMware ISO**—Displays the date and time when the VMware ISO component was added.
  - **Filename**—Displays the file name of the VMware ISO component added.

- **VMware Upgrade Depot**—Displays the date and time when the VMware patch depot file component was added.
- **Filename**—Displays the file name of the VMware patch depot file component added.

From this page, you can:

- Add a compliance file.

  A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

- Select **Default Version** and choose a version from the drop-down list to set it as the default for compliance.
- Select a compliance version from the list, and click the delete icon in the same row to remove the version.

  If you remove a compliance version, the version is deleted from the appliance not from the file share.

- Select a version from the list, and, in the right pane, click **View Bundles** to view the firmware and software bundles available in the compliance version.

Add a compliance file

A compliance file must be in an available state for PowerFlex Manager to add the file.

1. From the **Getting Started** page, click **Upload Compliance File**.

   Alternatively, click **Settings** > **Repositories**. On the **Repositories** page, select the **Compliance Versions** tab, and click **Add**.

2. In the **Add Compliance File** dialog, select one of the following options:
   - **Download from SupportAssist (Recommended)**—Select this option to import the compliance file that contains the firmware bundles you need.
     - **Download from local network path**—Select this option to download the compliance file from a CIFS file share. This option is intended for sites that do not have Internet accessibility to Dell Technologies connectivity services.
3. If you selected **Download from SupportAssist (Recommended)**, click the **Available Compliance Files** drop-down list, and select the file.

   Before downloading a compliance file from Dell Technologies connectivity services, you must configure Dell Technologies connectivity services. To configure Dell Technologies connectivity services, you must enable Dell Technologies connectivity services in the **Initial Configuration** wizard.

   If you are downloading a compliance file from Dell Technologies connectivity services, the file type is a ZIP or TAR.GZ file.

   If you are downloading a compliance file from the Dell Technologies support site, the file will be called `Software_Only_Complete_5.0.x_xxx.zip`

   NOTE: The XML file type is not supported as a download option.

4. If you selected **Download from local network path**, perform the following:
   - In the **File Path** box, enter the location of the compliance file. Use the following format:
     - CIFS share for ZIP file: **`\\host\share\filename.zip`**
     - CIFS share for TAR.GZ file: **`\\host\share\filename.tar.gz`**
   - If you are using a CIFS share, enter the **User Name** and **Password**.
   - Mark the target upgrade catalog as the default version for compliance checking.

   Download the 5.0.x software catalogs. PowerFlex requires the catalog for the version that the system is on and the target version.

5. Click **Save**.

   PowerFlex Manager unpacks the compliance file as a repository that includes the firmware and software bundles that are required to perform any updates.

   If the compliance file contains an embedded ISO file for an operating system image (such as for an VMware ESXi or PowerFlex image), the download process unpacks the file and adds it to the operating system image repository.

   A compliance file can be large and take some time to download and unpack. To help you monitor the progress of the downloading and unpacking operations, PowerFlex Manager provides percentage complete information for these operations in the **State** column on the **Compliance Versions** tab.

Add VMware ESXi files to the compliance file

This procedure is applicable only when using ESXi hosts. The vCenter Server Appliance (VCSA) can be hosted either in your environment or through the compute-only extension.

In this process, the ESXi ISO image and patch depot file that are compatible with the IC are uploaded. This step applies only to PowerFlex compute-only deployments.

NOTE: You must use the VMware standard ISO (not the Dell-customized ISO).

The VMware ESXi image and patch depot file are no longer available on the Dell support site. You must obtain these files from the Broadcom support portal and upload them to the compliance file before creating a template, performing a deployment, running a compliance check, or carrying out a non-disruptive upgrade with VMware ESXi. After downloading the appropriate image and patch depot file, you should add these components (ESXi ISO and patch depot file) to the compatible compliance file.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** tab, select the **IC** and click **Add Component**.
3. In the **Add Compliance Component** window, leave the default **VMware Components** setting for **Type of components being added**.
4. In the **Add VMware Installation ISO** section, enter the file path of the VMware installation ISO file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
5. In the **Add VMware Upgrade Depot Zip** section, enter the file path of the VMware patch depot file, username, and password and click **Test Connection** to verify that the file path is valid and can be accessed.
6. Click **Add**.

   The VMware ISO and patch depot files are tracked under the **Hardware** category on the **Monitoring** > **Events** page. The **Additional Components** in the **Compliance and OS Repositories** window displays the VMware ISO and patch depot file name, date, and time at which the VMware ISO and patch depot files are added.

Changing the compliance file

To perform upgrades against a different compliance version, you can change the compliance file.

1. To change the default compliance version, click **Change** on the **Compliance and OS Repositories** page.

   To change the target version for a resource group, click **Change Target** on the **Resource Group Details** page or click **Change** on the **Node Compliance Report**.
2. Choose the **Preferred Compliance File**.
3. Review the **Compatibility** setting. If it shows **Recommended** or **Supported**, you can proceed with the change. If it shows **Not Allowed**, PowerFlex Manager blocks the change. Review the alert for details about recommended or supported paths.
4. If the change is either recommended or supported, type **`CHANGE COMPLIANCE FILE`** if you are to proceed.
5. Click **Save**.

Firmware repository types

The **Resources** page displays devices that are validated against the default repository.

This capability is available on the PowerFlex appliance and PowerFlex rack offerings only.

There are two types of firmware repositories:

Table 41. Firmware repository types

Repository	Description
Default	The default firmware repository is applied to all devices that are not in a resource group. To set a default firmware repository, you must download a compliance file from either Dell Technologies connectivity services or an internal share through PowerFlex Manager. If you set a default compliance version, you can view compliance with this repository on the Resources page.

Table 41. Firmware repository types (continued)

Repository	Description
	Devices with firmware levels below the minimum firmware that is listed in the default compliance version are viewed as non-compliant (out of compliance).
Service level	This repository is applied only to nodes that are in service and assigned the service level firmware repository. Devices with firmware levels below the minimum firmware level that is listed in the service level repository are marked as non-compliant. When a service level firmware repository is assigned to a resource group, the firmware validation is checked only against the service level firmware repository and the default firmware repository checks are no longer applied to the devices associated with this resource group.

Viewing firmware and software bundles details

The details that are provided for bundles include name, version, and date, allowing for management and tracking of updates.

1. On the menu bar, click **Settings** and click **Compliance and OS Repositories**.
2. Click the **Compliance Versions** tab, select a repository, and then click **View Bundles**.
   A list of system and software bundles is displayed.
3. Click **Firmware Bundles** to view the firmware or system bundles.
4. Click **Software Bundles** to view the software bundles.

You can see the following information regarding bundles:

- Name—Name of the firmware or software update package.
- Version—Version of the firmware or software update package.
- Date and Time—Date and time when the firmware or software update package was downloaded.

Single component upgrade

The single component upgrade allows for upgrading the software and firmware on individual components, such as applying security patches, without performing a full RCM/IC upgrade. It is recommended to use this option for component-specific upgrades, such as patch updates, and to ensure compliance with the standard RCM/IC version.

Single Component upgrades are not engineering-validated upgrades. You should understand the implication and risks. Dell Technologies recommends aligning with RCM/IC as the official-validated catalog. If you consider deviating to align with the security or functionality requirements, consider testing the compatibility.

A single component upgrade involves the following steps:

1. Upgrading the firmware and software components.
2. Extracting the compliance report.
3. Modifying and publishing the existing template.

CAUTION: Since Dell Technologies only validates the released RCM/IC code catalogs, you must be aware of the implications and risks that are involved in a single component upgrade. If any issues are identified related to the RCM/IC deviation, you may be asked to modify the code or firmware.

The use of single component upgrade enables the PowerFlex platform to remain in compliance, using the deviated code/firmware, which means future upgrades will not be blocked due to non-compliance.

The single component upgrade option enables the PowerFlex platform to remain in compliance. The future upgrades are not blocked due to non-compliance.

The following limitations are associated with the single component upgrade:

- VMware ESXi version changes are restricted; upgrading from VMware ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade the firmware components

To upgrade firmware on a PowerFlex node, authenticate as an administrator, choose a bundle, modify the component, and apply the updates.

You must verify that you are using administrator credentials.

An upgrade of the following firmware components is supported:

- Drive firmware
- iDRAC Lifecycle Controller
- BIOS firmware
- Network Interface Card (NIC) firmware
- BOSS card firmware
- Switch firmware

The upgrade of the PowerFlex core is not supported.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. From the **Compliance Versions** page, select the current RCM/IC bundle and click **View Bundles**.
3. From the list of bundles in the **Firmware Bundles** tab, verify the bundles that you want to modify.
   To modify components on the PowerFlex R660 node, select **System Bundle (Windows x64) PER660 v850** or select the appropriate components such as switches, CloudLink, or operating system bundles.
4. Select the appropriate bundle to upgrade and click **View**.

   NOTE: Only one component can be modified at a time.

5. Select the specific component to upgrade and click **Modify**.
6. On the **Update Bundle Component** page, perform the following:
   a. Select either **Download from local network path** or **Upload from local**.
   b. Provide the **New Bundle Version** (if it is not picked automatically) to match the existing bundle version.
   c. In the **Type "MODIFY COMPLIANCE FILE" to confirm** field, enter `Modify Compliance File` and click **Proceed**.
7. Repeat the steps through 5 and 7 for the remaining components.
8. Click **Settings** > **Repositories** > **Compliance Versions** and verify that a wrench icon and `Modified` is appearing for the modified RCM/IC.
9. To update the resources, perform the following:
   a. On the menu bar, click **Resources** and select the PowerFlex nodes to update and click **Run Inventory**.
   b. Click **Lifecycle** > **Resource Groups**.
   c. Select the resource group to be modified and click **View Details**.
   d. Click **View Compliance Report**.
   e. Select the modified PowerFlex node or other components and click **Update Resources**.
   f. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now**.
   g. Select the appropriate maintenance mode and click **Apply**.
10. To monitor and confirm that the component is updated with the new version, perform the following:
    a. Click **Lifecycle** > **Resource Groups** to verify that the required component is upgraded.
    b. Select the resource group that you already modified and click **View Details**.
    c. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

       If an upgrade failure occurs, contact Dell Technologies Support for assistance.

Upgrade the software components

Use the following procedures to upgrade the software components using PowerFlex Manager. These procedures are applicable only when you consider deviating from the standard upgrade procedure.

NOTE: Ensure you upload identical versions of the VMware ESXi patch bundle and the ESXi ISO file to PowerFlex Manager.

A software component upgrade involves operating system updates such as the following:

- Applying ESXi patches
- Upgrading the CloudLink virtual machine

The following limitations are associated with the single component upgrade:

- ESXi version changes are restricted; upgrading from ESXi 8.x to 9.x or downgrading is not supported.
- Updates to embedded operating systems, such as SLES or CentOS are not supported.

Upgrade VMware ESXi patch and ISO images

You can upgrade the VMware ESXi patch and ISO images using PowerFlex Manager.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**.
2. Verify that the required RCM/IC is uploaded and is available.
3. Select the RCM/IC bundle to be modified and click **View Bundles**.
4. From the **Software Bundles** tab, select the **ESXI Software Bundle** checkbox, clear the checkbox for other software bundles, and click **View**.
5. Select **Upload Software Depot** and select either of the following options to upload the ESXi path depot file:
   - **Upload from local**: Select this option when the depot file is smaller than 500 MB.
   - **Download from local network path**: Select this option when the depot file is larger than 500 MB. Enter the CIFS file path and user credentials to access the file on the local network drive.
6. For the **Upload from local** option, complete the following:
   a. To provide a local file, click **Choose File**.
   b. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   c. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   d. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
7. For the **Download from local network path** option, complete the following:
   a. Enter the `CIFS` file or the `HTTPS` address to access the file.
   b. Enter the username and password.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. To confirm that you want to modify the existing file, enter **`Modify Compliance File`** and click **Proceed**.
   e. Wait for the compliance file to synchronize and verify that the status of the compliance file is `Available`.
   f. Verify that a wrench icon and `Modified` displays for the modified RCM/IC.
8. **NOTE:** You must complete steps 5 through 7 before you start step 8.

   To upload the `ESXI.depot.zip` and `ESXi.iso` files, complete the following:

   a. From **Repositories** > **Compliance Versions**, select the RCM/IC bundle, and click **Add**.
   The **Add Compliance File** dialog box displays.
   b. Enter the file path, username, and password for both the installation ISO and the depot .zip file. Ensure the files match with the existing VMware ESXi patch and ISO files.
   c. Select **Test Connection** to test the connection between the PowerFlex Manager user interface and the location of the CIFS file or the file that is at the HTTPS address.
   d. Click **Save** and wait for the synchronization and download to complete.
   e. On the **Compliance Versions** page, verify that the files are added and the RCM/IC is in `Available` state.

Update resource inventory and resource groups

Once you upgrade the VMware ESXi patch and ISO images, use this procedure to update the resource inventory and resource groups.

1. On the menu bar, click **Lifecycle** > **Resources**.
2. Select the PowerFlex node to update, and click **Run Inventory**.
3. Click **Lifecycle** > **Resource Groups** and select the service to upgrade.
4. Click **Non-Compliant** in the **Resource Group Information** tab.
5. From the **Software Components** tab, select all the listed nodes, and click **Update Resources**.
6. Select the appropriate **Maintenance Mode** and click **Apply**.
7. To monitor the job and verify that the required component is upgraded, perform the following:
   a. Click **Lifecycle** > **Resource Groups**.
   b. Select the resource group that you modified and click **View Details**.
   c. On the **Resource Group Information** tab, verify that the resource group compliance status is `Compliant`.

If the upgrade fails, contact Dell Technologies Support for assistance.

Upgrade CloudLink Center

Using the single component option, you can upgrade the CloudLink Center version.

CloudLink ISO is required to start the CloudLink version upgrade using the single component upgrade option.

You must verify that you are using the administrator credentials.

1. On the menu bar, click **Settings** > **Repositories** > **Compliance Version**.
2. Select the RCM/IC bundle, and click **View Bundles**.
3. From the **Firmware Bundles** tab, select the **System Bundle CloudLink** and click **View**.
4. Select the component **clc** that you want to upgrade and click **Modify**.
5. From the **Update Bundle Component** page, select **Download from local network path** option and upload the CloudLink ISO file.
6. Enter the **New Bundle Version** if it is not picked automatically, to match with the existing bundle version.
7. Type **`Modify Compliance File`** in the field and click **Proceed**.
8. Click **Repositories** > **Compliance Versions** and verify that the modified RCM/IC appears with a wrench icon and `Modified`.
9. To update the resources, perform the following:
   a. From **Lifecycle** > **Resources**, select both the CloudLink resources and click **Run Inventory**. Wait for two minutes to complete the inventory and refresh your browser.
   b. Select the CloudLink Center to upgrade and click **Non-Compliant** and click **Update Resource**.
   c. From the **Apply Resource Updates** page, select **Allow PowerFlex Manager to perform firmware and software updates now** and click **Apply Updates**. Wait till the deployment status becomes `Available`.
   d. On the **Resources** page, click **CloudLink Center** > **View Details** to verify the upgraded CloudLink Center version. Wait for three to four minutes. Both CloudLink Centers are updated with the new version.
10. To manually upgrade the SecureVM agent on all the PowerFlex nodes from CloudLink Center, perform the following:
    a. In your web browser, type the URL for CloudLink Center in the following format: **`https://clc_address`**
    b. Log in with the **`secadmin`** credentials.
    c. Click **Agents** > **Machines** and select the node.
    d. Click **Actions** > **Upgrade**. The upgrade option is disabled if an upgrade is not required for the SecureVM agent.
    e. Repeat step 10a through 10d for every PowerFlex node that requires a CloudLink agent upgrade.
    f. Click **Agents** > **Machines**, select the node, and verify the SecureVM agent version.
    g. After the agents are updated on the nodes, from CloudLink Center, click **Agents** > **Agent Download** to download the `64-bit SUSE RPM` package or `64-bit CentOS RPM` package depending on the operating system running on the nodes and modify the SecureVM agent component in the RCM/IC bundle.
11. To modify the SecureVM agent component in the RCM/IC bundle, perform the following:
    a. Verify that you are using the administrator credentials.
    b. On the menu bar, click **Settings** > **Repositories** > **Compliance Versions**, select the current RCM/IC bundle and click **View Bundles**.
    c. Click **Software Bundles** > **CloudLink SecureVM** > **View**. You will notice that there are two CloudLink SecureVM bundles with the same name. One bundle is for SUSE and the other bundle is for CentOS. Select the bundle and click **View** to differentiate the bundle. Select the bundle based on the operating system installed on the PowerFlex nodes.
    d. Click **Modify**.
    e. From the **Update Bundle Component** page, select the **Download from local network path** or **Upload from local** and select the downloaded agent.
    f. Enter the **New Bundle Version** to match the existing bundle version if it is not picked automatically.
    g. Type **`Modify Compliance File`** in the field and click **Proceed**.
12. To update the resource group details, perform the following:
    a. On the menu bar, click **Lifecycle** > **Resource Groups**.
    b. Select the resource group to modify and click **View Details**.
    c. Click **More Actions** > **Update Resource Group Details**.
    d. On the **Resource Group Information** page, verify that the resource group compliance status is `Compliant`.

    If the upgrade fails, contact Dell Technologies Support for assistance.

Exit the PowerFlex nodes from service mode.

Export the compliance report

You can export a compliance report using PowerFlex Manager. The compliance report identifies the list of modified components and their versions. It also compares component versions during upgrades from a modified RCM/IC to a standard RCM/IC.

Before exporting the compliance report, you must ensure that the default RCM/IC version is selected. To verify that the default version is selected, go to **Settings** > **Repositories** > **Compliance Versions**.

When transitioning from a modified RCM to a desired RCM, ensure the following:

- The component versions in the modified RCM are equal to or lower than the component versions in the desired RCM.
- If any component in the modified RCM has a higher version, use the **Extract Compliance** tool to identify the components.
- Use the single component upgrade to add those components with a higher version to the desired RCM to ensure version alignment and compatibility,

1. On the menu bar, click **Resources** > **Export Report**.
2. Select **Export Compliance PDF Report** or **Export Compliance CSV Report**.
3. Identify the modified components and verify the number of components.

Modify and publish the template

After completing a single component upgrade, you can modify and publish the template. This process involves updating the existing template by incorporating a modified ESXi ISO image. The updated template is then published and used for future system expansions.

Ensure the modified RCM/IC is the same version as the **Target Version** under the **Existing Resource Group** page. While adding a resource group on an existing PowerFlex system, ensure that you select the latest uploaded operating system images.

1. On the menu bar, click **Templates** > **My Templates**.
2. Select the template that you want to modify and click **Modify**.
3. Select the node and click **Modify**.
   The node details display.
4. Click **Continue**.
   The **OS Settings** display.
5. Under **OS Settings** > **OS Image**, ensure that the newly uploaded image is selected from the dropdown.
6. Click **Save** > **Publish Template**.
   A confirmation message displays asking whether you are sure that you want to publish the template, click **Yes**.

OS images

To add an OS image repository:

1. On the **Repositories** page, click **OS Images**, and then click **Add**.
2. In the **Add OS Image Repository** dialog box, enter the following:
   a. In the **Repository Name** box, enter the name of the repository. The repository name must be unique.
   b. In the **Image Type** box, enter the image type.
   c. In the **Source Path and Filename** box, enter the path of the OS image file name in a file share.

      To enter the CIFS share, see the format that is used in the following example: `\\host\lab\isos\filename.iso`

   d. If you are using the CIFS share, enter the **User Name** and **Password** to access the share.
3. Click **Add**.

After adding an OS image, you can modify it or remove it by selecting the image, then clicking **Modify** or **Remove**.

Resynchronizing an OS image repository

Use the resynchronize option to restore the OS image from the database after a backup and restore.

If an operating system image was uploaded as a part of an ISO file, you must resynchronize the OS image repository from the **OS Image Repositories** tab. However, if the operating system image was uploaded as a part of a compliance ZIP file, go to the **Compliance Versions** tab and resynchronize the ZIP file. The following procedure provides steps for performing the resynchronization from the **OS Image Repositories** tab.

To resynchronize the OS image repository for an operating system image that was uploaded as part of an ISO file:

1. On the **Repositories** page, click **OS Images**.
2. Click **Resynchronize** for a repository in an **Error** state.
   The **Resynchronize OS Repository** page is displayed.
3. Enter the user credentials and click **Test Connection** to test the network connection.

   NOTE: You cannot edit the **Source Path** and **Filename**.

4. Click **Resynchronize**.
   The repository state changes to `Copying state`.

Compatibility management

To facilitate upgrades, PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which upgrade paths are valid and which are not.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded are incompatible with the target compliance versions. PowerFlex Manager uses information that is provided in the compatibility matrix file to determine which paths are valid. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software.

When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any updates. You must upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

You can download the file from Dell Technologies connectivity services or upload it from a local directory path. The file has a GPG extension and an associated compatibility version number.

If you are uploading from a local directory path, ensure that you have access to a valid compatibility matrix file that has the GPG extension. If you are using Dell Technologies connectivity services, ensure that Dell Technologies connectivity services is correctly configured.

1. On the menu bar, click **Settings** and then click **Compatibility Management**.
2. Click **Edit Settings**.
3. Click **Download from Dell Technologies connectivity services** if you are using Dell Technologies connectivity services.
4. Click **Upload from Local** to use a local file. Then, click **Choose File** to select the GPG file.

Getting started

If you want to return to the **Getting Started** page, you can launch it from the **Settings** page.

The **Getting Started** page guides you through the common configurations that are required to prepare a new PowerFlex Manager environment. A green check mark on a step indicates that you have completed the step. Only super users have access to the **Getting Started** page.

Networking

On the **Networking** page, you can define, edit, or delete a network. You can also verify which IP addresses are in use before deployment.

Related information

Getting started
Configuring block storage
Deploying and provisioning

Networks

The **Networks** page displays information about networks that are defined in PowerFlex Manager, including:

- **Name**
- **Network Type**
- **VLAN ID**
- **IP Address Setting**
- **Starting IP Address**
- **Ending IP Address**
- **Role**
- **IP Address in Use**

On the **Networks** page, you can:

- Define or modify an existing network.
- Delete an existing network.
- Click **Export All** to export all the network details to a CSV file.
- Export network details for a specific network. To export the specific network details, select a network, and then click **Export Network Details**.
- Click a network to see the following details in the **Summary** tab:
  - Name of the user who created and modified the network.
  - Date and time that the network was created and last modified.
- Sort the column by network names, click the arrow next to the column header. You can also refresh the information about the page.

If you select a network from the **Networks** list, the network details are displayed.

For a static network, the following information is displayed:

- **Subnet Mask**
- **Gateway**
- **Primary DNS**
- **Secondary DNS**
- **DNS Suffix**
- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**
- **Static IP Details**

For a DHCP network, the following information is displayed:

- **Last Updated By**
- **Date Last Updated**
- **Created By**
- **Date Created**

You can filter the IPs by selecting any of the following options from the **View** drop-down list, under the **Static IP Address Details** section:

- **All IP Addresses**
- **IP Addresses in Use**
- **Available IP Addresses**

You can also select the links under the **IP Address in Use** column. The IPs are automatically filtered based on the **IP addresses In Use** criteria.

Defining a network

Adding the details of an existing network enables PowerFlex Manager to automatically configure nodes that are connected to the network.

Ensure that the following conditions are met before you define the network:

- PowerFlex Manager can communicate with the out-of-band management network.
- PowerFlex Manager can communicate with the operating system installation network in which the appliance is deployed.
- PowerFlex Manager can communicate with the hypervisor management network.

- The DHCP node is fully functional with appropriate PXE settings to PXE boot images from PowerFlex Manager in your deployment network.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Click **Define**. The **Define Network** page opens.
4. In the **Name** field, enter the name of the network. Optionally, in the **Description** field, enter a description for the network.
5. From the **Network Type** drop-down list, select one of the following network types:
   - **General Purpose LAN**
   - **Hypervisor Management**
   - **Hypervisor Migration**
   - **Hardware Management**
   - **PowerFlex Data**
   - **PowerFlex Data (Client Traffic Only)**
   - **PowerFlex Data (Server Traffic Only)**
   - **PowerFlex Management**

   For a PowerFlex configuration that uses a hyperconverged architecture with two data networks, you typically have two networks that are defined with the PowerFlex data network type. The PowerFlex data network type supports both client and server communications. The PowerFlex data network type is used with hyperconverged resource groups.

   For a PowerFlex configuration that uses a two-layer architecture with four dedicated data networks, you typically have two PowerFlex (client traffic only) VLANs and two PowerFlex data (server traffic only) VLANs. These network types are used with storage-only and compute-only resource groups.
6. In the **VLAN ID** field, enter a VLAN ID between **1** and **4094**.

   NOTE: PowerFlex Manager uses the VLAN ID to configure I/O modules that enable network traffic to flow from the node to configured networks during deployment.
7. Optionally, select the **Configure Static IP Address Ranges** check box, and then do the following:

   NOTE: The **Configure Static IP Address Ranges** check box is not available for all network types. For example, you cannot configure a static IP address range for the operating system installation network type. You cannot select or clear this check box to configure static IP address pools after a network is created.

   a. In the **Subnet** box, enter the IP address for the subnet. The subnet is used to support static routes for data networks.
   b. In the **Subnet Mask** box, enter the subnet mask.
   c. In the **Gateway** box, enter the default gateway IP address for routing network traffic.
   d. Optionally, in the **Primary DNS** and **Secondary DNS** fields, enter the IP addresses of primary DNS and secondary DNS.
   e. Optionally, in the **DNS Suffix** field, enter the DNS suffix to append for hostname resolution.
   f. To add an IP address range, click **Add IP Address Range**. In the row, indicate the role in PowerFlex nodes for the IP address range and then specify a starting and ending IP address for the range. For the **Role**, select either:
      - **Server or Client**: Range is assigned to the server and client roles.
      - **Client Only**: Range is assigned to the client role on PowerFlex hyperconverged nodes and PowerFlex compute-only nodes.
      - **Client Only**: Range is assigned to the client role on PowerFlex compute-only nodes.
      - **Server Only**: Range is assigned to the server role on PowerFlex hyperconverged nodes and PowerFlex storage-only nodes.
      - **Server Only**: Range is assigned to the server role on PowerFlex storage-only nodes.

      Repeat this step to add IP address ranges based on the requirement. For example, you can use one range for SDC and another for storage node.

      NOTE: IP address ranges cannot overlap. For example, you cannot create an IP address range of 10.10.10.1–10.10.10.100 and another range of 10.10.10.50–10.10.10.150.
8. Click **Save**.

Modifying a network

If a network is not associated with a template or resource group, you can edit the network name, the VLAN ID, or the IP address range.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
   The **Networks** page opens.
3. Select the network that you want to edit and click **Modify**. The **Modify Network** page opens.
4. Edit the information in any of the following fields: **Name**, **VLAN ID**, **IP Address Range**.

   For a PowerFlex data network, you can specify a **Subnet** IP address for a static route configuration. The subnet is used to support static routes for data networks.

5. Click **Save**.

You can also view details for a network or export the network details by selecting the network, then clicking **View Details** or **Export Network Details**.

(Optional) Deleting a network

You cannot delete a network that is associated with a template or resource group.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **Networks**.
3. In the **Networks** page, click the network that you want to delete, and click **Delete**.
4. Click **OK** when the confirmation message is displayed.

Adding an IP verification port number

You must verify what IP addresses are in use before they are deployed. PowerFlex Manager uses default ports 22, 80, and 135 for verification.

1. On the menu bar, click **Settings** and click **Networking**.
2. Under **IP Verification Port Number**, click **Edit**.
3. In the **Port Numbers Used for IP Verification** box, add or remove the port numbers used by PowerFlex Manager to verify the IP addresses.

   Port numbers must be entered in comma-separated list, and must be between 1 and 65535.

4. Click **Save**.

To remove a port number, click **X** to the right of the port number.

System data networks

The **System Data Networks** page displays information about system data networks for use with NVMe/TCP. This information includes:

- **Name**
- **Network**
- **Network Mask**

On the **System Data Networks** page, you can:

- Add a system data network.
- Rename a system data network.
- Remove a system data network.

Adding a system data network

Use this procedure to add a new data network for NVM/TCP host connectivity.

To add a system data network, complete the following steps:

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click **Add**. The **Create System Data Network** page opens.
4. In the **Name** field, enter the name of the network.
5. In the **Network** field, enter the IP address of the network.
6. In the **Network Mask** field, enter the network mask.
7. Click **Create**.

Renaming a system data network

After you have added a system data network, you can edit the network name.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Select the network that you want to edit and click **Rename**. The **Rename System Data Network** page opens.
4. Edit the information in the **System Data Network Name**.
5. Click **Apply**.

Removing a system data network

After you have added a system data network, you can delete it, if necessary.

1. On the menu bar, click **Settings** and then click **Networking**.
2. Click **System Data Networks**.
   The **System Data Networks** page opens.
3. Click the network that you want to delete, and then click **Remove**.
4. Click **OK** when the confirmation message is displayed.

Events and alerts

A source is used to configure the receiving of external events and syslog content.

A destination is used to configure the ability to send events and alerts information out. A destination is always external. Dell Technologies connectivity services, email, SNMP, and remote syslog are considered destinations.

Notification policies define what information is sent to each destination. Events and alerts exist irrespective of whether notification policies are created.

SNMP sources are not automatically discovered and must be configured to receive events about these sources. PowerFlex Manager is preconfigured and events, and alerts are automatically available. Resources in the PowerFlex rack, PowerFlex appliance, and Dell VxFlex Ready Node are automatically discovered. Any future resources, for example, switch replacements or additional nodes, are considered external and must be added manually as sources.

Related information

Enabling Dell Technologies connectivity services

Configuring an external source

You must define a source to enable PowerFlex Manager to receive an external event. You need not create a source to receive PowerFlex events.

External resources such as iDRAC, CloudLink, and Dell or Cisco Switches have to be discovered on the **Resources** page either in a managed or unmanaged state.

Alerts received from external resources like iDRAC, CloudLink, and Dell or Cisco Switches are seen only on **Events** Page.

PowerFlex related alerts can be seen on both the **Alerts** and **Events** pages.

A syslog source can only go to a syslog destination and does not display in events. An SNMP source, either V2 or V3, displays in events even without a defined notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click **Add**.
   The **Add Source** window opens.
3. Enter a source name and description.
4. Configure either SNMP, syslog, or audit log forwarding:
   - If you select **SNMPV2c**:
     a. Enter the community string by which the source forwards traps to destinations.
     b. Enter the same community string for the configured resource. During discovery, if you selected PowerFlex Manager to automatically configure iDRAC nodes to send alerts to PowerFlex Manager, enter the community string that is used in that credential here.
   - If you select **SNMP V3**:
     a. Enter the username, which identifies the ID where traps are forwarded on the network management system.

        NOTE: The username must be at most 16 characters.

     b. Select a security level from the following:

        Table 42. Security levels

Security Level	Details	Description	authPassword	privPassword
Minimal	noAuthNoPriv	No authentication and no encryption	Not required	Not required
Moderate	authNoPriv	Messages are authenticated but not encrypted (Password-based encryption with message digest (MD5) with at least eight characters)	Required	Not required
Maximum	authPriv	Messages are authenticated and encrypted (MD5 and Data Encryption Standard (DES) both with at least eight characters)	Required	Required

        PowerFlex supports SNMP v3 alerts only for iDRAC.

   - If you select **Syslog**, toggle **Enable Syslog** right to on.
   - If you select **Audit Log**, click **Enable Audit Log Retention**. You can retain audit logs locally for 15 days.
5. Click **Submit**.

Related tasks

Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Modifying an external source

You can edit the information about how PowerFlex Manager receives an event.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Sources** pane, click the source that you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Configuring a destination

Define a location where the event and alert data that PowerFlex Manager processes should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click **Add**.
   The **Create New Destination Protocol** window opens.
3. From the **Destination Properties** window:
   a. Enter the destination name and description.
   b. From the **Destination Type** menu, select either **SNMP V2c**, **SNMP V3**, **Syslog**, **Email (SMTP)**, **Webhook**, or **Audit Log**.
4. Click **Next**.
   The **Protocol Settings** window opens.
5. Depending on the destination type, enter the following information:

Destination type	Protocol settings
SNMP V2c	• Network Name/IP Address • Port • Community string
SNMP V3	• Network Name/IP Address • Port • Username • Security level: ○ Minimal - no more information required ○ Moderate - MD5 authentication password required ○ Maximum - MD5 authentication and DES privacy passwords required

Destination type	Protocol settings
Syslog	• Network Name/IP Address • Port • Protocol ○ UDP ○ TCP • Facility ○ All ○ Authentication ○ Security and authentication
Email (SMTP)	• Destination name • Description • Destination Type ○ Server Type: ■ SMTP ■ SMTP over SSL ■ SMTPS SMARTTLS ○ Server IP or FQDN ○ Port ○ Sender address and up to five recipient addresses ○ If you choose to use credentials enter: ■ Username, password, sender address and up to five recipient addresses • Send Test Email • Test Email Server Connection
Webhook	• Destination name • Description • Destination Type ○ Select Webhook. ○ You can have up to three webhook destinations. • Webhook Destination API URL ○ For example, if you want to configure a webhook destination for BigPanda, you must specify the BigPanda endpoint here. ○ You can get this URL from the BigPanda site. • Enable Credentials • Credentials ○ For example, for BigPanda, the credential you use must have an App Key and a Token string. You can generate these values on the BigPanda site and then copy them to the App Key and Token fields. • Test Webhook
Audit Log	• Destination name • Description • Destination Type ○ Select Audit Log. ○ You can have up to four audit log destinations. • Audit Log Network Name or IP Address • Port • Protocol ○ User Datagram Protocol (UDP) ○ Transmission Control Protocol (TCP)

6. Click **Save**.

Related tasks

Configuring an external source

Modifying an external source
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Modifying a destination

You can edit the information about where event and alert data that is processed by PowerFlex Manager should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. From the **Destinations** pane, click the destination whose information you want to modify.
   The **Edit Source** window opens.
3. Edit the information and click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Add a notification policy
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Add a notification policy

When you add a notification policy, you define the rules for processing events or alerts from sources, and to which destinations that information should be sent.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Click **Create New Policy**.
3. Enter a name and a description for the notification policy.
4. From the **Source Type** menu, select how you want events and alerts to be received. The source type options are:
   - **Snmpv2c**
   - **Snmpv3**
   - **Syslog**
   - **Powerflex_alerts**
   - **Powerflex_events**
   - **Auditlog**

   PowerFlex related alerts are generated and forwarded to a webhook destination when the notification policy is configured with the **Source Type** set to **Powerflex**.
5. From the **Resource Domain** menu, select the resource domain that you want to add a notification policy to. The resource domain options are:
   - **All**
   - **Management**
   - **Block (Storage)**
   - **Compute (Servers, Operating Systems, virtualization)**
   - **Network (Switches, connectivity etc.)**
   - **Security (RBAC, certificates, CloudLink etc.)**
6. Select the checkbox beside the severity levels that you want to associate with this policy. The severity indicates the risk (if any) to the system, in relation to the changes that generated the event message.
7. Select the required destination. You can choose one or more destinations. For a webhook, you can have up to three destinations defined. For audit logs, you can configure up to four destinations.

NOTE: The audit log policy is disabled by default. Click **Configure Now** to configure and activate the audit log policy.

8. Click **Submit**.

After adding a notification policy, you might need to perform additional configuration steps. For example, If you are setting up a notification policy for a webhook destination that uses BigPanda, you can optionally configure BigPanda to show the severity levels from PowerFlex Manager. To do this, you must configure the status mapping on the BigPanda site. Map the `major` and `minor` severity values from PowerFlex Manager to the `Warning` status for BigPanda.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Modify a notification policy
Delete a notification policy
Configuring an audit log policy

Related information

Enabling Dell Technologies connectivity services

Modify a notification policy

You can modify certain settings that are associated with a notification policy.

You cannot modify the source type or destination once it is assigned to a notification policy.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to modify.
3. You can choose to modify the notification policy in the following ways:
   - To activate or deactivate the policy, click **Active**.
   - To modify the policy, click **Modify**. The **Edit Notification Policy** window opens.
4. Click **Submit**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Delete a notification policy
Configuring an audit log policy

Delete a notification policy

Use this procedure to delete a notification policy.

NOTE: Once a notification policy is deleted, it cannot be recovered.

1. Go to **Settings** > **Events and Alerts** > **Notification Policies**.
2. Select the notification policy that you want to delete.
3. Click **Delete**.
   You receive a message to confirm if you want to delete the policy.
4. Click **Submit** and click **Dismiss**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Configuring an audit log policy

Configuring an audit log policy

Audit log is a collection of various events that occur in a system. Audit logs facilitate detection and response to cyber threats. By default, an audit log policy is disabled.

In PowerFlex Manager, the audit log contains details of the following events:

- User authentication and authorization
- System configuration and security policy changes
- Security events

Security Information and Event Manager (SIEM) manages log collection, log analysis, log forensics, real-time alerting, user activity monitoring and retention. About four SIEM can be sent over a secured connection.

Administrators can collect events and alerts from remote systems, also called sources and store them on a central server. The Audit Log Converter receives the events and alerts and forwards them in an encrypted form to the remote SIEM.

The Audit Logging Enabler is required to send logs in real-time on an automated basis. The audit logs are persistent for 10 to 90 days. Audit logs sent through SIEM have a retention period of 12 to 18 months.

1. Go to **Events and Alerts** > **Notification Policies** > **Policies** > **Audit Log Policy**.
2. Click **Configure Now**.
   The **Audit Log Policy** window opens.
3. Select and existing destination to forward the audit logs, from the **Destination** drop-down or click **Create Destination** to create a destination in the **Add Destination** window.

   You can create up to four audit log destinations.

   NOTE: You can skip this step if you do not want to forward the audit logs.
4. Click **Enable Audit Logs** to create a source of the type **Audit Log**.
5. Click **Save**.

Related tasks

Configuring an external source
Modifying an external source
Configuring a destination
Modifying a destination
Add a notification policy
Modify a notification policy
Delete a notification policy

License management

You can upload a single production license for the whole PowerFlex system. You can also upload other software licenses (for example, for CloudLink).

No license is required for the first 90 days of use. During this period, you are running PowerFlex in trial mode.

When you deploy a PowerFlex cloud-based cluster, you are provided with a free evaluation license. This license gives you 90 days to use the cloud-based storage cluster. After 90 days, the storage is preserved. However, you are not allowed to make configuration changes to the deployment. Once the evaluation license has expired, you must purchase a subscription or permanent license to continue using the cluster with the full range of capabilities.

PowerFlex Manager shows the start date and end date for both evaluation and subscription licenses on the **License Management** page. This allows you to see when your license will expire. Evaluation licenses apply only to cloud deployments, whereas subscription licenses apply to both cloud and on-premises deployments.

Related information

Getting started
Enabling Dell Technologies connectivity services

Upload a PowerFlex license

You can upload PowerFlex licenses in PowerFlex Manager.

If you have a PowerFlex appliance or PowerFlex rack, you can upload a production license file for a single system.

No license is required for the first 90 days of use. During this period, you are running PowerFlex in trial mode, and all features are enabled. PowerFlex Manager shows an alert on the **Monitoring** > **Alerts** page when you are running in trial mode.

Consider the following prior to uploading a PowerFlex license:

- You must deploy the MDM cluster before uploading a PowerFlex license.
- For information on obtaining and activating the PowerFlex capacity license, see the Dell PowerFlex Licensing Guide.

1. Log in to PowerFlex Manager.
2. On the menu bar, click **Settings** and then click **License Management**.
3. Click **PowerFlex License**.
4. To upload a production license for PowerFlex, click **Choose File to Browse** in the **Production License** section, select the license file, and click **Open**.
5. Click **Save**.

When you upload a license file, PowerFlex Manager checks the license file to ensure that it is valid.

After the upload is complete, PowerFlex Manager stores the license details and displays them on the **PowerFlex Manager License** page. You can see the installation ID, system name, and SWID for PowerFlex. In addition, you can see the total licensed capacity and the license capacity left.

Uploading other software licenses

You can also upload other software licenses, in addition to the licenses required for PowerFlex. For example, you might want to upload a license for CloudLink.

1. On the menu bar, click **Settings** and then click **License Management**.
2. Click **Other Software Licenses**.
3. Click **Add**.
4. Click **Choose File** and select the license file.
5. Choose the **Type** of license you want to upload.
6. Click **Save**.

To remove a license, select it and click **Remove**.

Add the CloudLink Center license

Use this procedure to add CloudLink Center in PowerFlex Manager.

1. On the menu bar, click **Settings** > **License Management**.
2. Click **Other Software Licenses**.
3. Click **Add**.
4. Under **Upload License**, click **Choose License**.
5. Browse and select the license to upload and click **Open**.
6. Select the **Type** as **CloudLink** and click **Save**.
7. From **Lifecycle** > **Resource**, select the CloudLink VMs, and click **Run inventory**.

Delete expired or unused CloudLink Center licenses

Use this procedure to delete expired or unused CloudLink Center licenses from PowerFlex Manager.

1. On the menu bar, click **Settings** > **License Management** > **Other Software Licenses**.
2. Select the license you want to delete, and click **Delete**.
3. From **Resource**, select the CloudLink VMs, and click **Run inventory**.

Security

On the **Security** page, you can upload SSL trusted certificates for connecting to Active Directory, as well as appliance SSL certificates that ensure data secure transmission for PowerFlex Manager. You can also define credentials for the resources that PowerFlex Manager accesses and manages.

Adding SSL trusted certificates

You can upload a trusted SSL certificate from a certificate authority (CA). You can use a trusted SSL certificate to establish a secure LDAP connection to Active Directory.

Before you upload a trusted SSL certificate, you must obtain the certificate file. The file must contain an X.509 certificate in PEM format. It must start with **`---BEGIN CERTIFICATE---`** and end with **`---END CERTIFICATE---`**.

1. On the menu bar, click **Settings** and then click **Security**.
2. Click **SSL Trusted Certificates**.
   The **SSL Trusted Certificates** page opens.
3. Click **Add**.
4. Click **Choose File** and select the SSL certificate.
5. Provide a **Name** for the certificate. The name must be a single word.
6. To upload the certificate, click **Save**.

To delete an SSL trusted certificate, select the certificate and click **Delete**.

Adding appliance SSL certificates

Uploading an SSL certificate for the appliance ensures secure transmission by encrypting data that PowerFlex Manager sends over the web. It also provides authentication and ensures that data is routed to its intended endpoint and prevents users from receiving browser security errors.

To upload an SSL certificate, you typically perform the steps in the process:

1. Generate a certificate signing request (CSR) from the PowerFlex Manager user interface. Generating a CSR using a third party certificate tool is unsupported.
2. Download the CSR.
3. Submit the CSR to a certificate authority (CA). The CA provides a valid SSL certificate.
4. Upload the SSL certificate to PowerFlex Manager.

Generating a certificate signing request

A certificate signing request (CSR) includes information (such as domain name, locale) that certificate authorities require to provide a valid SSL certificate. After generating the CSR, download the encrypted text, and then submit it to a certificate authority. The certificate authority provides a valid SSL certificate for you to upload.

1. On the menu bar, click **Settings** and then click **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Generate Certificate Signing Request**.
   a. In the **Distinguished Name (www.domain.com)** box, enter a distinguished name in the format **`www.domain.com`** (for example, example.com).

b. In the **Business Name** box, enter a business name where the certificate is recorded.
c. In the **Department Name** box, enter a department name of the organizational unit (for example, IT, HR, or Sales) for which the certificate is generated.
d. In the **Locality (Town/City)** box, enter a locality name in which the organization is located.
e. In the **State (Province/Region)** box, enter a state name in which the organization is located (do not abbreviate).
f. From the **Country** list, select a country in which the organization is located.
g. In the **Email** box, enter a valid email address.
h. Click **Generate**.

4. Click **Download Certificate Signing Request**, and then copy the text that is displayed. To receive a valid SSL certificate, submit this text to a certificate authority.

Downloading the certificate signing request

After generating the certificate signing request (CSR), download the resulting text and submit it to a certificate authority. The certificate authority provides an SSL certificate for you to upload to PowerFlex Manager.

1. On the menu bar, click **Settings** and then **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Download Certificate Signing Request**.
4. To receive a valid SSL certificate, copy the displayed text and then submit it to a certificate authority.

After the certificate authority provides the SSL certificate, upload it to PowerFlex Manager.

Uploading an SSL certificate

Before you upload an SSL certificate, generate and download a certificate signing request (CSR). To receive a valid SSL certificate, submit the CSR to a certificate authority. Save the certificate to a local network share. The file must contain an X.509 certificate in PEM format. The file must start with **`---BEGIN CERTIFICATE---`** and end with **`---END CERTIFICATE---`**.

1. On the menu bar, click **Settings** and then click **Security**.
2. Click **Appliance SSL Certificates**.
   The **Appliance SSL Certificates** page opens.
3. Click **Choose File** under **SSL Certificate** and select an SSL certificate.
4. Click **Choose File** under **Trusted CA Certificate** and select an SSL certificate.
5. To upload the certificate, click **Save**.

Adding resource credentials

PowerFlex Manager requires a root-level username and password to access and manage nodes, switches, VMware vCenter, element managers, PowerFlex gateway, and operating system resources.

The **Credentials Management** page displays the following information about the credentials:

- **Name**—A user-defined name that identifies the credentials.
- **Type**—A type of resource that uses the credential.
- **Resources**—The total number of resources to which the credential is assigned.

From the credential list, click a credential to view its details in the **Summary** tab:

- Name of the user who created and modified the credential.
- Date and time that the credential was created and last modified.

On the **Credentials Management** page, you can:

- Create credentials
- Edit existing credentials
- Delete existing credentials

Create credentials

Perform the following procedure to create credentials:

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create the credentials:
   - **Node**
   - **Switch**
   - **vCenter**
   - **Element Manager**
   - **PowerFlex Gateway**
   - **OS Admin**
   - **OS User**
   - **PowerFlex Management System**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager itself.
5. In the **Credential Name** field, enter the name to identify the credential.
6. Click **Enable Key Pairs** to enable login with SSH key pairs:

   To enable key pairs for the **Node** or **Switch** credential type:
   a. Import an existing key:
      i. Click **Import SSH Key Pair**.
      ii. Click **Choose File** and browse to the file that contains your public and private keys, and select the private key.
      iii. Type a name for the key pair.
      iv. Click **Import**.

   To enable key pairs for the **OS Admin** or **OS User** credential type:
   a. To create a key:
      i. Click **Create a new key**.
      ii. Click **Create & Download Key Pair.**.
      iii. Type a name for the key pair.
      iv. Click **Create**.

      The private key file (id_rsa) is downloaded to the downloads folder. Click the **Download Public Key** button to download the public key file (id_rsa.pub).

   b. To import an existing key:
      i. Click **Import existing key**.
      ii. Click **Import SSH Key Pair**.
      iii. Click **Choose File** and browse to the file that contains your public and private keys.
      iv. Type a name for the **Key Pair Name field**.
      v. Click **Import**.

   If you enable SSH key pairs for a **Node** or **Switch** credential and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a username and password. If you enable SSH key pairs for an **OS user** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

   NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. In this case, the username and password would be used instead for all communication.
7. In the **Domain** box, optionally specify an LDAP domain for the user.
8. In the **User Name** field, enter the username for the credential.

   `root` is the only valid username for root-level credentials on nodes (iDRAC). You can add iDRAC users with a username other than **`root`**.

   For the **OS User** credential type, you can enter a user other than root. For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the

administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a nonroot user.

The account on the SVM and/or storage-only nodes for the **OS User** credential type must have a `/home` directory and have the correct group permissions. For example, if the account were pfxm_admin, the home directory would be: `/home/pfxm_admin`

Here is an example showing the requirements for a pxflex user in the pxflex-grp group on an SVM:

```
[root@svm-dkim-hc-node1 home]# ls -alhtr
        total 0
        drwxr-xr-x.  3 root   root        20 Feb 26 16:03 .
        drwxr-xr-x   3 pxflex pxflex-grp  22 Feb 26 16:10 pxflex
        dr-xr-xr-x. 17 root   root       224 Feb 26 16:37 ..
        [root@svm-dkim-hc-node1 home]# cd pxflex/
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]#
        [root@svm-dkim-hc-node1 pxflex]# pwd
        /home/pxflex
```

For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

Provide two usernames for the PowerFlex gateway credential type:

- **Gateway Admin User Name**
- **Gateway OS User Name**

The Gateway admin user is the REST API administrator. The gateway operating system user is the SSH login user. The Gateway admin user must be the **`admin`** user, and the Gateway operating system user must be **`root`**.

9. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

   NOTE: When the SSH key pair feature is enabled, the switch credential does not require the **Password** option.

   For the PowerFlex gateway credential type, provide two passwords:

   - **Gateway Admin Password**
   - **Gateway OS Password**

   PowerFlex Manager allows you to supply different passwords for the Gateway admin and Gateway operating system users. The passwords are validated when you discover the PowerFlex gateway on the **Resources** page. The discovery fails if the passwords were not specified correctly on the **Credentials Management** page. PowerFlex Manager does not update the Gateway Admin and Gateway operating system passwords on the node itself. However, if you change the password manually for either of these users, you can update the password on the **Credentials Management** page, and PowerFlex Manager ensures that subsequent operations on the resource group use the new password.

   Optionally, provide additional settings for VMware vCenter and element manager, and for switch credentials:

   - For VMware vCenter and element manager, in the **Domain** box, optionally enter the domain ID.
   - For switch credentials, under **Protocol**, optionally click one of the following connection protocols that are used to access the resource from remote:
     - **Telnet**
     - **SSH**

       NOTE: SSH is enabled on supported switches by default.

10. To configure trap receiving for SNMPv2:
    a. Under **SNMP Configuration**, select **V2** as the SNMP type.
    b. Click **+** beside the **SNMP v2 Community String** box.
    The **SNMP v2 Community String** page opens.
    c. Enter the community string by which PowerFlex Manager receives traps from devices and by which it forwards traps to destinations.
    d. Click **Save**.

    NOTE: You can add more than one community string. For example, add more than one if the community string by which PowerFlex Manager receives traps differs from the community string by which it forwards traps to a remote destination.

11. To configure trap receiving for SNMPv3:

a. Under **SNMP Configuration**, select **V3** as the SNMP type.
b. Click **+** beside the **SNMP V3 User** box.
The **SNMP V3 User** page opens.
c. Enter the **Username**, which identifies the ID where traps are forwarded on the network management system.

NOTE: The username must be at least 16 characters.

d. Select a **Security Level** from the following:

- **Minimal** - No additional information is required.
- **Moderate** - Provide an **MD5 Authentication Password** which is at least eight characters.
- **Maximum** - Provide an **MD5 Authentication Password** and a **DES Privacy Password** which is at least eight characters.

You can add more than one user.

NOTE: You can only select SNMPv3 for nodes. Select the **Auto configure nodes to send alerts to PowerFlex Manager** checkbox to discover iDRACs.

12. Click **Save**.

Create credentials for root and non-root users

Use this procedure to create credentials for root and non-root users in PowerFlex Manager.

To import and create the SSH keys for a PowerFlex node, switch, **OS Admin**, **OS User**, ensure you generate SSH key pairs of RSA type without passphrase.

You can now use a non-root user instead of the root user for PowerFlex system administration functions. This enhances security by disabling the root user during node discovery, operating system installation, and non-disruptive updates. The default non-root user name is **`pflex`**.

NOTE: The credential type **OS Admin** is used for root users, and **OS User** is used for non-root user. **OS Admin** and **OS User** credential types apply for the deployment of resource groups.

For non-root user authentication, after the deployment of the resource group, the SSH access for the user root is disabled, but the password is still available for to use the console for troubleshooting.

PowerFlex Manager allows you to specify a non-root user when you configure a template for a compute-only, storage-only deployment.

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**. The Credentials Management page opens.
3. Click **Create**.
4. In the **Create Credentials** dialog box, from the **Credential Type** drop-down list, select one of the following resource types for which you want to create non-root credentials:
   - **Node**
   - **Switch**
   - **OS Admin**
   - **OS User**

   The **OS Admin** and **OS User** credential types apply to deployed items, not to PowerFlex Manager. If you are creating an OS user credential set for the management virtual machines on a PowerFlex management controller resource group, select **OS User**.
5. In the **Credential Name** field, enter the name to identify the credential.

   NOTE: If you are creating an **OS User** credential set for the management virtual machines on a PowerFlex management controller resource group, do the following:

   a. Enter **`MVM delladmin`** to identify the credential.
   b. In the **User Name** field, enter **`delladmin`**.
   c. Enter the delladmin account password in the **Password** and **Confirm Password** fields.

6. Click **Enable Key Pairs** to enable log in with SSH key pairs and perform the following:

Table 43. Key pairs options

To...	Do this...
Enable key pairs for the **Node** or **Switch credential**:	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the private key. c. Type a name for the key pair. d. Click **Import**.
Create keys using PowerFlex Manager for the **OS Admin** or **OS User** credential and enable key pairs:	a. Click **Create a new key**. b. Click **Create & Download Key Pair**. c. On **Key Pair Name**, type the name for key pair. d. Click **Create**. e. Click **Download Public Key**.
To manually generate and import an existing key pairs for the **OS Admin** or **OS User** credential	a. Click **Import SSH Key Pair**. **NOTE:** Manually generate the SSH keys pairs. b. Click **Choose File** and browse to the file that contains the public and private key. c. Type a name for the key pair. d. Click **Import**.

If you enable SSH key pairs for a **Node** or **Switch credential** and use that credential for discovery, PowerFlex Manager uses public or private RSA key pairs to SSH into your node or switch securely, instead of using a user name and password.

If you enable SSH key pairs for an **OS User** or **OS Admin** credential and use that credential for a deployment, PowerFlex Manager uses RSA public or private key pairs for the deployment operations.

NOTE: PowerFlex Manager does not consume SSH keys for all component types. For example, if you enable SSH key pairs for an admin credential, the SSH keys are not used for the deployment of a CloudLink Center VM. Instead, the user name and password are used instead for all communication.

7. In the **User Name** field, enter the username for the credential.

   For **Nodes** (iDRAC), `root` is the only valid username for root-level credentials. For a non-root user name, enter the default non-root user name.

   For the **OS Admin** credential type, the **User Name** field is disabled because the user is assumed to be root. You must use the root user for new deployments.

   For the **OS User** credential type, enter the default non-root user name.

   For the embedded operating system, this user account must have SSH enabled and have sudo access. For VMware ESXi, the account must be configured with the administrator role on the local server permission setting, which should enable SSH and other tools like esxcli. You can add existing resource groups with a non-root user. The account on the PowerFlex storage-only nodes for the **OS User** credential type must have a /home directory and have the correct group permissions.
8. In the **Password** and the **Confirm Password** boxes, enter the password for the credential.

Modify credentials

Perform this procedure to modify credentials:

1. On the menu bar, click **Settings** and click **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. Select a credential that you want to edit, and click **Modify**.
4. Modify the credential information in the **Modify Credentials** dialog box.
5. Click **Save**.

Remove credentials

Perform this procedure to remove credentials:

1. On the menu bar, click **Settings** > **Security**.
2. Click **Resource Credentials**.
   The **Credentials Management** page opens.
3. On the **Credentials Management** page, select the credential that you want to delete, and click **Remove**.
4. Click **OK** when the confirmation message appears.

Converting a private key from PuTTY format to OpenSSH format

Perform this procedure to convert a private key from PuTTY to OpenSSH format.

1. Navigate to the folder where PuTTY was installed.
2. Double-click `puttygen.exe` to open the **PuTTY Key Generator**.
3. Click **File** > **Load private key** on the menu bar.
4. Select the private key that you want to convert to OpenSSH format and click **Open**.
5. Click **Conversions** > **Export open SSH key** on the menu bar.
6. Click **Save** to save the key.
   The key should now be in OpenSSH format.
7. Copy the public key content that is displayed on the **Key** tab in the **PuTTY Key Generator**.
   You need this content so you can copy it to the nodes.

Configuring SSH key pairs

Perform this procedure to configure SSH key pairs for nodes and switches.

1. To configure an SSH key pair for a software-only node, run these commands:

```
# below command will generate SSH Key pair in /tmp directory
ssh-keygen -m PEM -b 2048 -t rsa -f /tmp/sshkey-key1 -q -N ""

# To copy public key to software only node
ssh-copy-id -i /tmp/sshkey-key1.pub root@100.68.105.233

# Keys can be copied for both root and non-root users
```

2. To configure an SSH key pair for a Cisco Nexus switch, run these commands to add the public key:

```
username pfxmsvc password <password> role priv-15

username pfxmsvc sshkey ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7

username pfxmsvc passphrase  lifetime 99999 warntime 14 gracetime 3
```

3. To configure an SSH key pair for an OS10 switch, run these commands to add the public key:

```
username pfxmsvc password <password> role sysadmin priv-lvl 15

username pfxmsvc sshkey "ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQDWxvVQ9PgfFI3NMsg3TsR9G89ykZ3UEyciz47hcXH/
YXNakva5PV5l7ubHY4nUxKj3zpVHXg/Bq9zRK5lQFs4KGC5eaYB9nm2xEwhuhpvsCvROo8SzjjTNRcpok2EMD/
3VhuZAWdaXITSukq2D1Ek5W7qeE0LiLSC+2s0u6iPTU5KHs3mrELbRlxpVxnHQtrgUZnsLvkxwzuEnBKWNIEZ0
6HoDLP1vBCaYJslE5iDseUCZstWvFbvkaeE4FOrqohllAhVGVH19B6Q38ijVjLAwjXX03a6IQqeWj+JkpHpB61
vMsAqqje4pm9VFzAY2NAcvoBo7Oab+52qA1vJTCbG7"
```

Note that the quotes must be used as shown in the command line above.

4. To configure an SSH key pair for iDRAC, perform these steps:
   a. Connect to iDRAC.
   b. Choose **iDRAC Settings** > **Users** > **Edit/Add User** > **SSH Key Configuration** > **SSH Key (Edit)**.
   c. Add the public key.

Serviceability

On the **Serviceability** page, you can generate a troubleshooting bundle and also perform a backup of the appliance. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Generating a troubleshooting bundle

You can generate a troubleshooting bundle which is a compressed file that contains logging information for PowerFlex Manager managed components.

If necessary, download the bundle and send it to Dell Technologies Support for issue debugging.

The troubleshooting bundle includes the following logs:

- ASM deployer
- iDRAC life cycle
- Dell PowerSwitch switch
- Cisco Nexus switch
- VMware ESXi
- CloudLink Center
- Standard output logs from all pods
- Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes
- PowerFlex Block logs
- PowerFlex data server and device gateway target logs

NOTE: You can also generate troubleshooting bundles from the **Resource Groups** page.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Generate Troubleshooting Bundle**.
3. Select **Include PowerFlex Block Core Dump logs** to include core dump logs.
4. Select a **PowerFlex Log Level**:
   - Default Node Logs
   - Default Node Logs plus additional MDM information
   - Latest Logs only (Most recent copy of all logs)
5. Select a **Node Selection**:
   - Logs from all nodes
   - Select Specific Nodes

   If you select the **Select Specific Nodes** option and select the number of nodes for which you want to generate the log, the **View/Select Nodes** button is displayed. Click the button to view the list of nodes in the **Node Selection** window. Select the required nodes from the **Available Nodes** list and click **>>** to view them in the **Selected Nodes** list. Click **Save** to return to the **Generate Troubleshooting bundle** page.

   For the **Select Specific Nodes** option **Generate** is enabled only if a node is selected in the **Node Selection** window.
6. Choose a **Delivery Option**:

Option	Description
Send to Configured SupportAssist	If you are using Dell Technologies connectivity services, **Send to Configured SupportAssist** is selected by default. Leave this default setting. If Dell Technologies connectivity services is not configured, this option is disabled.

Option	Description
Download to Share	Provide a path using the following format with credentials: `CIFS:\ \1.1.1.1\uploadDirectory`. Enter a username and password, and click **Test Connection** to verify the connection to the CIFS share before generating the bundle.
Download Locally	Download the troubleshooting bundle to a local file: `\\IP Address\Any folder`

7. Click **Generate**.

Sometimes, the troubleshooting bundle does not include log information for all the nodes. The log collection may appear to succeed, but the log for one or more of the nodes may be missing. You may see an error message in the scaleio.trace.log file that says `Could not run get_info script`. If you see this message, you must generate the troubleshooting bundle again to include information for all the logs.

Back up and restore

Creating a backup saves all user-created data to a remote share from which it can be restored. To restore from a backup, you need to run a script outside of PowerFlex Manager. The user interface does not support the ability to restore from a backup.

Perform frequent backups to guard against data loss and corruption. The best practice is to take a snapshot of PowerFlex Manager every time you perform a restore.

The **Backup** page displays information about the last backup operation that was performed on PowerFlex Manager. The information provided applies to both manual and automatically scheduled backups and includes the following:

- Last backup date
- Last backup status
- Back up directory path to a CIFS share
- Back up directory username

The **Backup** page also displays information about the status of automatically scheduled backups (enabled or disabled).

On this page, you can:

- Manually start an immediate backup
- Edit general backup settings
- Edit automatically scheduled backup settings

After performing a backup operation, you need to run a script outside of PowerFlex Manager to restore from the backup. The user interface does not support the ability to restore from a backup.

Editing backup settings and details

You can change the location where backup files are saved or the password that is required to access a backup file.

1. On the menu bar, click **Settings** and then click **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Settings**.
4. Indicate the network share location where the backup file is saved, enter a backup directory path in the **Backup Directory Path** box.

   Use the following format:

   - CIFS—`\\host\share`
5. If the username and password are required to access the network share, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes.
6. Click **Test Connection** to confirm that the backup settings you provided are correct.
7. To open the backup file, enter a password in the **Encryption Password** box.
8. To verify the encryption password, enter the password in the **Confirm Encryption Password** box.
9. To schedule automatic backups, next to **Scheduled Backups**, select **Enabled**. To discontinue automatically scheduled backups, deselect **Enabled**.
10. To specify the days on which backup must occur, select the days under **Frequency**.
11. From the **Run Time** drop-down list, select the time.

12. Click **Save**.

Backing up

In addition to automatically scheduled backups, you can manually run an immediate backup.

1. On the menu bar, click **Settings** > **Serviceability**.
2. Click **Backup**.
   The **Backup** page opens.
3. Click **Backup Now**.
4. Select one of the following options:
   - To use the general settings that are applied to all backup files, select **Use Backup Directory Path and Encryption Password from Settings and Details**.
   - To use custom settings:
     a. In the **Backup Directory Path** box, enter a file path where the backup file is saved. Use this format:
        - CIFS—`\\host\share`

     b. Optionally, enter a username and password in the **Backup Directory User Name** and **Backup Directory Password** boxes, if they are required to access the location you provided.
     c. Click **Test Connection** to confirm that the backup settings you provided are correct.
     d. In the **Encryption Password** box, enter a password that is required to open the backup file, and verify the encryption password by entering the password in the **Confirm Encryption Password** box.
5. Click **Backup Now**.

Software upgrade

When a new version of the management software becomes available, you can upgrade to that version from the **Software Upgrade** page.

Upgrading PowerFlex Manager from a local repository path

You can upgrade the management software from a .TGZ file on a local repository path.

Before you begin, ensure that you:

- Take a backup of PowerFlex Manager settings.
- Upload the latest compatibility matrix file to ensure that PowerFlex Manager has access to the latest upgrade information.

When you attempt an upgrade, PowerFlex Manager warns you if the current version of the software is incompatible with the target version, or if any of the RCM or IC versions that are currently loaded on the management software are incompatible with the target compliance versions. To determine which paths are valid and which are not, PowerFlex Manager uses information that is provided in the compatibility matrix file. The compatibility matrix file maps all the known valid and invalid paths for all previous releases of the software. Before proceeding with an upgrade, PowerFlex Manager notifies you if any of the following situations is detected:

- No upgrade is required. This situation occurs when the current and target versions are the same. You still can force an upgrade, if needed.
- The upgrade path that you are attempting to perform is a valid path. The source version of PowerFlex Manager is compatible with the target version and the source version of the RCM or IC is also compatible with the target version.
- The source version of PowerFlex Manager is compatible with the target version. However, the source version of the RCM or IC is not compatible with the target version. Your resource groups may go into lifecycle mode until you perform an RCM or IC upgrade. PowerFlex Manager puts a resource group in lifecycle mode if the resource group's RCM or IC is not within a year of the target PowerFlex Manager launch. You can mitigate a resource group that is in lifecycle mode by upgrading to a compliance version that is within a year of the current PowerFlex Manager launch.
- The upgrade path that you are attempting to perform is not supported.

The compatibility matrix file is required for all upgrades. When you first install PowerFlex Manager, the software does not have the compatibility matrix file, so you must upload the file before performing any upgrades.

1. If you are using PowerFlex rack, log in to Dell Technologies Download Center. Then, go to the section for PowerFlex Manager.

2. If you are using PowerFlex appliance, log in to Dell Technologies Support. Then, enter your service tag to be taken to available downloads.
3. Download the PowerFlex management platform .TGZ file.
4. Copy the PowerFlex management platform .TGZ file to a share drive location.
5. On the menu bar, click **Settings**, and then click **Software Upgrade**.
6. Click **Management Software Upgrade**.
7. On the **Management Software Upgrade** page, you can see the **Current Management Software Version** and check the **Available Management Software Version** field to see if a newer version of PowerFlex Manager is available.
8. Click **Edit Settings**.
   a. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.
   b. Enter the username and password.
   c. Click **Save**.
9. At the top of the **Management Software Upgrade** page, click **Upgrade Now**.
   After you click **Upgrade Now**, a dialog box displays with a warning message indicating whether the upgrade path is valid.
10. Under **Management Credentials**, optionally override the default username and password for the nodes hosting PowerFlex Manager. The upgrade requires access to the nodes hosting PowerFlex Manager. Provide a single user that has superuser privileges for all nodes. You can enter the username for the first row in the table, and the changes you make will be applied to all the nodes. Enter the password for each of the nodes.
11. Type **`UPDATE POWERFLEX MANAGER`** if you want to proceed with the update. If you want to perform an update even if none is required, type **`FORCE UPDATE`**.
12. Click **Yes** to update PowerFlex Manager.
    The update process displays messages indicating the progress of the update. Depending on which services were upgraded, you may be required to login once again, after the upgrade process is complete.

After upgrading to PowerFlex Manager, you may notice that resource groups that are tied to the PowerFlex gateway are no longer in compliance. To bring the resource groups into compliance, you may need to upgrade them. In this case, you will need to change to a later RCM or IC and upgrade the resource groups.

If you do not want to upgrade the resource groups, you do not need to perform the upgrade.

Editing the upgrade settings

To edit the upgrade settings:

1. On the menu bar, click **Settings** and then click **Software Upgrade**.
2. Click **Management Software Upgrade**.
3. Click **Edit Settings**.
   a. To update to the latest version using Dell Technologies connectivity services, select **Update Software from configured Dell SupportAssist**.
   b. To upgrade from a specific version on a local repository path, select **Update Software from local repository path**, and enter the path in the **Repository Path** box.

      You can upgrade from a .tgz file on a network share. This option provides an easy way to update the software in a dark site. Provide a path using one of the following formats:

      Table 44. Path for network shares

For...	Use this path...
CIFS	`\\IP Address\Any folder\file-name.tgz`
FTP	`ftp://IP Address/Any folder/file-name.tgz`
HTTP	`http://IP Address/Any folder/file-name.tgz`
HTTPS	`http://IP Address/Any folder/file-name.tgz`

   c. Optionally, provide a **User Name** and **Password**.

      The username and password are only used with CIFS. All other local repository paths must specify public share locations.

   d. Click **Save**.

4. On the **Management Software Upgrade** page, you can view the updated settings for **Current Management Software Version**, **Available Management Software Version**, and **Repository Path**.
5. To perform the upgrade, click **Upgrade Now**. A message displays confirming that you want to update the software.
6. Click **Upgrade Software**.
7. Click **Yes**.
8. Depending on which services were upgraded, you may be required to log in again once the upgrade process is complete.

13

PowerFlex Manager user interface reference

This section provides reference details for various parts of the user interface for PowerFlex Manager.

The topics in this section are organized according to the main tabs within the user interface.

Lifecycle

This section provides reference information for the **Resource Groups**, Templates, and **Storage Nodes** pages.

Related information

Managing components
Monitoring system health
Managing external changes
Deploying a resource group
Adding an existing resource group

Resource groups

This section provides reference information for the **Resource Groups** page.

Related information

Managing components
Managing external changes

Resource group states

Table 45. Resource group states

Icon	Resource group state	Description
	**Healthy**	The resource group was deployed successfully, and the resources are firmware compliant and healthy.
	**Warning**	One or more resources in the resource group require corrective action. This state does not affect overall system health. For example, the firmware running on the resource is not at the required level or not compliant.
	**Critical**	Resource group deployment has failed due to some issues.
	**Pending**	Resource group deployment is scheduled for a later time or date.
	**In Progress**	Resource group deployment is in progress.
	**Canceled**	Resource group deployment was canceled.
	**Incomplete**	The resource group is not fully functional because it has no associated volumes. Click **Add Resources** to add volumes.

Table 45. Resource group states (continued)

Icon	Resource group state	Description
	**Service Mode**	The resource group has been placed in service mode because one of the nodes within the resource group was put into service mode. If a node is in instant maintenance mode, PowerFlex Manager detects this situation and automatically places the node in service mode and also ensures that the resource group itself goes into service mode. If a node is in instant maintenance mode for more than 30 minutes, PowerFlex Manager sends a critical alert to Dell Technologies connectivity services.
	**Lifecycle Mode**	The resource group supports health and compliance monitoring, service mode, and non-disruptive updates. All other resource group operations are blocked. Lifecycle mode controls the operations that can be performed for configurations that have limited support.
	**Managed Mode**	The resource group supports health and compliance monitoring, non-disruptive updates, automated resource addition, and automated resource replacement features.

Existing resource groups

If you already have a cluster deployed that includes Dell-based hardware, you can discover and import these hardware resources as an existing resource group. Even though these resources were not originally deployed with PowerFlex Manager, you can take advantage of the health monitoring, compliance, and update features.

When you add an existing resource group for a PowerFlex cluster, PowerFlex Manager matches up the hosts, vCenter, and other items it finds with discovered resources in the resource list.

Only managed or reserved resources are included in the resource group. Any resource discovered as **Unmanaged** is missing from the resource group. If a component is missing, you can change your resource inventory, and update the resource group to reflect these changes. Go to the resources list, select the component, and mark it as **Managed** by selecting **Change resource state to Managed**. Then perform an **Update Resource Group Details** operation on the resource group to pull in the missing component.

Each node is identified as a PowerFlex host if the inventory for the node shows that it is present in a gateway, and it is properly configured as a storage data client (SDC) and storage node. A PowerFlex host has the **Use Node for Dell PowerFlex** check box set to **True** in the operating system settings for the node within the resource group deployment settings.

You can perform most of the actions on an existing resource group that are available within PowerFlex Manager for a new resource group. For some configurations, PowerFlex Manager displays a yellow banner for the resource group to indicate that some features are not available.

PowerFlex Manager enables you to add an existing resource group for an environment using VMware NSX-T or NSX-V. Most resource group actions are disabled for an NSX-T or NSX-V configuration, except the ability to update the firmware and software components, remove resources (or the resource group as a whole), and update resource group details.

When you add an existing resource group for a CloudLink configuration, you choose a single CloudLink Center as the target for the resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.

Lifecycle mode

If you add an existing resource group that includes an unsupported configuration, PowerFlex Manager might put the resource group in lifecycle mode. This mode limits the actions that you can perform within the resource group.

Lifecycle mode allows the resource group to perform only monitoring, service mode, and compliance upgrade operations. All other resource group operations are blocked when the resource group is in lifecycle mode. Lifecycle mode is used to control the operations that can be performed for configurations that have limited support.

When you add an existing resource group, PowerFlex Manager puts the resource group in lifecycle mode if the configuration you want to import includes any of the following:

- Invalid server inventory
- Missing network settings

  The minimum required IP addresses are one data IP and one PowerFlex IP for a storage-only resource group,

- Unsupported NIC (a 1 GB QLogic NIC, for example)
- Unsupported server configurations (an unsupported SD Boot device, for example)
- Unsupported NIC teaming policies, an unsupported switch port configuration, or an unsupported access facing port configuration
- Network configuration without a PXE VLAN setting
- No switch configuration
- PowerFlex MDM cluster without virtual IPs
- vSphere Cluster Services (vCLS) VMs on local storage

When you add an existing resource group, PowerFlex Manager checks to see whether there are any vSphere Cluster Services (vCLS) VMs on local storage. If it finds any, it puts the resource group in lifecycle mode and lets you migrate the VMs to shared storage.

PowerFlex Manager also puts a resource group in lifecycle mode if you select a minimal compliance version that includes firmware only for the resource group.

When PowerFlex Manager must put a resource group in lifecycle mode, the **Summary** page for the **Add Existing Resource Group** wizard displays a warning message indicating the reason.

In some situations, an imported configuration might not meet the minimal requirements for lifecycle mode. In this case, PowerFlex Manager does not allow you to add the resource group.

For missing virtual IPs or no switch connectivity, you can correct the error and use **Update Resource Group Details** to take the resource group out of lifecycle mode.

If you have manually added NSX to a cluster outside of PowerFlex Manager, click **Update Resource Group Details** on the resource group, so it will correctly reflect the environment with NSX, which will be lifecycle mode.

Templates

This section provides reference information for the **Templates** page.

Related information

Getting started
Managing components

Managing templates

On the **Templates** page, you can view information about templates in a list or tile view. To switch between views, click the tile icon or list icon at the top of the **Templates** page. When in tile view, you can view the templates under a category by clicking the graphic that represents the category.

Users with standard permissions can view details of only those templates for which the administrator has granted permission.

Templates - My Templates List View

The **My Templates** page displays the details of the templates that you have created.

Table 46. My Templates page

Field	Description
+ Add a Template	Allows you to add a template. After you add a template, you can add node, cluster, and VM components on the template builder page.
Export All	Allows you to export all templates to a CSV file.
Filter By	Allows you to filter and view templates based on the template category.

Table 46. My Templates page (continued)

Field	Description
State	Displays the state of the template - **Draft** or **Published**.
Category	Displays the template category.
Name	Displays the template name.
Last Deployed On	Displays the date when the template was last deployed.
Components	Displays the components in the template.

Templates - Sample Templates

The **Sample Templates** page displays the default templates that you can use in your environment.

Table 47. Templates - Sample Templates page

Field	Description
Name	Displays the template name.
Components	Displays the components in the template.

Templates

On the **Templates** page, the right pane displays the name of the template, icons of components in the template, and the following details for a selected template:

Table 48. Templates page

Field	Description
Edit	Click to edit the template.
Delete	Click to delete the template.
View Details	Click to view the details of the template, such as components in the template.
Clone	Click to clone the template.
Export Template	Click to export the template to a GPG file. You can use the exported template to duplicate the settings.
Created On	Displays the date and time when the template was created.
Created By	Displays the name of the user who created the template.
Updated On	Displays the date and time when the template was updated.
Updated By	Displays the name of the user who updated the template.

Building a template overview

The template builder allows you to build a customized template by configuring both physical and virtual components. On the template builder page, you can set the component properties. For example, you can create a template that provisions only physical nodes with operating systems on them.

NOTE: A newly created or a cloned template appears in a draft state on the **Template** page and remains in the same state until published.

You can configure node, cluster, and VM components in a template.

The template builder page displays a graphical representation of the topology that is created within a particular template. From this page, you can:

- Add node, cluster, and VM components to a template
- Build and publish a template
- Delete a template

- Import a template
- Deploy a resource group (this feature is available only on published templates)

Component types

Components (physical or virtual or applications) are the main building blocks of a template.

PowerFlex Manager has the following component types:

- Node (Software/hardware and software only)
- Cluster
- VM

Related information

Adding components to a resource group
Edit a template
Deploying a resource group
View template details

Node settings

This table describes the following node settings: hardware, BIOS, operating system, and network.

Table 49. Node settings

Setting	Description
Full network automation	Allows you to perform deployments with full network automation. This feature allows you to work with supported switches, and requires less manual configuration. Full network automation also provides better error handling since PowerFlex Manager can communicate with the switches and identify any problems that may exist with the switch configurations.
Partial network automation	Allows you to perform switchless deployments with partial network automation. This feature allows you to work with unsupported switches, but requires more manual configuration before a deployment can proceed successfully. If you choose to use partial network automation, you give up the error handling and network automation features that are available with a full network configuration that includes supported switches. For a partial network deployment, the switches are not discovered, so PowerFlex Manager does not have access to switch configuration information. You must ensure that the switches are configured correctly, since PowerFlex Manager can not configure the switches for you. If your switch is not configured correctly, the deployment may fail and PowerFlex Manager is not able to provide information about why the deployment failed. For a partial network deployment, you must add all the interfaces and ports, as you would when deploying with full network automation. However, you do not need to add the operating system installation network, since PXE is not required for partial network automation. PowerFlex Manager uses virtual media instead for deployments with partial network automation. The **Switch Port Configuration** must be set to **Port Channel (LACP enabled)**. In addition, the LACP fallback or LACP ungroup option must be configured on the port channels.
Component Name	Indicates the node component name
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The encryption options are: • Software encryption • Self-encrypting drive (SED)

Table 49. Node settings (continued)

Setting	Description
Number of Instances	Enter the number of instances that you want to add. If you select more than one instance, a single component representing multiple instances of an identically configured component are created. Edit the component to add extra instances. If you require different configuration settings, you can create multiple components.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Import Configuration from Reference Node	Click this option to import an existing node configuration and use it for the node component settings. On the **Select Reference Node** page, select the node from which you want to import the settings and click **Select**.
OS Settings	
Host Name Selection	If you choose **Specify At Deployment Time**, you must type the name for the host at deployment time. The hostname must not exceed 15 characters. If you choose **Auto Generate**, PowerFlex Manager displays the **Host Name Template** field to enable you to specify a macro that includes variables that produce a unique hostname. For details on which variables are supported, see the context-sensitive help for the field. If you choose **Reverse DNS Lookup**, PowerFlex Manager assigns the hostname by performing a reverse DNS lookup of the host IP address at deployment time.
OS Image	Specifies the location of the operating system image installation files. You can use the image that is provided with the target compliance file, or specify your own location, if you created additional repositories. To deploy a compute-only or storage-only resource group with the Linux image that is provided with a compliance file, choose **Use Compliance File Linux image**. To deploy a storage-only resource group with Red Hat Enterprise Linux, you must create a repository on the **Settings** page and specify the path to the Red Hat Enterprise Linux image on a file share. Dell Technologies recommends that you use one of your own images that are published from the customer portal at Red Hat Enterprise Linux. For Linux, you may include one node within a resource group. For ESXi, you must include at least two nodes. **NOTE:** If you select an operating system from the **OS Image** drop-down menu, the field **NTP Server** displays. This field is optional, but it is highly recommended that you enter an NTP server IP to ensure proper time synchronization with your environment and PowerFlex Manager. Sometimes when time is not properly synchronized, resource group deployment failure can occur.
OS Credential	Select an **OS Admin** or **OS User** credential that you created on the **Credentials Management** page. Alternatively, you can create a credential while you are editing a template. If you select a credential that was created on the **Credentials Management** page, you do not need to type the username and password, since they are part of the credential definition. PowerFlex Manager allows to specify a non-root user instead of the root user when you configure a template for a compute-only or storage-only deployment.
NTP Server	Specifies the IP address of the NTP server for time synchronization.

Table 49. Node settings (continued)

Setting	Description
	If adding more than one NTP server in the operating system section of a node component, be sure to separate the IP addresses with commas.
Use Node For Dell PowerFlex	Indicates that this node component is used for a PowerFlex deployment. When this option is selected, the deployment installs the MDM, storage node, and SDC components, as required for a PowerFlex deployment in a VMware environment. The MDM and storage node components are installed on a dedicated PowerFlex VM (SVM), and the SDC is installed directly on the ESXi host. To deploy a PowerFlex cluster successfully, include at least three nodes in the template. PowerFlex Manager uses the following logic to determine the MDM roles for the nodes: 1. Checks the PowerFlex gateway inventory to see how many primary MDMs, secondary MDMs, and tiebreakers are present, and the total number of storage node components. 2. Adds the number of components being deployed to determine the overall PowerFlex cluster size. For example, if there are three storage node components in the PowerFlex gateway inventory, and you are deploying two more, you will have a five node cluster after the deployment. 3. Adds a single primary MDM and determines how many secondary MDMs and tiebreakers should be in the cluster by looking at the overall cluster size. The configuration varies depending on the size of the cluster: • A three-node cluster has one primary, one secondary, and one tiebreaker. • A five-node cluster has one primary, two secondaries, and two tiebreakers. 4. Determines the roles for each of the new components being added, based on the configuration that is outlined above, and the number of primary, secondary, and tiebreakers that are already in the PowerFlex cluster. For each storage node in the cluster, the deployment adds all the available disks from the nodes to the storage pools created. For each compute-only, the deployment installs the SDC VIB. When you select this option, the teaming and failover policy for the cluster are automatically set to **Route based on IP hash**. Also, the uplinks are configured as active and active, instead of active and standby. Teaming is configured for all port groups, except for the PowerFlex data 1 and PowerFlex data 2 port groups. If you select the option to **Use Node For Dell PowerFlex**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the **Target Boot Device** under **Hardware Settings**.
PowerFlex Role	Specifies one of the following deployment types for PowerFlex: • **Compute Only** indicates that the node is only used for compute resources. • **Storage Only** indicates that the node is only used for storage resources. If you select an ESXi image type in the **OS Image** field, the **PowerFlex Role** must be set to **Compute Only**. If you add a compute-only node, only the SDC is added. If you select a Red Hat Enterprise Linux image type in the **OS Image** field, the **PowerFlex Role** must be set to **Storage Only**. If you add a storage-only node, only the storage node is added. The only prerequisites for a storage-only node are that the iDRAC must have an IP address and a credential. PowerFlex Manager takes care of all other configuration steps that are required for the node. For each node, PowerFlex Manager configures the MDM roles as needed and configures the storage node RPMs. Once the

Table 49. Node settings (continued)

Setting	Description
	cluster is set up, PowerFlex Manager adds every node as an storage node. Then, it adds all available disks for the storage node, adds a storage pool, and adds all disks to the storage pool. If you are creating a compute-only template, be sure to include both the **VMware Cluster** and **PowerFlex Cluster** components in the template builder. If you are creating a storage-only template, do not include a **VMware Cluster** component in the template builder. Only the **PowerFlex Cluster** component is required for a storage-only template.
Protection Domain Type	Erasure coding is the protection domain default type.
Persistent Memory	Specifies whether persistent memory is required.
Client Storage Access	Determines how clients access storage. For a storage-only role, select one of the following options: ● Storage Data Client (SDC) Only ● SDC and NVMe/TCP Initiator For a compute-only role the Client Storage Access control is not displayed, and the client access is set to SDC automatically.
Enable Compression	Enables compression on the protection domain. The default setting is for compression to be enabled. You can enable compression on a storage-only resource group. Compression is supported for new resource group deployments, and existing resource groups. You can also enable compression for storage-only nodes when performing a scale up of a resource group. **NOTE:** Irrespective of whether you have persistent memory,(SDPM or NVDIMM), you can enable compression.
Enable Encryption	Enables disk encryption on the node. This option allows you to take advantage of CloudLink encryption. You can enable CloudLink encryption on a storage-only resource group. Encryption is supported for new resource group deployments, and existing resource groups. You can also enable encryption for storage-only nodes when performing a scale up of a resource group. A resource group cannot mix encrypted and unencrypted nodes. Scale up of a storage-only resource group is supported, with the new nodes using the same encryption settings as the nodes already in the resource group. If you select this option, PowerFlex Manager looks for servers with 12 cores or more for CloudLink deployments. After encryption is enabled, the template displays a warning message indicating that the template is missing some data. The template adds the **Cloud Link Center Settings** section to the PowerFlex cluster component to allow you to specify the required data. Some PowerFlex nodes might not be selected for deployment, depending on the encryption type that is selected. For example, if you choose **Software Encryption**, you cannot include a PowerFlex node with only SEDs. If you choose **Self Encrypting Drive**, you cannot include a PowerFlex node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers do not typically have this drive combination, but PowerFlex Manager verifies the drives and uses only servers of the specified type. **Validate Settings** detects PowerFlex nodes that do not match the specified **Drive Encryption Type**.

Table 49. Node settings (continued)

Setting	Description
Drive Encryption Type	Specifies the type of encryption to use when encryption is enabled. The options are: • **Software Encryption** • **Self Encrypting Drive (SED)** Some nodes might not be selected for deployment, depending on the encryption type selected. For example, if you choose **Software Encryption**, you cannot include a node with only SEDs. Similarly, if you choose **Self Encrypting Drive**, you cannot include a node with only software encryption drives. PowerFlex Manager does not allow you to mix SEDs and software encryption drives in the same protection domain. Servers should not typically have this mix, but PowerFlex Manager checks for this and uses only servers of the type you specify. **Validate Settings** detects nodes that do not match the specified **Drive Encryption Type**.
Switch Port Configuration	Specifies whether Cisco virtual PortChannel (vPC) or Dell Virtual Link Trunking (VLT) is enabled or disabled for the switch port. For storage-only templates, the options are: • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that use a Linux operating system image, the options are: • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT. For compute-only templates that uses an ESXi operating system image, the **Switch Port Configuration** setting includes these options: • **Port Channel** turns on vPC or VLT. • **Port Channel (LACP enabled)** turns on vPC or VLT with the link aggregation control protocol enabled. • **Trunk Port** enables a networking configuration that does not use vPC or VLT.
Teaming And Bonding Configuration	The teaming and bonding configuration options depend on the switch port configuration selected. For compute-only templates, the following options are available: • If you choose **Port Channel (LACP enabled)** or **Port Channel** as the switch port configuration, the only teaming and bonding option is **Route Based on IP hash**. • If you choose **Trunk Port** as the switch configuration, there are three teaming and bonding options: ○ **Route Based on originating virtual port** ○ **Route Based on physical NIC load** ○ **Route Based on Source MAC hash** For storage-only templates, the following options are available: • If you choose **Port Channel (LACP enabled)** as the switch port configuration, the only teaming and bonding option is **Mode 4 (IEEE 802.3ad policy)**.

Table 49. Node settings (continued)

Setting	Description
	• If you choose **Trunk Port** as the switch configuration, you have the following teaming and bonding options: ○ **Mode 1 (active-backup policy)** ○ **Mode 6 (ALB Adaptive policy)** For a non-bonded storage-only template, you need to select **Trunk Port** for **Switch Port Configuration** setting, not **Port Channel (LACP enabled)**. For the **Teaming and Bonding Configuration** setting, you typically select **Mode 1** for most deployments.
Transmit Hash Policy	The option is available for storage-only and compute-only Linux nodes only when the **Teaming And Bonding Configuration** is set to Mode 4 (EEE 802 3ed policy). Mode 4 is the bond with LACP and supports the following load balancing options: • LACP in layer 2 & 3 • LACP in layer 3 & 4 LACP in layer 3 & 4 is the default option
Hardware Settings	
Target Boot Device	Specifies the target boot device. • **Local Flash Storage**: Installs the operating system to either the SATADOM or the BOSS flash storage device present in the node. With the **Local Flash Storage** option, only nodes with a BOSS storage controller and two attached hard drives or SATADOM are selected to be deployed as part of the resource group, depending on the Dell PowerEdge servers used. For PowerEdge servers that support BOSS, during deployment PowerFlex Manager creates RAID 1 with the two hard drives attached to the BOSS controller. • **Local Flash Storage for Dell PowerFlex**: Installs the operating system to either the SATADOM or the BOSS flash storage device that is present in the node and configures the node to support PowerFlex. If you select the option to **Use Node for Dell PowerFlex** under **OS Settings**, the **Local Flash Storage for Dell PowerFlex** option is automatically selected as the target boot device.
Node Pool	Specifies the pool from which nodes are selected for the deployment.
BIOS Settings	
System Profile	Select the system power and performance profile for the node.
User Accessible USB Ports	Enables or disables the user-accessible USB ports.
Number of Cores per Processor	Specifies the number of enabled cores per processor.
Virtualization Technology	Enables the additional hardware capabilities of virtualization technology.
Logical Processor	Each processor core supports up to two logical processors. If enabled, the BIOS reports all logical processors. If disabled, the BIOS reports only one logical processor per core.
Execute Disable	Enables or disables execute disable memory protection.
Node Interleaving	Enable or disable the interleaving of allocated memory across nodes. • If enabled, only nodes that support interleaving and have the read/write attribute for node interleaving set to enabled are displayed. Node interleaving is automatically set to enabled when a resource group is deployed on a node.

Table 49. Node settings (continued)

Setting	Description
	• If disabled, any nodes that support interleaving are displayed. Node interleaving is automatically set to disabled when a resource group is deployed on a node. Node interleaving is also disabled for a resource group with NVDIMM compression. • If not applicable is selected, all nodes are displayed irrespective of whether interleaving is enabled or disabled. This setting is the default.
Network Settings	
Multi-Network Selection	Select the check-box to include multiple management networks of the same type. If you select multiple networks of the same type without selecting the check box, an error is displayed when you publish the template. The multiple network selection is supported on the following networks: • Hypervisor Management • PowerFlex Management • Hypervisor Migration
Add New Interface	Click **Add New Interface** to create a network interface in a template component. Under this interface, all network settings are specified for a node. This interface is used to find a compatible node in the inventory. For example, if you add **`Two Port, 100 gigabit`** to the template, when the template is deployed PowerFlex Manager matches a node with a two-port 100-gigabit network card as its first interface. To add one or more networks to the port, select **Add Networks to this Port**. Then, choose the networks to add, or mirror network settings defined on another port. To see network changes that are previously made to a template, you can click **View/Edit** under **Interfaces**. Or, you can click **View All Settings** on the template, and then click **View Networks**. To see network changes at resource group deployment time, click **View Networks** under **Interfaces**.
Add New Static Route	Click **Add New Static Route** to create a static route in a template. To add a static route, you must first select **Enabled** under **Static Routes**. A static route allows nodes to communicate across different networks. A static route requires a **Source Network** and a **Destination Network**, and a **Gateway**. The source and destination network must each be a PowerFlex data network that has the **Subnet** field defined. If you add or remove a network for one of the ports, the **Source Network** drop-down list does not get updated and still shows the old networks. In order to see the changes, save the node settings and edit the node again.
Validate Settings	Click **Validate Settings** to determine what can be chosen for a deployment with this template component. The **Validate Settings** wizard displays a banner to when one or more resources in the template do not match the configuration settings that are specified in the template. The wizard displays the following tabs: • **Valid (*number*)** lists the resources that match the configuration settings. • **Invalid (*number*)** lists the resources that do not match the configuration settings. The reason for the mismatch is shown at the bottom of the wizard. For example, you might see `Network Configuration Mismatch` as the reason for the mismatch if you set the port layout to use a 100-Gb network architecture, but one of the nodes is using a 25 GB architecture.

Table 49. Node settings (continued)

Setting	Description
	If you set the encryption method to use self-encrypting drives (SEDs), but the nodes do not have these drives, you might see `Self Encrypting Drives are required but not found on the node, or software encryption requested but only available drives are SED.`

After entering the information about operating system Installation (PXE) network in the respective field as described in the table above, PowerFlex Manager untags vLANs entered in the operating system installation network on the switch node facing port. For vMotion and hypervisor networks, PowerFlex Manager tags these networks on the switch node-facing ports for the entered information. For rack node, PowerFlex Manager configures the vLANs on node facing ports (untag PXE vLANs, and tag other vLANs).

If you select **Import Configuration from Reference Node**, PowerFlex Manager imports basic settings, BIOS settings, and advanced RAID configurations from the reference node and enables you to edit the configuration. Some BIOS settings might no longer apply once new BIOS settings are applied. PowerFlex Manager does not correct these setting dependencies. When setting advanced BIOS settings, use caution and verify that BIOS settings on the hardware are applicable when not choosing **Not Applicable** as an option. For example, when disabling SD card, the settings for internal SD card redundancy become not applicable.

You can edit any of the settings visible in the template, but keep in mind that many settings are hidden when using this option. For example, only ten out of many BIOS settings that you can see and edit when using a template are displayed. However, you can configure all BIOS settings. If you want to edit any of the settings that are not visible through the template feature, edit them before importing or uploading the file.

For storage only nodes, the memory required by the node for SDPM or NVDIMM is specified in the following table::

Table 50. SDPM or NVDIMM storage capacity

Node memory (in GB)	SDPM/NVDIMM capacity required (in GB)
512	128 or 64

Cluster component settings

This table describes the cluster component settings.

Table 51. Cluster component settings

Field name	Description
Select a Component	Select **VMware Cluster** or **PowerFlex Cluster**.
Component Name	Indicates the cluster component name.
Related Components	Select **Associate All** or **Associate Selected** to associate all or specific components to the new component.
Cluster settings (for the VMware cluster component)	
Target Virtual Machine Manager	Select virtual machine manager from the **Target Virtual Machine Manager** list.
Data Center Name	Select the data center name from **Data Center Name** list.
Cluster Name	Select a new cluster name from **Cluster Name** list.
New Cluster Name	Select new cluster name from **New Cluster Name** list.
Cluster HA Enabled	Enables or disables a highly available cluster. You can either select or clear (default) the check box.
Cluster DRS Enabled	Enables or disables distributed resource scheduler (DRS). You can either select or clear (default) the check box.

Table 51. Cluster component settings (continued)

Field name	Description
PowerFlex settings (for the PowerFlex cluster component)	
Target PowerFlex Gateway	Select a target PowerFlex gateway, which acts as an endpoint for PowerFlex API calls. During the provisioning process, PowerFlex Manager connects to the PowerFlex gateway and uses its APIs to configure all storage node and SDC parameters.
Protection Domain Name	Provide a name for the protection domain. You can automatically generate the name (recommended), or specify a new name explicitly. A protection domain is a logical entity that contains a group of storage nodes that provide backup for each other. Each storage node belongs to only one protection domain. Each protection domain is a unique set of storage nodes. A protection domain may also contain SDTs. You can automatically generate the protection domain name or specify a new name explicitly for a storage-only template. The PowerFlex cluster must have at least three nodes before you can publish the template and deploy it as a resource group. However, if you select an existing protection domain that is associated with another previously deployed storage-only resource group, and this protection domain has at least three nodes, PowerFlex Manager recognizes the new template as valid if the cluster has fewer than three nodes. You can publish the template successfully and deploy it as a resource group, since the protection domain it uses has enough nodes. **NOTE:** A compute-only resource group only requires a minimum of two nodes, since it does not have a protection domain. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name** field.
New Protection Domain Name	Specify a new name for the protection domain.
Protection Domain Name Template	If you choose to generate the protection domain name automatically, PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field. If you choose **Compute Only** as the **PowerFlex Role** for the node component, the **PowerFlex** settings do not include the **Protection Domain Name Template** field.
Device Group Name	Provide a name for the device group. You can automatically generate the name (recommended), or specify a new name explicitly. A device group contains devices from four or more nodes in a single protection domain. The nodes that belong to a device group have the same drive type or media type. The total capacity of the device group is derived from the capacity of the storage devices that it contains. Storage pools are allocated through the device group and consume the capacity of the device group.
Device Group Name Template	Define a template for generating the device group name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Persistent Memory Device Group Name	Provide a name for the persistent memory device group. You can automatically generate the name (recommended), or specify a new name explicitly.

Table 51. Cluster component settings (continued)

Field name	Description
Persistent Memory Device Group Template	Define a template for generating the persistent memory device group name automatically. PowerFlex Manager fills this field with a default template that combines static text with variables for several pieces of the autogenerated name. If you modify the template, be sure to include the *${num}* variable to ensure that the name is unique. For details on the rules for defining a template, see the contextual help that appears when you hover over the field.
Device Group Spare Capacity	The device group spare capacity is the additional storage reserved within the system to handle failures and ensure data is available, You must set the spare capacity limit of the device group in terms of nodes for this release. The default spare capacity is one node, and the maximum is two nodes.
Custom Spare Node Count	The spare capacity is set to the size of the largest device in the node multiplied by the number of devices in the node multiplied by the number of nodes specified. For example, if the largest device within a PowerFlex node is 900 GB and there are five devices in the PowerFlex node and two PowerFlex nodes in total, the custom spare node count is 9000 GB (900 GB x 5 devices per node x 2 nodes in total = 9000 GB).
Storage Pool Name	Provide a name for the storage pool. You can automatically generate the name (recommended), select from a list of existing storage pools for the selected protection domain, or specify a unique storage pool name. If you choose to automatically generate a name, you are prompted to define a storage pool name template. Storage pools allow the generation of different storage tiers in PowerFlex. A storage pool is a set of physical storage devices in a protection domain. Each storage device belongs to one (and only one) storage pool. The number of storage pools that are created at deployment time depends on the number of disks available.
CloudLink Center Settings	
Target CloudLink Center	Specifies the target CloudLink Center for an encrypted node. The **Cloud Link Center Settings** are only available when the **Enable Encryption** option is selected in the **OS Settings** for the node component. For any CloudLink Center that has encryption, you must either add an existing resource group to PowerFlex Manager, or use PowerFlex Manager to create the CloudLink encryption machine groups and keystores. PowerFlex Manager is only able to create the machine groups and keystores if the CloudLink Center exists in the management cluster. Ensure that CloudLink Center is available on the resource page. If you select an existing protection domain under the **PowerFlex Settings**, you must choose a **Target CloudLink Center**. The other CloudLink Center settings are not available, and PowerFlex Manager uses the machine group and keystore settings for the selected target. If you automatically generate the protection domain name or specify a new protection name explicitly, you can set the **Machine Group Name** and **Keystore Name** settings with the template. When you deploy a template with CloudLink settings, PowerFlex Manager creates the machine groups and keystores, and then assigns the machine groups to keystores and networks. Next, it adds the machines to the machine groups, and encrypts the drives. After deployment, you can see the machines that are added to machine groups by clicking **View Details** on the **Resources** page.

Table 51. Cluster component settings (continued)

Field name	Description
	When defining a template, you choose a single CloudLink Center as the target for the deployed resource group. If the CloudLink Center for the resource group shuts down, PowerFlex Manager loses communication with the CloudLink Center. If the CloudLink Center is part of a cluster, PowerFlex Manager moves to another CloudLink Center when you update the resource group details.
Machine Group Name	Allows you to provide a name for the machine group. The **Machine Group Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the machine group name or specify a new machine group name explicitly.
New Machine Group Name	Specify a new name for the machine group.
Keystore Name	Allows you to provide a name for the keystore. The **Keystore Name** is only available if you automatically generate the protection domain name or specify a new protection name explicitly. You can automatically generate the keystore name or specify a new keystore name explicitly. Alternatively, you can choose an existing keystore from the list.
Keystore Name Template	Specify a new name for the keystore.

Related information

Add cluster component settings to a template

VM settings

The table describes VM settings for the CloudLink Center and the PowerFlex gateway.

Table 52. VM settings

Field name	Description
Select a Component	Select **CloudLink Center** or **PowerFlex Gateway**.
Number of Instances	Select the number of instances of the VM that you want to deploy.
Related Components	Select **Associate Selected** and select the **VMware Cluster** checkbox to associate the selected VMware cluster with the CloudLink Center.
VM Settings	
Specify Datastore	Specify the storage location.
Specify Network	Select the network type.
CloudLink Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options: • Auto Generate - Select to autogenerate the hostname at deployment time. • Specify at Deployment Time - Select to provide a unique hostname at deployment time.
OS Credential	Select the operating system credential used to set the username and password on the operating system that is installed.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.
Secadmin Credential	Select the secadmin credential, an element manager credential, which is used to set the password for the secadmin user of the CloudLink Center. The password should be minimum of ten characters with at least one special character.

Table 52. VM settings (continued)

Field name	Description
Vault Password	Specify up to three unique passwords for the CloudLink Vault. You must specify at least one. The password should be minimum of ten characters with at least one special character.
Confirm Vault Password	Retype the vault password.
License File	Upload a new or additional software license file. If additional, the resource is added to your current total resources.
Additional CloudLink Settings	
Configure Syslog Forwarding	Select the checkbox to send syslog messages from the CloudLink to a remote network management system.
Syslog Facility	Select the remote server where the syslog messages are forwarded to.
Configure Email Notifications	Select the checkbox to configure email alerts.
Server Address	Specify the IP address of the email server.
Port	Specify the port number for the email server. The default port is 25. The port numbers must be entered in comma-separated list and must be 1–65535.
Sender Address	Specify the email address for the sender.
User Name	Specify the required username.
Password	Specify the required password.
Confirm Password	Retype the password.
PowerFlex Gateway Settings	
Host Name Selection	Defines how the hostname is selected at deployment time. There are two options: • Auto Generate - Select to autogenerate the hostname at deployment time. • Specify at Deployment Time - Select to provide a unique hostname at deployment time.
PowerFlex Credential	Select the operating system credential used to set the username and password.
NTP Server	Specify the IP address of the NTP server for time synchronization. If adding more than one NTP server, ensure that you separate the IP addresses with commas.

Support for full and partial network automation

This topic lists the supported configurations for full and partial network automation.

Table 53. Supported configurations for full and partial network automation

Template components	Full network automation	Partial network automation (Requires manual switch configuration)
OS Images	ESXi, PowerFlex embedded operating system For more information, see PowerFlex Operating System OS Support Matrix	ESXi, PowerFlex embedded operating system For more information, see PowerFlex Operating System OS Support Matrix
PowerFlex roles	Compute-only, storage-only	Compute-only, storage-only (CentOS only)
Switch Port Configuration	• Port channel (LACP enabled) • Port channel	• Port channel (LACP enabled) • Trunk Port

Table 53. Supported configurations for full and partial network automation (continued)

Template components	Full network automation	Partial network automation (Requires manual switch configuration)
	• Trunk port	
Target Boot Device	• Local flash storage for Dell PowerFlex	Local flash storage for Dell PowerFlex
Network Settings	• Static bonding NIC port design or LACP bonding NIC port design • 10 GB, 25 GB, 100 GB • Required PXE network • Network Automation Type: Full	• LACP bonding NIC port design • 25 GB • No PXE network (using iDRAC virtual media) • Network Automation Type: Partial

Related information

Adding an existing resource group
Build and publish a template

Storage nodes

This topic provides reference information for the **Storage Nodes** page.

Add a storage node

You can add a storage node either through a resource group or the PowerFlex Installer.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. Click **+ Add Storage Node**.
   An **Add Storage Node** dialog box opens.
3. If there are resource groups that are installed on your PowerFlex system, add a storage node through the resource groups.
   a. In the **Add Storage Node** window, go to the resource groups by clicking **Continue to Resource Groups**.
   You are redirected to the **Resource Groups** page.
   b. Select the resource group that you want to add a storage node and click **View Details** > **Add Resources** > **Add Nodes**.
4. If the nodes in your environment are not deployed through the resource groups, add a storage node through the getting started page.
   a. In the **Add Storage Node** dialog box, click **Continue to Getting Started**.
   The **PowerFlex Installer** wizard opens.
   b. Use the **PowerFlex Installer** wizard to add a storage node.

Related information

Getting started
Adding nodes to a resource group

View objects related to a storage node

You can view the protection domain, device groups, and devices associated with a specific storage node. You can view the overall health of the storage node and devices.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node whose related objects you want to view.
3. In the right hand pane, click on any of the objects related to the storage node to view the object details in the storage system.

View component details of a storage node

You can view the details of the MDM, PowerFlex data server (PDS), device gateway target (DGWT), and devices associated with a storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node whose component details you want to view, and click **View Details**.
   A **Storage Nodes** page opens to display the details of the MDM, PowerFlex data server (PDS), device gateway target (DGWT), and devices.
3. To view the MDM, click **View Details**.
   You are redirected to **Resources**, where you can view the MDM cluster details, including the MDM role, name, management IP address, IP addresses, port, and status.
4. If you want to view details of other components, you must go back to **Lifecycle** > **Storage Nodes** and select the check box beside the storage node whose component details you want to view, and click **View Details**.

Add a device to a storage node

You can add a device to a storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to add a device to, and click **More Actions** > **Add Devices**.
   A **Add Device to Storage Node** dialog box opens.
3. From the **Add Device to Storage Node** dialog box, enter the following information:
   - **Path**: Enter a path for the device.
   - (Optional)**Name**: Enter a name for the device.
   - **Device Group**: From the drop-down, select the device group that the device will be part of.
   - **Media Type**: The media type is automatically populated based on the device group that you selected.
4. Click **Add devices**.
5. (Optional) You can add additional devices by clicking **Add row**.
6. (Optional) You can choose to set up device tests through **Advanced Options**. You can choose from the following settings:
   - **Test and activate device**: This operation tests whether the device is readable and writable. If it is readable and writable, the device is activated.
   - **Test only**: This operation tests whether the device is readable and or writable.
   - **Activate without test**: This operation activates the device without testing the devices.
   - **Timeout**: This setting specifies the time-out for each operation.
   - **Device Takeover**: This setting enables the MDM to use the device for the operations.
7. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a device from a storage node

You can remove a device from a storage node. However, removing a device reduces the capacity, and potentially enters the storage node into a state where the storage node does not have the same amount of capacity for failover and resiliency.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to remove a device from and click **More Actions** > **Remove Device**.
3. In the **Remove Device** dialog box, verify you selected the wanted device and that the removal impacts the correct device group.
4. Type **`REMOVE DEVICE`** in the box and click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Remove a storage node

When you remove a storage node, the devices that are associated with the storage node are removed.

You cannot remove a storage node with an active MDM role.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to remove , and click **More Actions** > **Remove Storage Node**.
3. In the **Remove Storage Node** dialog box, verify that you selected the wanted storage node.
4. Type **REMOVE STORAGE NODE** in the box and click **Remove**.
5. Verify that the operation is finished and is successful, and click **Dismiss**.

Related information

Enter a storage node into maintenance mode

Abort a storage node removal

You can stop the removal of a storage node during the remove operation.

You must start the remove storage node operation before you can stop the removal of the storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to stop the removal of, and click **More Actions** > **Abort Remove Storage Node**.
3. In the **Abort Removal for Storage Node** dialog box, verify that you want to stop the removal of the storage node and click **Abort**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Enter a storage node into maintenance mode

When a storage node is put into maintenance both the PowerFlex data server and the device gateway target go into maintenance mode.

If the storage node that is entering into maintenance mode is serving as an NVMe host, you must go to NVMe hosts and put the node into maintenance mode before starting a reboot of this storage node.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the check box beside the storage node that you want to enter into maintenance mode and click **More Actions** > **Enter Maintenance Mode**.
3. In the **Enter Storage Node into Maintenance Mode** dialog box, verify that you are selecting the wanted storage node, and click **Enter Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Related information

Enter an NVMe target into maintenance mode

Exit a storage node from maintenance mode

If the devices associated with the storage node are error free, you can exit a storage node from maintenance mode.

1. On the menu bar, click **Lifecycle** > **Storage Nodes**.
2. From the list of storage nodes, select the checkbox beside the storage node that you want to exit from maintenance mode and click **More Actions** > **Exit Maintenance Mode**.
3. In the **Exit Maintenance Mode on Storage Node** dialog box, verify that you are selecting the wanted storage node, and click **Exit Maintenance Mode**.
4. Verify that the operation is finished and is successful, and click **Dismiss**.

Resources

This section provides reference information for the **Resources** page.

Related information

Deploying and provisioning
Managing components
Monitoring system health

Resource health status

PowerFlex Manager assigns health status to the resources based on the conditions described in the following table.

Table 54. Resource health status

Icon	Health status	Description
	Healthy	Indicates that there is no issue with the resource and it is working as expected.
	Warning	Indicates that the resource is in a state that requires corrective action, but does not affect overall system health. For example, the firmware running on the resource is not at the required level or not compliant. For supported switches, a warning health status might be because the SNMP community string is invalid or not set correctly. In this case, PowerFlex Manager is unable to perform health monitoring for the switch. **NOTE:** If the resource group has VMware ESXi and is set to maintenance mode, a message is displayed in the **View Details** pane.
	Critical	Indicates that an issue requiring immediate attention exists in one of the following hardware or software components in the device: • Battery • CPU • Fans • Power supply • Storage devices • Licensing For supported switches, a critical health status might indicate that the power supply is not working properly or a CPU has overheated. **NOTE:** If the resource group is in a power off state, a message displays in the **View Details** pane.
	Unknown	Indicates that the resource status is unknown. For example, for a supported switch, an unknown status might indicate that the switch has been turned off or is still being discovered.
	Pending	Indicates that the resource status is pending.
	Service Mode	Indicates that the resource is in **Service Mode**. **NOTE:** If the resource fails to exit **Service Mode**, an informational message displays in the recent activity.

Related information

Discover a resource
Removing resources

Resource operational state

After initiating resource discovery, PowerFlex Manager assigns one or more of the states to the resources. These operational states display on the **Resources** page, in the **Deployment Status** column of the **All Resources** tab.

Table 55. Deployment status

Deployment status	Description
Not In Use	Resource is available for deployment.
Available	Resource is available. Applies only to switches, virtual machine, or gateway.
Deploying	Resource is in the process of being deployed in a resource group.
In Use	Resource is deployed in a resource group.
Pending Updates	One or more of the following tasks are in progress: • Discovering resource • Determining resource details, including firmware version • Applying template to the resource • Updating firmware • Removing resource from PowerFlex Manager inventory
Deployment Failed	Resource group deployment failed.
Cancelled	Resource group deployment is cancelled.

PowerFlex Manager keeps track of which resources it is managing. These operational states display on the **Resources** page, in the **Managed State** column of the **All Resources** tab.

Table 56. Managed state

Managed state	Description
Managed	Indicates that PowerFlex Manager manages the firmware on that node, and the node can be used for deployments.
Unmanaged	Indicates that the resource is not managed by PowerFlex Manager. By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, you need to select the node and change the operational state to **Managed**. If you have not yet uploaded a license file, PowerFlex Manager is configured for monitoring and alerting only. All resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.
Reserved	Indicates that PowerFlex Manager only manages the firmware on that particular node, but that node cannot be used for deployments. You can assign a host to the reserved state only if the host has been discovered, but is not part of a resource group.

Compliance status

PowerFlex Manager assigns one of the following firmware statuses to the resources:

Table 57. Compliance status

Firmware status	Description
Compliant	The firmware running on the resource is compliant with the firmware version that is specified in the default compliance version.
Non–Compliant	The firmware running on the resource is less than or greater than the firmware version specified in the default compliance version. Indicates that firmware update is required.
Update Required	The firmware running on the resource is less than the minimum firmware version recommended in the default compliance version. Indicates that firmware update is required.

Related information

Discover a resource
Removing resources

Discovery overview

You can discover new resources or existing resources that are already configured within your environment. After discovery, you can deploy resource groups on these resources from a template. Only administrator-level users can discover resources.

By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, select the node and change the operational state to **Managed**.

If you have not yet uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, all the resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

For some resources such as nodes, the default credentials are prepopulated in PowerFlex Manager. If the credentials are changed from the defaults, add the credential to PowerFlex Manager with the new login information.

Element Manager discovery

The IPI cabinet is discovered as an element manager. Once PowerFlex Manager discovers the IPI cabinet, you can view cabinet details or you can open the IPI appliance management application to view more details.

The CloudLink Center is discovered as an element manager. If you create your own username in CloudLink Center, you need the SecAdmin access role and the local user type for the new user to complete discovery. The client user type does not work for discovery in PowerFlex Manager. If one member of a cluster of CloudLink Center is shut down, the inventory still succeeds because PowerFlex Manager falls back to inventorying the other CloudLink Center members if the first one is down.

Node discovery

PowerFlex Manager supports PowerFlex node discovery and allows you to onboard nodes by configuring the initial management IP address and iDRAC credentials. To perform initial discovery and configuration, verify that the management IP address is set on the node and that PowerFlex Manager can access the IP address through the network. While configuring IP addresses on the node, verify that PowerFlex Manager can access any final IP address in a range used for hardware management, to complete discovery of these nodes.

PowerFlex Manager also allows you to use name-based searches to discover a range of nodes that were assigned IP addresses by DHCP to iDRAC. You can search for a range of DNS hostnames or a single hostname within the **Discovery Wizard**. After you perform a name-based discovery, PowerFlex Manager operations to the IDRAC continue to use name-based IP resolution, since DHCP may assign alternate addresses.

Switch discovery

If you attempt to discover a Cisco switch with terminal color configured, the discovery fails. To discover the switch successfully, disable the terminal color option by running `configure no terminal color persist`.

VM manager discovery

A VMware vCenter is discovered as a VM manager. PowerFlex Manager users with the administrator role can discover a vCenter in PowerFlex Manager. A vCenter read-only user can discover a vCenter in PowerFlex Manager only after the following requirements are met:

- The vCenter user who is specified in the vCenter credential is granted the `VirtualMachine.Provisioning.ReadCustSpecs` and `StorageProfie.View` privileges.
- The permission containing these privileges is granted to that user on the root vCenter object and the `Propogate to children` property is set to `True`.

PowerFlex Manager allows you to deploy new hyperconverged and compute-only resource groups and add existing resource groups to a vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple vCenter servers, allowing you to search across servers and perform other vCenter management functions. ELM does not provide clustering or redundancy.

If you are working with a vCenter that has an enhanced linked mode configuration, you must discover only the vCenter to which you want to deploy or add an existing resource group. You do not need to discover the other vCenters that are connected through the enhanced linked mode configuration.

Discovery overview

You can discover new resources or existing resources that are already configured within your environment. After discovery, you can deploy resource groups on these resources from a template. Only administrator-level users can discover resources.

By default, the operational state for all discovered nodes is **Unmanaged**. If you want to perform firmware updates or deployments on a discovered node, select the node and change the operational state to **Managed**.

If you have not yet uploaded a license, PowerFlex Manager is configured for monitoring and alerting only. In this case, all the resources are restricted to the **Unmanaged** resource state, and you cannot change the state to **Managed** or **Reserved**.

For some resources such as nodes, the default credentials are prepopulated in PowerFlex Manager. If the credentials are changed from the defaults, add the credential to PowerFlex Manager with the new login information.

Element Manager discovery

The IPI cabinet is discovered as an element manager. Once PowerFlex Manager discovers the IPI cabinet, you can view cabinet details or you can open the IPI appliance management application to view more details.

The CloudLink Center is discovered as an element manager. If you create your own username in CloudLink Center, you need the SecAdmin access role and the local user type for the new user to complete discovery. The client user type does not work for discovery in PowerFlex Manager. If one member of a cluster of CloudLink Center is shut down, the inventory still succeeds because PowerFlex Manager falls back to inventorying the other CloudLink Center members if the first one is down.

Node discovery

PowerFlex Manager supports PowerFlex node discovery and allows you to onboard nodes by configuring the initial management IP address and iDRAC credentials. To perform initial discovery and configuration, verify that the management IP address is set on the node and that PowerFlex Manager can access the IP address through the network. While configuring IP addresses on the node, verify that PowerFlex Manager can access any final IP address in a range used for hardware management, to complete discovery of these nodes.

PowerFlex Manager also allows you to use name-based searches to discover a range of nodes that were assigned IP addresses by DHCP to iDRAC. You can search for a range of DNS hostnames or a single hostname within the **Discovery Wizard**. After you perform a name-based discovery, PowerFlex Manager operations to the IDRAC continue to use name-based IP resolution, since DHCP may assign alternate addresses.

Switch discovery

If you attempt to discover a Cisco switch with terminal color configured, the discovery fails. To discover the switch successfully, disable the terminal color option by running `configure no terminal color persist`.

VM manager discovery

A VMware vCenter is discovered as a VM manager. PowerFlex Manager users with the administrator role can discover a vCenter in PowerFlex Manager. A vCenter read-only user can discover a vCenter in PowerFlex Manager only after the following requirements are met:

- The vCenter user who is specified in the vCenter credential is granted the `VirtualMachine.Provisioning.ReadCustSpecs` and `StorageProfie.View` privileges.
- The permission containing these privileges is granted to that user on the root vCenter object and the `Propogate to children` property is set to `True`.

PowerFlex Manager allows you to deploy new compute-only resource groups and add existing resource groups to a vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple vCenter servers, allowing you to search across servers and perform other vCenter management functions. ELM does not provide clustering or redundancy.

PowerFlex Manager allows you to deploy new compute-only resource groups and add existing resource groups to a vCenter that has an Enhanced Linked Mode (ELM) configuration. ELM connects multiple vCenter servers, allowing you to search across servers and perform other vCenter management functions. ELM does not provide clustering or redundancy.

If you are working with a vCenter that has an enhanced linked mode configuration, you must discover only the vCenter to which you want to deploy or add an existing resource group. You do not need to discover the other vCenters that are connected through the enhanced linked mode configuration.

Node pools

In PowerFlex Manager, a node pool is made up of nodes that are grouped for specific use cases such as business units or workload purposes. An administrator can specify which users can access these node pools.

Standard users can view details only for node pools for which they have permission.

From the **Node Pools** tab, you can view existing node pools.

Users with an administrator role can create, edit, or delete node pools.

Click a node pool from the list to view detailed information about the following tabs:

- **Nodes**: Displays the number of nodes that are associated with the node pool.
- **Users**: Displays the number of users with access rights to the node pool.

Configuration checks

When you initiate a resource group action, PowerFlex Manager performs critical configuration checks to ensure that the system is healthy before proceeding. A configuration check failure may result in the resource group action being blocked. You can export a PDF report at any time that shows detailed system health and configuration data. When you generate the report, PowerFlex Manager initiates the full set of configuration checks and reports the results.

Configuration checks for switches

Dell Networking OS9 switches are not covered by the configuration checks.

Table 58. Configuration checks

Configuration check	Resource group actions blocked if the check fails
PowerFlex gateway is powered off	None
PowerFlex cluster state is not normal	• Add node • Add volume

Table 58. Configuration checks (continued)

Configuration check	Resource group actions blocked if the check fails
	• Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
Secondary MDMs have a status not equal to normal	• Add nodes • Add volumes • Delete node • Delete resource group • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Enter service mode • Drive replacement
storage node registered data network IP address count does not equal 2 or 4	• Compliance upgrade • Enter service mode • Drive replacement • Reconfigure MDM roles
storage node has a state not equal to connected	• Add volume • Compliance upgrade • Reconfigure MDM roles
storage node interfaces have incorrect MTU sizes	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade • Reconfigure MDM roles
VMware ESXi vmks are not configured with an MTU size of 9000	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
Interfaces are not configured with an MTU size of 9216, and QOS is enabled with MTU size	• Add node • Add resource group with an existing PowerFlex cluster • Compliance upgrade
storage node has a Performance Profile not equal to High Performance	None
SDC has a Performance Profile not equal to High Performance	None

Related information

Exporting a configuration report for all resources and resource groups

Settings

This section provides reference information for the **Settings** page.

Backup details

PowerFlex Manager backup files include the following information:

- Activity logs
- Credentials

- Deployments
- Resource inventory and status
- Events
- Initial setup
- IP addresses
- Jobs
- Licensing
- Networks
- Templates
- Users and roles
- Resource module configuration files
- Performance metrics

Network types

You can manage various network types in PowerFlex Manager.

- **General Purpose LAN**—Used to access network resources for basic networking activities.
- **Hypervisor Management**—Used to identify the management network for a hypervisor or operating system that is deployed on a node.
- **Hypervisor Migration**—Used to manage the network that you want to use for live migration. Live migration enables you to move running virtual machines from one node of the failover cluster to different node in the same cluster.
- **OS Installation**—Allows static or DHCP network for operating system imaging on nodes.
- **Hardware Management**—Used for out-of-band management of hardware infrastructure
- **PowerFlex Data**—Used for data traffic between storage node and storage data clients (SDC)
- **PowerFlex Data (Client Traffic Only)**—Used for storage data client traffic only
- **PowerFlex Data (Server Traffic Only)**—Used for storage node traffic only
- **PowerFlex Management**—Used for PowerFlex system management